ADVANCE TRAUMA

Clinical Trial Protocol

Effects of Advanced Trauma Life Support® Training Compared to Standard Care on Adult Trauma Patient Outcomes: A Stepped-Wedge Cluster Randomised Trial

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1 Visual abstract

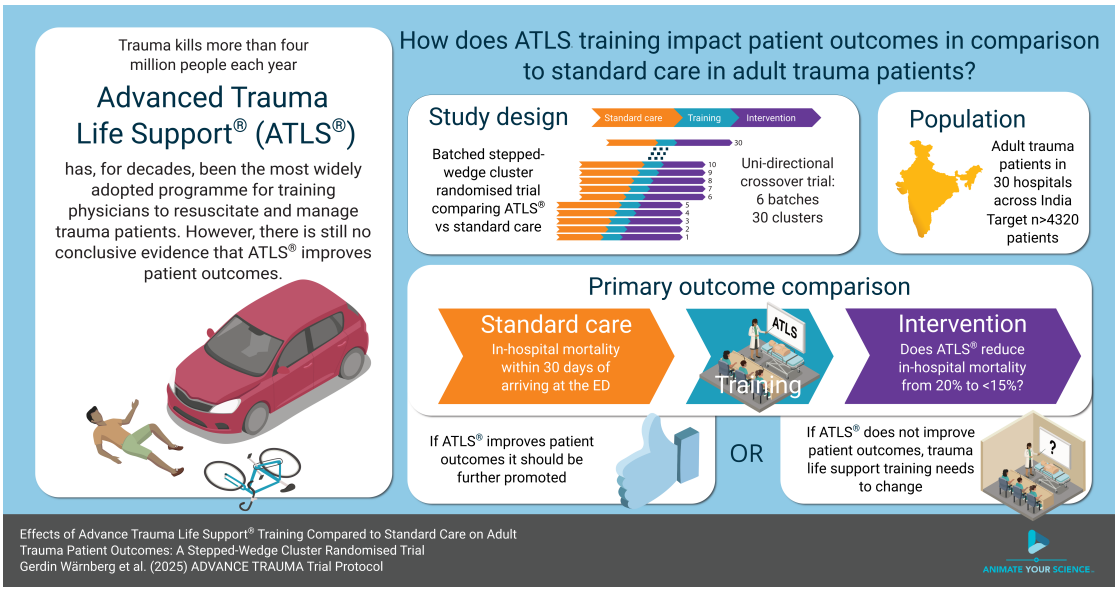


Figure 1: ADVANCE TRAUMA visual abstract.

2 Administrative information

2.1 Changelog

Version	Date	Details
1.6.0	?	• Update title page and text formatting • Updated visual abstract • Shortened very long lists of variable categories • Added references to published systematic review, feasibility study and community consultations pre-print • Removed forest plot after adding the systematic review reference • Remove note on randomisation
1.5.0	2025-06-12	• Clarified the definition of clusters and units, based on discussions during the writeup of the protocol for publication • Clarified the patient exclusion criterion of isolated limb injury as isolated closed extremity fracture • Included “stepped-wedge” in the title • Corrected Eligibility criteria heading numbering • Added citation for the nested staircase design • Revise intention to treat terminology
1.4.0	2025-04-30	• Updated systematic review results and references • Added visual abstract • Added details on the nested staircase design for measuring adherence, quality of life and disability • Removed return to work as a secondary outcome to be measured using the nested staircase design, this should be measured using the main stepped-wedge design
1.3.0	2024-11-15	• Updated names of events in the table of procedures • Added new references • Added nested staircase design for measuring adherence, quality of life, disability and return to work • Updated small sample correction to be based on best available evidence closer to the time of analysis • Added contributors • Removed reassessment of the sample size calculation from the interim analysis • Revised details on measuring ATLS adherence
1.2.0	2024-08-26	• Added details on measuring ATLS adherence • Clarified the section describing the consent process • Fixed minor issues with how the variables were listed • Indicated non-routinely recorded data in the list of variables • Added Administrative information section with contributors • Added CTRI registration number

Version	Date	Details
1.1.0	2024-05-09	Updated the primary outcome to in-hospital mortality and spelling corrections. The primary outcome was updated following a voting procedure in the Trial Management Group.

2.2 Study identifiers

- ClinicalTrials.gov identifier: NCT06321419
- Clinical Trials Registry - India identifier: CTRI/2024/07/071336

2.3 Contributors

The following have contributed to the design and implementation of the trial:

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Abbreviations: TMG, Trial Management Group; TT, Trial Team.

3 Synopsis

Title Effects of Advanced Trauma Life Support® Training Compared to Standard Care on Adult Trauma Patient Outcomes: A Stepped-Wedge Cluster Randomised Trial

Rationale Trauma is a massive global health issue. Many training programmes have been developed to help physicians in the initial management of trauma patients. Among these programmes, Advanced Trauma Life Support® (ATLS®) is the most popular, having trained over one million physicians worldwide. Despite its widespread use, there are no controlled trials showing that ATLS® improves patient outcomes. Multiple systematic reviews emphasise the need for such trials.

Aim To compare the effects of ATLS® training with standard care on outcomes in adult trauma patients.

Primary Outcome In-hospital mortality within 30 days of arrival at the emergency department.

Trial Design Batched stepped-wedge cluster randomised trial in India.

Trial Population Adult trauma patients presenting to the emergency department of a participating hospital.

Sample Size 30 clusters and at least 4320 patients.

Eligibility Criteria

Clusters are secondary or tertiary hospitals providing trauma care in India that admit or refer/transfer for admission at least 400 patients with trauma per year. Within each cluster, one or more units of physicians providing initial trauma care in the emergency department will be trained.

Participants are adult trauma patients who present to the emergency department of participating hospitals and are admitted or transferred for admission.

Intervention The intervention will be ATLS® training, a proprietary 2.5 day course teaching a standardised approach to trauma patient care using the concepts of a primary and secondary survey. Physicians will be trained in an accredited ATLS® training facility in India.

Ethical Considerations We will use an opt-out consent approach for collection of routinely recorded data. We will obtain informed consent for collection of non-routinely recorded data, such as quality of life and disability outcomes. Patients who are unconscious or lack a legally authorized representative will be included under a waiver of informed consent. Note that consent here refers to consent to data collection.

Trial Period February 2025 to October 2029

4 Background and rationale

Each year, 4.9 million people die from trauma[1]. Among people aged 10-49 years trauma is the largest cause of disability adjusted life years[2]. Most deaths from trauma occur within the first 24-48 hours[3]. Traumatic brain injury and exsanguination are the most common causes of trauma deaths[4,5]. Most preventable trauma deaths are caused by clinical judgement errors during initial resuscitation or early care including airway management and haemorrhage control, even though the deaths occur later during the hospital stay[4,6].

Several trauma life support training programmes have been developed to improve the early management of patients in the hospital by providing a structured framework for assessment and treatment[7–11]. The proprietary Advanced Trauma Life Support® (ATLS®) is the most established trauma life support training programme and more than one million physicians in over 80 countries have been trained in the programme since the first course in 1978[12]. In the US and many other countries training in ATLS® is virtually mandatory for trauma care physicians[13]. Uptake in low- and middle income countries (LMIC) has been slow, potentially due to high costs[9].

There are three randomised studies showing that ATLS® improves knowledge and clinical skills[14–16], but there are no randomised controlled trials or high-quality quasi-experimental trials indicating that ATLS® improves patient outcomes[7,8,10,11,17]. We conducted an updated systematic review[18], and estimated a pooled odds ratio of 0.51 (95% CI 0.37-0.69) from 10 heterogeneous (I^2 69.1) observational studies on the effect of ATLS on mortality[19–28].

We conducted a pilot cluster randomised controlled trial (ClinicalTrials.gov NCT05417243) between April 2022 and February 2023 as part of our network grant to assess the feasibility of a full scale trial[29,**GerdinWärnberg2022?**]. Our pilot study enrolled 376 patients from seven hospitals across India and shows that it is feasible to conduct the proposed trial with a high percentage of patients consenting to out of hospital follow up (78%), low loss to follow-up rate (1%), and low missingness in key variables (mean 0.8%).

To involve patients and the public in the planning of this trial we conducted 19 semi-structured interviews with trauma patients, caregivers, and community representatives[30]. The aim of these interviews was to understand their views on the trial and important outcomes and the interviews showed high acceptability of our research and emphasised the importance of better recovery before discharge and functional outcomes at and after discharge, including pain, mobility and self-care activities. The interviews also highlighted return to work as an important outcome.

5 Benefit-risk evaluation

The direct risks includes integrity violations and data leakage. We will mitigate these risks by employing rigorous data collection and storage mechanisms. The procedures that we will use to collect data will be direct observation of care, routine physical examinations, questionnaires, and extraction of already collected data from patient records, which are often seen as involving only minimal risk.

The long-term risks of the research and the risk that the research will be used in detrimental ways are minimal. Our trial will assess the effect of Advanced Trauma Life Support® (ATLS®) on patient outcomes. Training in ATLS® is standard in many health care systems and it is unlikely that training physicians in this programme induces any harm to participants.

We consider these risks weighed up by the potential direct benefit for the participants in the intervention phase, if ATLS® is found to improve patient outcomes, and by the potential for improved care for the trauma patient population.

6 Trial aim

To compare the effects of ATLS® training with standard care on outcomes in adult trauma patients.

7 Regulatory approvals and trial registration

We will submit this trial to the Health Ministry Screening Committee at the Indian Council for Medical Research for their approval. We will apply for ethical approvals from each participating hospital, The George Institute for Global Health in India and the Swedish Ethical Review Authority. We will register this trial with Clinical Trials Registry-India and ClinicalTrials.gov.

8 Trial design and procedures

8.1 Overall trial design

We will conduct a batched stepped-wedge cluster randomised controlled trial (see Figure 2). The stepped-wedge trial is a uni-directional cross-over trial but the time point when clusters cross-over from standard care to the intervention is randomised[31]. In this trial, the unit of randomisation is the hospital: that is, the "clusters" randomised to the sequences of the stepped wedge design are the hospitals. Within each hospital, one or more units of physicians performing initial resuscitation of trauma patients in the emergency department will be trained. The number of units that will be trained in each hospital will depend on the sizes of these units and the volumes of patients the physicians attend. If more than one unit is trained in the same hospital, these are all considered to be part of the same cluster (i.e. hospital) for the purpose of randomisation. We choose this approach for two reasons: 1) it will not be logistically or financially feasible to train all physician in a given hospital; and 2) we need to balance cluster size with the number of clusters. We will conduct this trial in India because physicians providing initial trauma care in India are so far not routinely trained in ATLS® or similar programmes.

We will roll out the interventions to 30 clusters over six batches, so there will be five clusters in each batch. The clusters in each batch will be randomised to one of five implementation sequences, with one hospital randomised to each implementation sequence. All clusters will transition through three phases, first a standard care phase, then a one month transition phase during which the training is delivered, and finally an intervention phase, for a total of 13 months. The implementation sequence determines

how long the phases of standard care and intervention are. Patient participants will be followed up for a total of three months.

Within the main stepped-wedge design, we will nest a staircase design to measure a range of secondary outcomes (see Figure 3)[32]. The staircase design will include a random subset of patients presenting during the three months preceding the transition phase, and the three months following the transition phase.

8.2 Design justification

We use the cluster randomised design because the intervention cannot be randomised at the individual patient level. We use the stepped-wedge design for two reasons. First, this design is statistically more efficient than the parallel cluster design when the number of clusters is limited[33]. In this trial, the number of clusters is limited because of the costs associated with ATLS® training and the available slots for ATLS® training in India. Second, the stepped-wedge design is likely to enhance participation and engagement because all clusters receive the intervention. The batched stepped-wedge design further improves feasibility as it does not require all clusters to start at the same time, and it is robust to potential delays in cluster recruitment[34]. We nest a staircase design within the main design because some of the secondary outcomes will be considerably more labour intensive to collect than the primary outcome, and collecting these for all patient participants would be unfeasible.

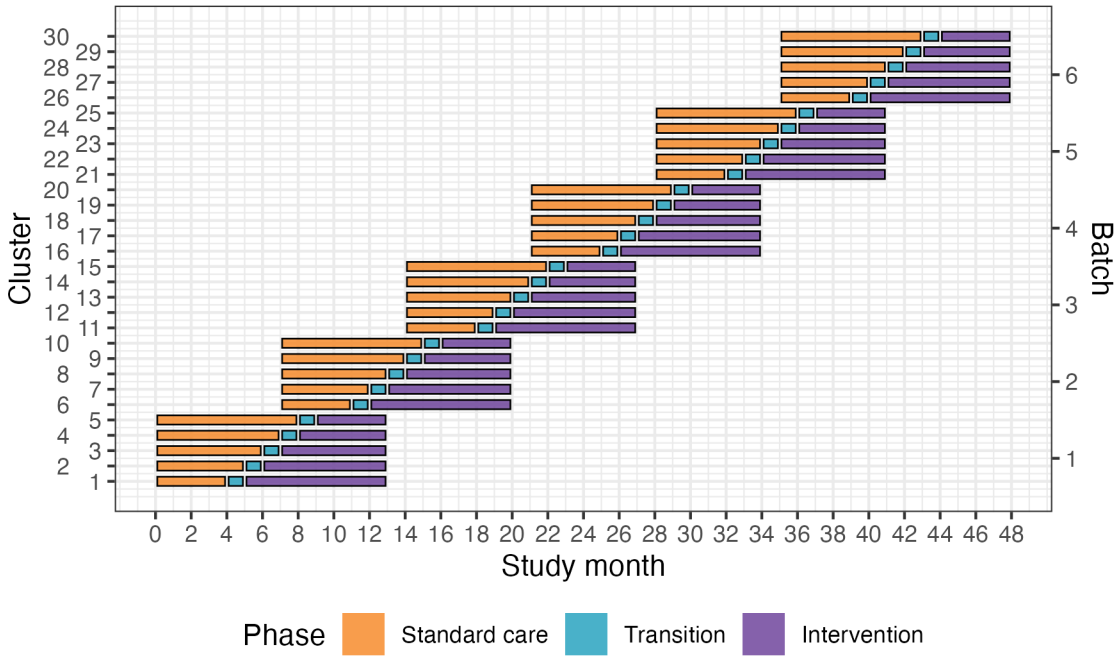


Figure 2: Trial design. Lines represent the duration of patient enrolment across clusters and phases. Clusters will be sequentially allocated to a batch based on when they enter the study. Within each batch clusters will then be randomised to an intervention implementation sequence.

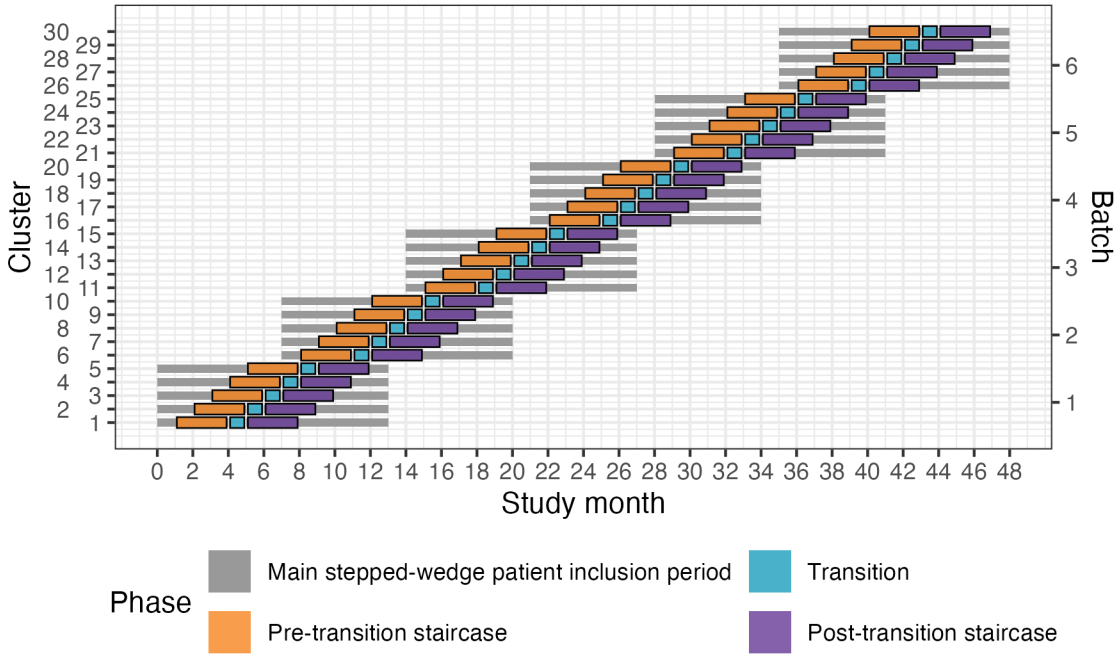


Figure 3: Nested staircase design. The design includes a random subset of patients presenting during the three months preceding the transition phase, and the three months following the transition phase.

8.3 Eligibility criteria

Our trial has eligibility criteria on the cluster and patient participant level.

8.3.1 Cluster selection

Clusters will be secondary or tertiary hospitals providing trauma care in India. This will be the unit of randomisation. Within each cluster, one or more units of physicians providing initial trauma care in the emergency department will be trained. These units already exist in the hospitals and rotate through the emergency department on specific days of the week.

8.3.1.1 Inclusion criteria

Clusters must meet the following criteria:

- admit or refer/transfer for admission at least 400 patients with trauma per year or 35 patients with trauma per month for at least the last six months;
- provide surgical and orthopaedic emergency services around the clock; and
- have at most 25% of physicians providing initial trauma care trained in a formalised trauma life support training programme, like ATLS® or Primary Trauma Care (PTC).

Additionally, the units to be trained within the cluster must meet the following criteria:

- admits or refers/transfers for admission at least 12 patients with trauma per month for at least the last six months; and
- no more than 25% of physicians providing initial trauma care trained in a formalised trauma life support training programme.

8.3.1.2 Exclusion criteria

Hospitals are excluded if they meet any of the following criteria:

- implement a formalised trauma life support training programme [1] during the trial period; or
- plan to implement or implement other major interventions[2] that affects trauma care during the trial period.

[1] These include but are not limited to the National Emergency Life Support (NELS) programme, the Basic Trauma Life Support (BTLS) programme, the Pre-Hospital Trauma Life Support (PHTLS) programme, the Trauma Nursing Core Course (TNCC) and the Advanced Trauma Care for Nurses (ATCN) programme.

[2] These include but are not limited to implementing of a trauma team approach, opening a trauma centre and implementing a trauma quality improvement programme.

8.3.1.3 Screening

The trial management group will compile a list of hospitals with potentially eligible clusters and reach out to them to assess their interest in participating in the trial. We will then screen hospitals for eligibility based on the criteria above, using a two-step procedure. First, we will approach hospitals to complete an initial hospital screening instrument (see Appendix Initial hospital screening instrument (Section 20.1)). We will then discuss each eligible hospital individually in the Trial Management Group before deciding whether to include it in the trial. We have this discussion because we strive to include hospitals that to a large extent conducts primary resuscitation of trauma patients, rather than hospitals that primarily receives transferred patients from other hospitals, but this is difficult to formalise in the eligibility criteria. We will then perform a more in-depth interview with selected hospitals (See Appendix In-depth hospital screening interview instrument (Section 20.2)). To avoid excluding hospitals we will also discuss plans to implement other potentially competing interventions during the trial period, and take these plans into account when assigning clusters to batches. For example, we are aware of the ongoing implementation of the National Emergency Life Support (NELS) programme in India, and will therefore not include hospitals that plan to implement this programme during the trial period. All screening steps and decisions will be logged using REDCap[35,36].

8.3.2 Patient participant selection

Patient participants are adult trauma patients who presents to the emergency department of participating hospitals and are admitted or transferred for admission.

8.3.2.1 Inclusion criteria

Patients participants must meet the following criteria:

- age of at least 15 years;
- trauma occurred less than 48 hours before arrival at the hospital;
- present to the emergency department of participating hospitals, with a history of trauma defined as having any of the reasons listed in the International Classification of Diseases chapter XX as the reason for presenting;
- admitted, or died between arrival at the hospital and admission, or referred/transferred from the emergency department of a participating hospital to another hospital for admission; and
- managed by a participating unit in the emergency department.

8.3.2.2 Exclusion criteria

Patients participants are excluded if they meet the following criteria:

- present with isolated closed extremity fracture; or
- are directly admitted to a ward without being seen by a physician in the emergency department.

8.3.2.3 Screening

Clinical research coordinators will screen patient participants either as they arrive to the emergency department or using emergency department registers. The patients or their representatives will receive written information about the study before they are discharged, including about their right to opt out at any time before final analysis. Phone numbers for out of hospital follow up will be extracted from the emergency department registers, and will be securely held only by the clinical research coordinators at each sites.

8.3.2.4 Withdrawal criteria

Patient participants can choose to withdraw their consent for collection of non-routinely recorded data at any time before the final analysis. If they withdraw their consent for this data collection the clinical research coordinator will not collect any more of this data, which also means that no further follow-ups will be conducted. They can also choose to have the data already collected about them removed from the trial at any time before final analysis of the data. Withdrawal of consent or removal of data from the trial will not affect their care in any way. If the patient participant withdraws consent, follow-up of this participant will be performed according to the participating hospitals routine.

8.4 Procedures

Table 3 shows an overview of trial procedures before and during patient admission, and Table 4 shows an overview of trial follow-up procedures. Clinical research coordinators will follow up patients daily until discharge to capture injury information. They will also follow up patients at 24 hours, 30 days and 90 days after arrival to the emergency department to capture mortality outcomes, and at 30 days and 90 days after arrival to the emergency department to capture functional outcomes and return to work. If patient participants are discharged before any of these follow-up time points, clinical research coordinators will follow up patients by phone.

8.5 Biological sampling procedures

This trial does not include biological sampling.

8.6 End of trial

The trial ends when the last patient participant has completed the last follow-up. The trial may be prematurely terminated if it this is necessary for safety reasons affecting the risk-benefit balance or if the recruitment of subjects cannot be met within reasonable time limits. If the trial is prematurely terminated or suspended, the investigator should immediately inform the subjects about this and ensure appropriate treatment and follow-up. Decisions on premature termination are taken by the joint Trial Steering and Data Monitoring Committee and Trial Management Group.

Table 3: Overview of trial procedures before and during patient admission

Procedure	Screening	Consenting	Initial assessment	In-hospital care
Eligibility criteria	√			
Study information[1]		√		
Informed consent[1]		√		
Baseline data collection			√	
Prehospital data collection			√	
ATLS adherence[2]			√	
ED data collection[3]			√	
Hospital data collection				√
Surgery data collection				√
Imaging data collection				√
Transfusion data collection				√
Injury data collection				√
Mortality data collection				√
Assessment of safety events				√

[1]Clinical research coordinators will inform patient participants about the study, including that they are free to withdraw their data from the study at any time, and approach them for informed consent for collection of non-routinely recorded data in person or telephonically.
[2]ATLS adherence will be assessed by observing the care provided to a random sample of patient participants.
[3]Emergency Department

Table 4: Overview of trial follow-up procedures

Procedure	Within 7 days of discharge	30 days	90 days
Mortality data collection[1]	√	√	√
EQ-5D/WHODAS	√	√	√
Return to work		√	√
End of study			√

[1]Will be ascertained daily from when the patient participant arrive to hospital until they leave the hospital, are discharged or die.

8.7 Intervention and control treatment

The intervention will be ATLS® training. The control will be standard care, meaning no formal trauma life support training. We will train the physicians that initially resuscitate and provide trauma care during the first hour after patient arrival at the emergency department. These physicians can be casualty medical officers, surgical residents, or emergency medicine residents, depending on the setup at each participating centre. The training will occur during the transition phase in each cluster. Our experience from our pilot study is that study sites adhere to the training slot alloted to them through the trial, so we judge the risk of clusters implementing ATLS® before their randomised implementation sequence as very low.

We will train the number units of physicians needed to reach the required patient sample size, but estimate that this will require training an average of ten physicians per hospital, which on average should be mean that we can train one to two units per hospital. This is possible because many hospitals in India organise physicians staffing their emergency departments in units, and the physicians in the same unit work together in the emergency department on the same days of the week. We will therefore collect data only on the days when these units work. The units selected to constitute a cluster from each hospital will be a convenience sample out of all eligible units in those hospitals.

Advanced Trauma Life Support® (ATLS®)[12] is a proprietary 2.5 day course teaching a standardised approach to trauma patient care using the concepts of a primary and secondary survey. The programme was developed by the Committee of Trauma of the American College of Surgeons. The course includes intial treatment and resuscitation, triage and interfacility transfers. Leaning is based on practical scenario-driven skill stations, lectures and includes a final performance proficiency evaluation. Physicians will be trained in an accredited ATLS® training facility in India. We will assess adherence to ATLS principles before and after implementing ATLS training.

Standard care varies across hospitals in India, but trauma patients are initially managed by casualty medical officers, surgical residents, or emergency medicine residents. They are mainly first- or second-year residents who resuscitate patients, perform interventions and refer patients for imaging or other investigations. Compared with other settings where a trauma team approach is adopted, nurses and other healthcare professionals are only involved to a limited extent during the initial management.

8.7.1 Description of investigational medicinal products

This trial does not include any investigational medicinal products.

8.7.2 Auxiliary medicinal products

This trial does not include any auxiliary medicinal products.

8.7.3 Concomitant use of other medications or treatments

Other than implementing another formalised trauma life support training programme or other major interventions to change the care of trauma patients as specified in the exclusion criteria, concomitant use of other medications and treatments may be provided at the discretion of the investigators and will not be considered an exclusion criterion.

8.8 Randomisation

We will assign clusters to batches as they are found to be eligible and receive ethical approval. Batches will include clusters from hospitals in different regions to optimize trial logistics. We will randomise the clusters alloted to each batch to the different intervention implementation sequences within that batch. We will balance the randomisation within each batch on cluster size, defined as monthly volume of eligible patient participants, using covariate constrained randomisation. The cluster sizes are expected to vary between 12 and 20 patients per month, based on our previous experiences. We will conceal the randomisation order for as long as it is logistically possible, considering that arrangements for sending physicians to ATLS® training need to be made in advance.

8.9 Blinding

It is not possible to blind a stepped-wedge trial, because all clusters receive the intervention.

8.10 Treatment after trial end

When the trial ends, the intervention will have been implemented in all clusters.

8.11 Outcomes

8.11.1 Primary outcome

The primary outcome will be in-hospital mortality within 30 days of arrival at the emergency department. We define this outcome as any death occurring in the hospital during the initial admission, up to 30 days post arrival at the emergency department. Clinical research coordinators will extract information on death from patient hospital records. If the patient has been transferred to another hospital, the clinical research coordinators will collect data on this outcome by calling the patient or a patient representative, or by contacting the hospital to which the patient was transferred. Data on this outcome will be collected continuously during the trial.

8.11.2 Secondary outcomes

8.11.2.1 Collected using the main stepped-wedge design

- All cause mortality within 24 hours, 30 days and three months of arrival at the emergency department. In contrast to the primary outcome, all-cause mortality includes all deaths regardless of setting. Data on this outcome will be collected in the same way as for the primary outcome.
- Length of emergency department stay. Data on this outcome will be collected from patient hospital records.
- Length of hospital stay. Data on this outcome will be collected from patient hospital records.
- Intensive care unit admission. Data on this outcome will be collected from patient hospital records.

- Length of intensive care unit stay. Data on this outcome will be collected from patient hospital records.
- Return to work at 30 days and three months after arrival at the emergency department. Data on this outcome will be collected in person if the patient is still in hospital, or by phone if the patient has been discharged.

8.11.2.2 Collected using the nested staircase design

- Adherence to ATLS® principles during initial patient resuscitation, up to one hour after the physician has first seen the patient. This assessment will be done using a 14 item checklist covering the key steps of the ATLS® primary survey, which was modelled based on previous work on ATLS® adherence[37]. We will consider completion of all 14 steps as 100% adherence. The clinical research coordinators collecting the data will be trained by the trial team to do this, prior to the start of the trial. We will collect this data by observing the care of a random sample of patients. The sampling will be designed as a nested staircase design.
- Quality of life within seven days of discharge, and at 30 days and three months of arrival at the emergency department, measured by the official and validated translations of the EQ5D5L. This outcome is only relevant to alive patients. Data on this outcome will be collected in person if the patient is still in hospital, or by phone if the patient has been discharged. We will collect this data using a nested staircase design.
- Disability within seven days of discharge, and at 30 days and three months of arrival at the emergency department, assessed using the WHO Disability Assessment Schedule 2.0 (WHODAS 2.0). This outcome is only relevant to alive patients. Data on this outcome will be collected in person if the patient is still in hospital, or by phone if the patient has been discharged. This data will also be collected using a nested staircase design.

8.12 Handling of adverse and safety events

8.12.1 Definitions

8.12.1.1 Adverse event

Any untoward medical occurrence in a clinical trial subject and, which does not necessarily have a causal relationship with the treatment, can be an unfavorable and unintended sign (including an abnormal laboratory discovery), symptom or disease temporally associated with the inclusion in the trial, whether or not related to the trial.

8.12.1.2 Serious adverse event

Any untoward medical occurrence in a trial participant that:

- leads to death
- is life-threatening
- requires inpatient hospitalization or prolongation of existing hospitalization
- results in persistent or significant disability or incapacity

- results in a congenital anomaly/malformation

8.12.1.3 Safety event

Any unexpected serious complication that might occur as a consequence of the trial and that are not part of the natural history of trauma.

8.12.2 Reporting and assessment of adverse and safety events

In alignment with other current trials including critically ill patients[38], we will not collect adverse events or serious adverse events, because many of these events are expected in this patient population and we already collect many of these events, for example mortality, as part of our outcomes.

We will only report safety events, if they are life-threatening, prolong hospitalisation or result in meaningful harm to the participant. We cannot pre-define a comprehensive list of events that can be considered safety events, but will actively assess the presence of the following safety events:

- Prolonged mechanical ventilation (> 7 days)
- Initiation of renal replacement therapy
- Prolonged (> 2 days) or renewed (restart after at least 2 days without) use of vasopressors such as norepinephrine or vasopressin

These events are considered safety events because they suggest pulmonary, renal, septic or bleeding complications and an increase in their occurrence following ATLS® training could indicate that the intervention is harmful. These events therefore need to be tracked during the standard care phase as well as the intervention phase, but will only be considered indicative of harm related to the intervention if they occur more often during the intervention phase than during the standard care phase.

We will also report any other safety events that we identify during the trial, and the reporting of such will have to be based on the intuition of the clinical research coordinators and local investigators. Examples of such safety events could include missed injuries or missed investigations, which could be suspected if certain injuries or investigations were identified or conducted more often during the standard care phase than during the intervention phase.

All safety events will be recorded in the Case Record Form (CRF) and reported to the trial management team within 24 hours of its occurrence. The trial management team will then assess if the event can be considered related to the trial or the intervention within 24 hours of it being reported. Events that are considered probably related will be reported immediately to the joint Trial Steering and Data Monitoring Committee.

8.12.3 Follow up of safety events

All safety events should be followed up by the local investigator until they are fully evaluated.

8.13 Statistics

8.13.1 General principles

The primary analysis set will include all observations within clusters with available data and irrespective of the receipt of the intervention. Clusters and observations within clusters will be considered exposed to the intervention after the date at which the cluster was scheduled to transition. All data will be included with the exception of the transition phases. We will not adjust for multiplicity of analyses because none of the secondary outcomes will be singularly more important. However, all secondary outcomes will be interpreted with due consideration for how all are affected by the intervention without putting any undue emphasis on a single outcome that might be statistically significant but where all others appear to have remained unchanged.

We will use a two-sided significance level of 5% and estimate 95% confidence intervals. The primary subgroup analyses will be based on geographical region because demonstrating the consistency of any effect across multiple regions will enhance the generalisibility of the results[3]. Additional subgroup analyses will include age across the groups older adolescents (15-19 years), young adults (20-24 years), adults (25-59 years), and older adults (60 years and older)[39]; sex; and the clinical cohorts blunt multisytem trauma, penetrating trauma, and severe isolated traumatic brain injury.

8.13.2 Analysis models

There are a number of requirements for the analysis model. Firstly, all analysis will consider the clustered nature of the design. Secondly, as the trial has only 30 clusters, it will be essential that the model allows for a correction due to the small number of clusters. Thirdly, as the design is a stepped-wedge study, we will adjust for temporal confounding using categorical effects for period of the study (month). Full details on how each of these will be undertaken, with justification is provided below[40].

For binary outcomes, a mixed effects binomial regression with a logit link will be used to estimate the odds ratio; and a binomial model with identity link used to estimate the risk difference. These models will be fitted using residual pseudo-likelihood estimation based on linearization with subject-specific expansion (RSPL). If the binomial model with the identity link does not converge then only a odds ratio will be reported.

We will include fixed effects for period and a fixed effect for intervention exposure. The primary analysis will allow for clustering by as a random cluster and random cluster by period effect. To correct the potential inflation of the type I error rate due to small number of clusters, a correction for a small number of clusters will be applied, but the correction that will be selected will be based on the best available evidence available closer to the time, and it may differ for the outcomes collected via the complete and incomplete designs. In a sensitivity analysis we will explore if models with more complicated correlation structures are a better fit to the data. These models are not being used as our primary analysis models as there is limited understanding as to when such models will converge and how to choose between the various different correlation structures which might be plausible.

[3] **Note:** Batches will not be based on regions because it will be logistically more feasible to include clusters from different regions in each batch.

To this end we will additionally fit generalised linear mixed models (with same link functions and fixed effects as described above) to include a discrete time decay correlation structure including a random cluster effect with auto-regressive structure (AR(1)). To allow for the randomisation by batches, a different secular trend will be included for each batch (interaction between batch and period). For continuous, count and prevalence outcomes similar model-based approaches will be used but with appropriate links and distribution functions, using transformations where appropriate.

8.13.3 Additional sensitivity analyses

To additionally explore if the fixed period effect is both parsimonious and adequate to represent the extent of any underlying secular trend, we will model the time effect using a spline function. Models will also be extended to include random cluster by intervention effects (with a non-zero covariance term) to examine if results are sensitive to the assumption of no intervention by cluster interaction. Models will also be extended to include an interaction between treatment and number of periods since first treated, to examine if there is any indication of a relationship between duration of exposure to the intervention and outcomes.

This will allow us to different lag effects (whereby it takes time for the intervention to become embedded within the culture before its impact can properly start to be realised); as well as weaning effects (whereby the effect of the intervention starts to decrease – or fade). This type of analysis attempts to disentangle how some clusters end up having a long exposure to the intervention and others have a much shorter exposure time. A fully adjusted covariate analysis will additionally adjust for a set of pre-specified individual-level covariates of known prognostic importance.

8.13.4 Estimation and reporting of within cluster correlations

We will report time adjusted within-cluster correlations for all outcomes with 95% confidence intervals. We will report correlations from the different assumed correlation structures (so we will report intra-cluster correlations (ICC); within and between-period correlations; and within-period correlations and exponential decay). As well as reporting correlations we will additionally report all variance components. For all outcomes we will report correlations on the latent scale (i.e. proportions scale for binary outcomes) as is appropriate to inform future sample size calculations.

8.13.5 Sample size calculations

8.13.5.1 Main stepped wedge design and primary outcome

With 30 clusters across 6 batches and a total sample size of 4320 our study has ~90% power across different combinations of cluster autocorrelations (CAC) and intra-cluster correlations (ICC) to detect a reduction in the primary outcome of in-hospital mortality within 30 days from 20% under standard care to 15% after ATLS® training (see Figure 4)[41]. This 5% absolute reduction in mortality is a clinically meaningful and conservative estimate based on our updated systematic review and pilot study[18,29]. We allowed for the clustered design, incorporated the one month transition period, and assumed: a discrete time decay correlation structure, an ICC of 0.02 (but considered sensitivity across the range 0.01-0.05), and a CAC of 0.9 (but considered sensitivity across the range 0.8-1.0), based on our pilot study and current guidance[42–45]. We included the CAC to allow for variation in clustering over time. We assume

that each cluster will contribute approximately 12 observations per month to the analysis, but allowed for substantial variation in cluster sizes, based on our previous work.

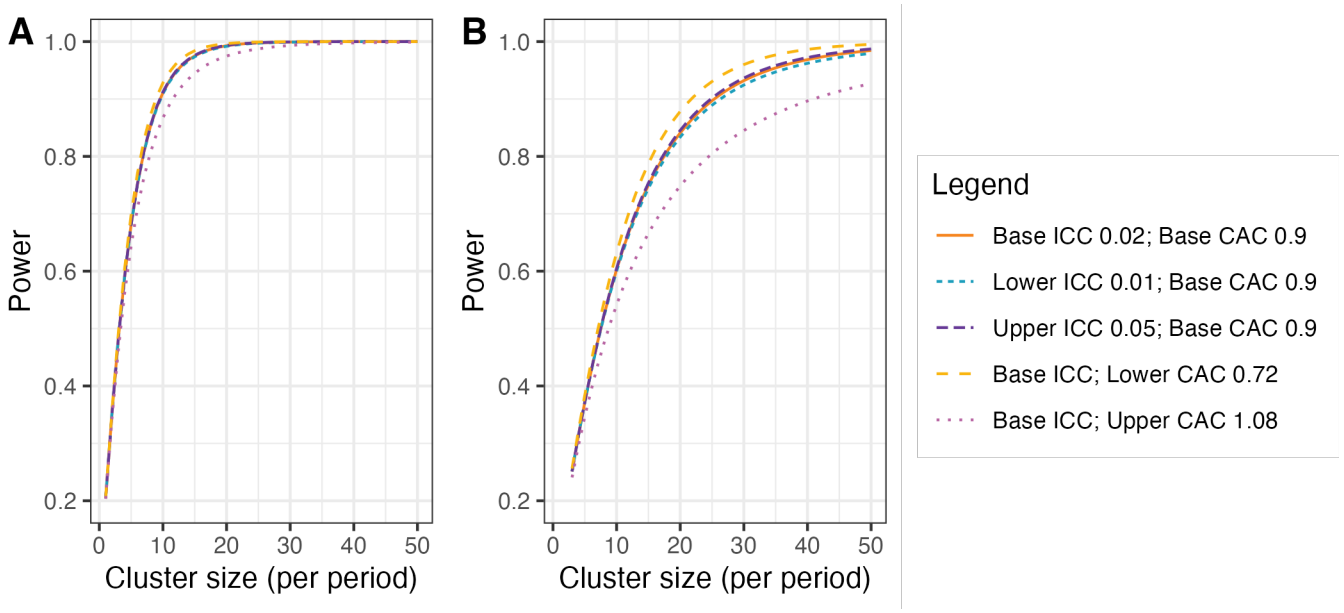


Figure 4: Power curves for different combinations of cluster autocorrelations (CAC) and intra-cluster correlations (ICC). **A)** Shows power curves assuming a reduction in the primary outcome of in-hospital mortality within 30 days from 20% under standard care to 15% after ATLS® training. **B)** Shows power curves assuming a reduction in the primary outcome from 10% under standard care to 7.5% after ATLS® training. Under this scenario, we would need to increase the sample size per month to around 30 observations to achieve 90% powere under most combinations of CAC and ICC.

8.13.5.2 Nested staircase design and secondary outcomes

The secondary outcomes that will be measured using the nested staircase design are adherence to ATLS® principles during initial patient resuscitation, quality of life, and disability. The expected effect of the intervention on each of these outcomes are an improvement in adherence from 50% during standard care to 70% after training[46], an increase in EQ5D5L health status from 70 during care to 75 after training[47], and finally a decrease in disability from a baseline value of 25 during standard care to 22.5 after training[48]. For quality of life and disability, these effects correspond to standardised effect sizes, expressed as Cohen's d, of 0.5. With 30 clusters in total, six per sequence, for a discrete time decay correlation structure, with ICCs of 0.01 to 0.15 and a CAC of 0.8, there is >80% power to detect these effects by including four patients in each cluster in each period. Accounting for loss to follow-up, we will include at least six patients per cluster per month. These patients will be a random subset of patients included in the primary outcome data collection during the staircase months. The random subset will be selected using simple random sampling on the shift level, meaning that the timing of the clinical research coordinator's shift will be randomised to cover approximately eight hours during the

morning, afternoon, or night shift. Adherence to ATLS® principles, quality of life, and disability will be measured in all patients included during these shifts. In each hospital, the number of shifts that will be randomised will be determined by the volume of patients included during the months preceding the staircase months.

8.13.6 Interim analysis

There will be one interim analysis after half of the batches have completed the trial. The interim analysis will be assessed by the joint Trial Steering and Data Monitoring Committee. The purposes of this interim analysis will be to:

- assess the trial's feasibility and recommend stopping the trial if the trial is not feasible, for example if hospitals fail to adhere to the randomisation schedule or if there are substantial missing data in outcomes;
- compare characteristics across intervention conditions to monitor for differential recruitment/ascertainment between intervention and control.

8.14 Quality control and quality assurance

The George Institute for Global Health - India will ensure proper conduct of the trial through quality control measures including on-site training of personnel, standard operating procedures, ongoing quality metrics assessment, review of missing data and outliers, and round-the-clock availability of coordinating center personnel and Principal Investigators. The trial will strictly follow ICH GCP principles, Indian regulations, and George Institute procedures. The trial operations staff from the George Institute India will train local investigators, and trial site staff, before the trial, with continuous documentation in the site master file. All documentation will be stored securely and retained according to regulatory requirements.

8.15 Quality assurance and oversight

The Trial Management Group and Trial Team, comprising key project leaders and managers, will play a pivotal role in ensuring the highest standards of quality assurance and effective sponsor oversight throughout the trial. These groups will be responsible for facilitating consistent communication, maintaining fidelity in study implementation, and overseeing the quality of data collection.

To achieve these objectives, the groups will implement a comprehensive communication plan and provide extensive training to site personnel. The training will cover not only the study protocol but also practical aspects of various systems, supplemented by both written and electronic materials designed to educate study and clinical emergency staff.

The trial's quality assurance systems will be meticulously designed based on a thorough risk analysis. A key component of our quality assurance strategy will include the development and implementation of detailed operational manuals and regular meetings. These tools and interactions will ensure that all trial personnel will be used to uphold the trial's quality standards.

Central to our oversight approach will be a comprehensive monitoring and auditing plan. This plan will be tailored based on the identified risks associated with the trial. Through these comprehensive measures, the trial management group, in conjunction with the hospital staff, will ensure that the trial is conducted with the utmost rigor, adhering to the highest standards of quality assurance and effective sponsor oversight.

8.16 Monitoring

We will implement a multi-tiered monitoring strategy, including centralized data consistency checks, statistical monitoring, and selective on-site evaluations. Key integrity measures include source data verification, data entry validation, and regular audits. Any protocol deviations will be thoroughly documented, with serious breaches promptly addressed to ensure data integrity. Monitors from coordinating centres will assist investigators in maintaining high ethical, scientific, technical, and regulatory quality. Monitoring visits will review protocol adherence, participant recruitment, adverse event reporting, compliance with study procedures, and regulatory adherence. Regular remote monitoring of the web-based database will be conducted to ensure data integrity, using validation and consistency rules and regular data cleaning. The Trial Team and Trial Management Group will monitor baseline characteristics, opt-in consent rates and differential opt-in consent rates across trial arms, follow-up rates, CRF return and completeness rates, and safety data.

9 Deviations, serious breaches and other reporting obligations

The responsible investigator shall, without delay, report to the sponsor any serious breaches and deviations from the trial protocol, ICH-GCP and other regulations that significantly and directly affect, or with high likelihood could affect, the subjects' safety and integrity or the reliability and robustness of the data generated in the trial. The sponsor should assess the suspected serious breach and the consequences of deviations that have occurred. Minor deviations that do not affect subjects' integrity or safety, nor significantly affect the trial's scientific value, are documented in the trial documentation of the principal investigator and the sponsor and appropriate measures shall be taken. The deviations must be recorded in the clinical trial report.

10 Audits and inspections

Authorized representatives for the sponsor and Competent Authorities (CA) may carry out audits or inspections at the trial site, including source data verification. The investigator must ensure that all source documents are available for audits and inspections. The purpose of an audit or inspection is to systematically and independently review all trial-related activities and documents, to determine whether these activities were performed, registered, analyzed and reported correctly according to protocol, ICH-GCP and applicable regulations.

11 Ethics

11.1 Compliance to the protocol, ICH-GCP and regulations

The trial will be performed in compliance with this clinical trial protocol, the Declaration of Helsinki, ICH-GCP (Good Clinical Practice), and current national regulations governing this clinical trial. This is to ensure the safety and integrity of the trial subjects as well as the quality of the data collected.

11.2 Ethical review of the trial

The final protocol will be submitted for ethical review at all participating hospitals, where possible, as well as the The George Institute for Global Health in India and Swedish Ethical Review Atuhortiy.

11.3 Procedure for obtaining consent

In this trial, consent refers to data collection, as patients cannot opt out of the intervention. This is because the intervention is implemented at the cluster level, involving training physicians in ATLS®. It is unreasonable to expect these physicians to temporarily disregard their training. Patient participants will be included in this trial under the following modes of consent:

- Opt out consent for **routinely recorded data and measurement of adherence to ATLS® principles**. Consent for the collection of routinely recorded data, either through interviews or by extracting information from medical records, as well as for the measurement of adherence to ATLS® principles, will be presumed unless explicitly declined. This approach is justified because the trial is considered to pose minimal risk and because data collection will be non-invasive. Additionally, obtaining consent specifically for the measurement of adherence to ATLS® principles could interfere with the provision of care and cause undue stress for the patient and their representatives. Patients, or their legally authorized representatives, will be provided with written information about the study upon their arrival at the hospital. The variables assumed to be routinely recorded are listed in Variables (Section 18.2).
- Opt in consent and assent for **non-routinely recorded data**. Informed consent for non-routinely recorded data will be actively sought from patient participants or their legally authorized representative. For participants who are between 15 and 18 years of age we will obtain both the assent of the participant as well as the consent of their guardian or legally authorized representative. The clinical research coordinators will approach patient participants and their representatives after admission. The consent and assent will be written for patient participants who are admitted to the hospital and verbal for participants who are transferred or discharged before the clinical research coordinators have had an opportunity to approach them. The verbal consent will be audio recorded.
- Waiver of informed consent for patients who are unconscious or otherwise unable to provide consent and do not have a legally authorized representative. This group represents the most severly injured patients and they have to be included to make the trial representative of the entire

population of trauma patients. Patients participants who regain consciousness will be informed about the study and asked for consent for collection of non-routinely recorded data.

11.4 Data protection

All data will be handled according to the Indian Council of Medical Research's guidelines and standard operating procedures of the George Institute for Global Health India on data security and protection. Trial data will be shared via the trial electronic CRF (eCRF) throughout the trial. The eCRF will be accessible via VPN with a two-factor authentication and the data will be held on a secure server. All investigators and trial site staff involved in this trial must comply with the requirements of the ICMR Guidelines on data security and protection. The participant information sheet provided to participants, will inform them how:

- the trial data will be collected, used and disclosed;
- how trial data are stored to maintain confidentiality in accordance with national data legislation; and
- for verification of the data, representatives delegated by the sponsor, as well as relevant authorities, may require access to parts of medical records or trial records that are relevant to the trial, including the patient participant's medical history.

12 Insurances

The George Institute for Global Health, India is responsible for ensuring that any insurance cover required to cover the set-up, management and conduct of the study in India has been obtained. The George Institute for Global Health, India is also responsible for ensuring that India Sites have been obtained and/or will obtain insurance prior to the opening of the study in India and shall be maintained for the duration of the study and for an appropriate period thereafter. This includes being responsible for ensuring that there is appropriate insurance for the duration of the study to cover against claims for compensation by participants arising out of their participation in the trial in India. Compensation in case of injury or death will be provided by the George Institute for Global Health, India according to the regulations outlined in rules 39, 40 and 42 of the New Drugs and Clinical Rules (2019). x

13 Substantial changes to the trial

Substantial changes to the signed clinical trial protocol are only possible through approved protocol amendments and by agreement between the sponsor and the principal investigator.

14 Trial organisation

Trial management and oversight is governed by three trial committees and groups: the Trial Team (TT), the Trial Management Group (TMG), the joint Trial Steering and Data Monitoring Committee

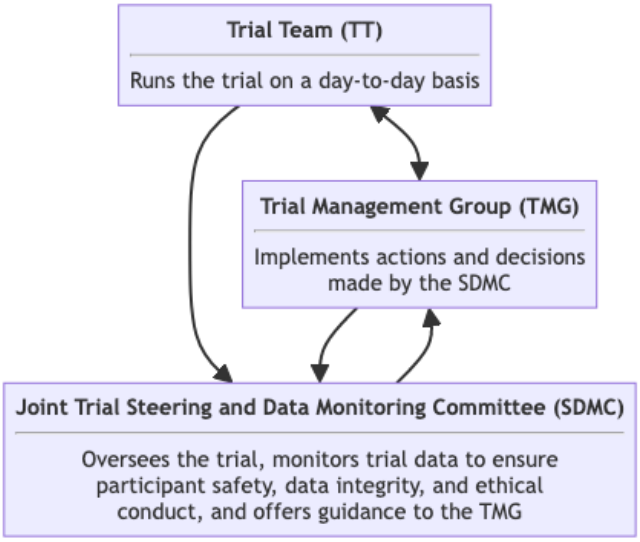


Figure 5: Trial organisation overview.

(SDMC). These groups and their relationships are briefly described in Figure 5. Details about each committee and group are available in their respective charter.

14.1 Trial team

Responsibility

To run the trial on a day-to-day basis, maintain trial databases, randomise clusters, ensuring complete and correct data, preparing reports for meetings (including those of the TMG and SDMC) and dealing with research governance and, if appropriate, regulatory matters.

Composition

Includes the project manager, clinical research associates, principal investigator and co-investigators as needed.

Relationships

Reports to the TMG and SDMC. Operationalises decisions made by the TMG.

Meeting frequencies

As often as needed, often weekly or bi-weekly.

14.2 Trial Management Group (TMG)

Responsibility

To manage the trial, including its clinical and practical aspects.

Composition

Includes members with broad expertise appropriate to the trial. The TMG will be chaired by the Principal Investigator.

Relationships

Receives reports from TT. Provides input to the SDMC. Implements decisions made by the SDMC.

Meeting frequencies

Monthly to every six months.

14.3 Joint Trial Steering and Data Monitoring Committee (SDMC)

Responsibility

The SDMC's responsibility is to oversee the trial, review results of interim analyses and safety events reported by the TMG, and review trial data for each batch, assessing data quality, completeness, cluster performance in recruitment and loss to follow-up rates, and external factors affecting trial validity, safety, or ethics. This committee also offer guidance to the TMG.

Composition

A majority of independent members, including a chair and three additional external experts specializing in the clinical area, biostatistics, and a community or patient representative, as well as and a minority of members with a direct interest in the trial, including the principal investigator. The chair should be independent of the trial, and the coordinating institutions Karolinska Institutet and The George Institute for Global Health.

Relationships

Receives reports from the trial team and TMG.

Meeting frequencies

After the completion of each batch, but may be more frequent if needed.

15 Funding

- Swedish Research Council (reg. no. 2023-03128)
- Laerdal Foundation (reg. no. 2023-0297)

16 Special considerations

16.1 Funding

This trial is not yet fully funded. The Trial Management Group has decided to proceed with the trial with the expectation that additional funding will be secured. The Trial Steering Committee will be informed of the funding status at each meeting. If funding is not secured, the trial will be stopped. This will likely result in an underpowered trial. The justification for this decision is that the intervention is considered standard of care in many countries and the data collection is considered minimal risk. There is therefore a very small risk of harm to patient participants, but a potential direct benefit to those patient participants who receive the intervention. The benefit-risk ratio is therefore considered to be favourable, even in the case of an underpowered trial.

17 Notification of trial completion, reporting, and publication

The trial will be reported to the Funders within a year of completion. The results of the trial will also be prepared as manuscripts for publication. Authorship on trial manuscripts will be based on the International Committee of Medical Journal Editors (ICMJE) criteria[49]:

- Substantial contributions to the conception or design of the work; or the acquisition, analysis, or interpretation of data for the work; AND
- Drafting the work or reviewing it critically for important intellectual content; AND
- Final approval of the version to be published; AND
- Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

In addition to being accountable for the parts of the work done, an author should be able to identify which co-authors are responsible for specific other parts of the work. In addition, authors should have confidence in the integrity of the contributions of their co-authors.

The most recent version of the ICMJE criteria will be adhered to. We will also use the ICMJE criteria for non-author contributorship.

Before work on a trial manuscript is initiated, a writing group will be formed and first and last authors will be designated. This writing group will be formed by discussion in the Trial Management Group.

18 Collection, handling, and archiving of data

Clinical research coordinators will collect data using a paper based CRF (see Appendix Case Record Form (Section 20.3)), which is then transferred to an eCRF. All trial data in the CRF must be extracted from and be consistent with the relevant source documents. The eCRF will be accessible to trial

coordinators, data managers, the Investigators, Clinical Trial Monitors, Auditors, and Inspectors as required. All data will be registered, managed, and stored in a manner that enables correct reporting, interpretation, and verification. The complete Trial Master File, as well as source documents, will be archived for at least 10 years after the trial is completed. Source data in the medical records system are stored and archived in accordance with national regulations. Metadata will be publicly accessible via a persistent DOI, and anonymised data will be released upon project completion. A detailed data management plan is available here https://doi.org/10.5281/zenodo.7748764.

18.1 Source data

The source data for each variable is given in Variables (Section 18.2). Whenever medical records are the source data, this includes imaging and lab reports. Whenever an interview is given as the source, the CRF will constitute the source data, as this is where the responses to questions will be recorded. The local investigator must keep source documents for each patient participant in the trial. A document describing what has been classified as source data in the trial (source data reference document) will be included in the Investigator Site File (ISF). The investigator must ensure that all source documents are accessible for monitoring and other quality control activities. Source data is further defined before trial start at each individual site and can, in cases where source data is not registered in another document, consist of the CRF. This should be decided in consultation with the monitor and clearly stated in the source data reference document. Access to trial-related documentation, such as patient participants' medical records, CRFs, other source data and other trial documentation will be provided for monitoring and auditing purposes. Access will also be granted in the context of regulatory inspections.

18.2 Variables

18.2.1 Screening

- **Screening ID**
- **1. Date of screening**
- **2. Date of data entry**
- **1. Is the patient at least 15 years old?** Source: Medical record or interview

    1. Yes
    2. No

- **2. Did the patient present with a history of trauma defined as having any of the reasons listed in the International Classification of Diseases chapter XX as the reason for presenting? Please see https://icd.who.int/browse10/2019/en#/XX for a complete list of ICD-10 codes** Source: Medical record or interview

    1. Yes
    2. No

- **3. Did the trauma occur less than 48 hours before arrival to the hospital?** Source: Medical record or interview

    1. Yes
    2. No

- **4. Was the patient admitted?** Source: Medical record

    1. Yes
    2. No

- **5. Did the patient die after arrival but before admission?** Source: Medical record

    1. Yes
    2. No

- **6. Was the patient transferred to another hospital for admission?** Source: Medical record

    1. Yes
    2. No

- **1. Did the patient present with isolated closed extremity fracture?** Source: Medical record

    1. Yes
    2. No

- **2. Was the patient directly admitted to a ward without being seen by a physician in the emergency department?** Source: Medical record

    1. Yes
    2. No

- **Is the patient eligible? (Eligible=1; Not Eligible=0)**
- **Any other comments?**

18.2.2 Consent

- **1. Is this patient included under the waiver of informed consent because the patient is unconscious or otherwise unable to provide consent and do not have a legally acceptable representative?**

    1. Yes
    2. No

- **1. Did the participant/ or legally acceptable representative (LAR) provided consent for collection of non-routinely recorded data**

    1. Yes

2. No

- **2. Who gave consent for collection of non-routinely recorded data?**

1. Patient participant
2. Legally acceptable representative

- **3. What is the LAR's relationship with the participant?**
- **4. Why was Legally acceptable representative (LAR) approached for consent for collection of non-routinely recorded data?**

1. The participant is incapacitated because of the trauma
2. The participant is younger than 18 years

- **5. Date when participant or legally acceptable representative (LAR) gave consent for collection of non-routinely recorded data.**
- **6. How did the participant or legally acceptable representative (LAR) consent for collection of non-routinely recorded data?**

1. In writing
2. Verbally

- **7. Reconsenting done?**

1. Yes
2. No

- **7.1 Date when the participant was reconsented?**
- **7.2 Why consenting was not done?**
- **1. Did the minor give assent for collection of non-routinely recorded data?**

1. Yes
2. No

- **2. Date when the minor gave assent for collection of non-routinely recorded data.**
- **3. In case the minor refused to participate, date when minor refused**
- **1. Is the participant or LAR wants to opt out from study?**

1. Yes
2. No

- **2. Who opted-out of the routinely recorded data (in-hospital)?**

1. Patient participant

2. Legally acceptable representative (LAR)

- **3. Date when participant or legally acceptable representative (LAR) opted-out.**
- **4. Did the participant or legally acceptable representative (LAR) suggested to delete all the previously recorded data?**

1. Yes
2. No

- **Any other comments?**

18.2.3 Consent withdrawn

- **1. Does the participant or legally acceptable representative (LAR) want to withdraw the consent?**

1. Yes
2. No

- **2. Date of consent withdrawal for follow-up data collection.**
- **3. Procedure(s) for which consent has been withdrawn**

1. Data collection prior to withdrawal
2. All data collection after withdrawal
3. Both

- **Any other comments?**

18.2.4 Baseline

- **1. Age in years** Source: Medical record of interview
- **2. Sex** Source: Medical record of interview

1. Female
2. Male
3. Other
4. Not known

- **3. Current marital status** Requires opt-in consent, not routinely recorded. Source: Interview

1. Never married
2. Currently married
3. Separated
4. Divorced
5. Widowed
6. Cohabiting

7. Not known

- **4. Education level** Requires opt-in consent, not routinely recorded. Source: Interview

1. Not attended school
2. Primary school
3. Secondary school
4. Higher secondary school
5. Graduate
6. Post graduate and above
7. Other
8. Not known

- **5. If other, please specify** Requires opt-in consent, not routinely recorded. Source: Interview
- **6. Main work status** Requires opt-in consent, not routinely recorded. Source: Interview

1. Paid work, such as daily wage earner, teacher, factory worker and government employee
2. Self-employed, such as own your business or farming
3. Non-paid work, such as volunteer or charity
4. Student
5. Keeping house/homemaker
6. Retired
7. Unemployed (health reasons)
8. Unemployed (other reasons)
9. Other
10. No income
11. Not known

- **7. If other, please specify** Requires opt-in consent, not routinely recorded. Source: Interview
- **8. Income level in INR per month** Requires opt-in consent, not routinely recorded. Source: Interview

1. Below 10,000
2. 10,001-20,000
3. 20,001-30,000
4. 30,001-50,000
5. 50,001-80,000
6. 80,001-1,00,000
7. Above 1,00,000
8. No income
9. Not known

- **9. Mechanism of injury description**

- **10. Mechanism of injury ICD 10 code** Coded using ICD 10. Please see https://icd.who.int/browse10/2019/en#/XX for a complete list of ICD-10 codes. Source: Medical record

1110. V01 Pedestrian injured in collision with pedal cycle
1111. V01.0 Pedestrian injured in collision with pedal cycle : nontraffic accident
1112. V01.1 Pedestrian injured in collision with pedal cycle : traffic accident
1113. V01.9 Pedestrian injured in collision with pedal cycle : unspecified whether traffic or nontraffic accident
1114. V02 Pedestrian injured in collision with two- or three-wheeled motor vehicle
1115. V02.0 Pedestrian injured in collision with two- or three-wheeled motor vehicle : nontraffic accident
1116. V02.1 Pedestrian injured in collision with two- or three-wheeled motor vehicle : traffic accident
1117. V02.9 Pedestrian injured in collision with two- or three-wheeled motor vehicle : unspecified whether traffic or nontraffic accident
1118. V03 Pedestrian injured in collision with car, pick-up truck or van
1119. V03.0 Pedestrian injured in collision with car, pick-up truck or van : nontraffic accident
1120. V03.1 Pedestrian injured in collision with car, pick-up truck or van : traffic accident
1121. V03.9 Pedestrian injured in collision with car, pick-up truck or van : unspecified whether traffic or nontraffic accident
1122. V04 Pedestrian injured in collision with heavy transport vehicle or bus
1123. V04.0 Pedestrian injured in collision with heavy transport vehicle or bus : nontraffic accident
1124. V04.1 Pedestrian injured in collision with heavy transport vehicle or bus : traffic accident
1125. V04.9 Pedestrian injured in collision with heavy transport vehicle or bus : unspecified whether traffic or nontraffic accident
1126. V05 Pedestrian injured in collision with railway train or railway vehicle
1127. V05.0 Pedestrian injured in collision with railway train or railway vehicle : nontraffic accident
1128. V05.1 Pedestrian injured in collision with railway train or railway vehicle : traffic accident
1129. V05.9 Pedestrian injured in collision with railway train or railway vehicle : unspecified whether traffic or nontraffic accident
1130. V06 Pedestrian injured in collision with other nonmotor vehicle
1131. V06.0 Pedestrian injured in collision with other nonmotor vehicle : nontraffic accident
1132. V06.1 Pedestrian injured in collision with other nonmotor vehicle : traffic accident
1133. V06.9 Pedestrian injured in collision with other nonmotor vehicle : unspecified whether traffic or nontraffic accident
1134. V09 Pedestrian injured in other and unspecified transport accidents
1135. V09.0 Pedestrian injured in nontraffic accident involving other and unspecified motor vehicles
1136. V09.1 Pedestrian injured in unspecified nontraffic accident
1137. V09.2 Pedestrian injured in traffic accident involving other and unspecified motor vehicles
1138. V09.3 Pedestrian injured in unspecified traffic accident
1139. V09.9 Pedestrian injured in unspecified transport accident

1140. V10 Pedal cyclist injured in collision with pedestrian or animal
1141. V10.0 Pedal cyclist injured in collision with pedestrian or animal : driver injured in non-traffic accident
1142. V10.1 Pedal cyclist injured in collision with pedestrian or animal : passenger injured in nontraffic accident
1143. V10.2 Pedal cyclist injured in collision with pedestrian or animal : unspecified pedal cyclist injured in nontraffic accident
1144. V10.3 Pedal cyclist injured in collision with pedestrian or animal : person injured while boarding or alighting
1145. V10.4 Pedal cyclist injured in collision with pedestrian or animal : driver injured in traffic accident
1146. V10.5 Pedal cyclist injured in collision with pedestrian or animal : passenger injured in traffic accident
1147. V10.9 Pedal cyclist injured in collision with pedestrian or animal : unspecified pedal cyclist injured in traffic accident
1148. V11 Pedal cyclist injured in collision with other pedal cycle
1149. V11.0 Pedal cyclist injured in collision with other pedal cycle : driver injured in nontraffic accident
1150. V11.1 Pedal cyclist injured in collision with other pedal cycle : passenger injured in non-traffic accident
1151. V11.2 Pedal cyclist injured in collision with other pedal cycle : unspecified pedal cyclist injured in nontraffic accident
1152. V11.3 Pedal cyclist injured in collision with other pedal cycle : person injured while boarding or alighting
1153. V11.4 Pedal cyclist injured in collision with other pedal cycle : driver injured in traffic accident
1154. V11.5 Pedal cyclist injured in collision with other pedal cycle : passenger injured in traffic accident
1155. V11.9 Pedal cyclist injured in collision with other pedal cycle : unspecified pedal cyclist injured in traffic accident
1156. V12 Pedal cyclist injured in collision with two- or three-wheeled motor vehicle
1157. V12.0 Pedal cyclist injured in collision with two- or three-wheeled motor vehicle : driver injured in nontraffic accident
1158. V12.1 Pedal cyclist injured in collision with two- or three-wheeled motor vehicle : passenger injured in nontraffic accident
1159. V12.2 Pedal cyclist injured in collision with two- or three-wheeled motor vehicle : unspecified pedal cyclist injured in nontraffic accident
1160. V12.3 Pedal cyclist injured in collision with two- or three-wheeled motor vehicle : person injured while boarding or alighting
1161. V12.4 Pedal cyclist injured in collision with two- or three-wheeled motor vehicle : driver injured in traffic accident
1162. V12.5 Pedal cyclist injured in collision with two- or three-wheeled motor vehicle : passenger injured in traffic accident

1163. V12.9 Pedal cyclist injured in collision with two- or three-wheeled motor vehicle : unspecified pedal cyclist injured in traffic accident
1164. V13 Pedal cyclist injured in collision with car, pick-up truck or van
1165. V13.0 Pedal cyclist injured in collision with car, pick-up truck or van : driver injured in nontraffic accident
1166. V13.1 Pedal cyclist injured in collision with car, pick-up truck or van : passenger injured in nontraffic accident
1167. V13.2 Pedal cyclist injured in collision with car, pick-up truck or van : unspecified pedal cyclist injured in nontraffic accident
1168. V13.3 Pedal cyclist injured in collision with car, pick-up truck or van : person injured while boarding or alighting
1169. V13.4 Pedal cyclist injured in collision with car, pick-up truck or van : driver injured in traffic accident
1170. V13.5 Pedal cyclist injured in collision with car, pick-up truck or van : passenger injured in traffic accident
1171. V13.9 Pedal cyclist injured in collision with car, pick-up truck or van : unspecified pedal cyclist injured in traffic accident
1172. V14 Pedal cyclist injured in collision with heavy transport vehicle or bus
1173. V14.0 Pedal cyclist injured in collision with heavy transport vehicle or bus : driver injured in nontraffic accident
1174. V14.1 Pedal cyclist injured in collision with heavy transport vehicle or bus : passenger injured in nontraffic accident
1175. V14.2 Pedal cyclist injured in collision with heavy transport vehicle or bus : unspecified pedal cyclist injured in nontraffic accident
1176. V14.3 Pedal cyclist injured in collision with heavy transport vehicle or bus : person injured while boarding or alighting
1177. V14.4 Pedal cyclist injured in collision with heavy transport vehicle or bus : driver injured in traffic accident
1178. V14.5 Pedal cyclist injured in collision with heavy transport vehicle or bus : passenger injured in traffic accident
1179. V14.9 Pedal cyclist injured in collision with heavy transport vehicle or bus : unspecified pedal cyclist injured in traffic accident
1180. V15 Pedal cyclist injured in collision with railway train or railway vehicle
1181. V15.0 Pedal cyclist injured in collision with railway train or railway vehicle : driver injured in nontraffic accident
1182. V15.1 Pedal cyclist injured in collision with railway train or railway vehicle : passenger injured in nontraffic accident
1183. V15.2 Pedal cyclist injured in collision with railway train or railway vehicle : unspecified pedal cyclist injured in nontraffic accident
1184. V15.3 Pedal cyclist injured in collision with railway train or railway vehicle : person injured while boarding or alighting
1185. V15.4 Pedal cyclist injured in collision with railway train or railway vehicle : driver injured in traffic accident

1186. V15.5 Pedal cyclist injured in collision with railway train or railway vehicle : passenger injured in traffic accident
1187. V15.9 Pedal cyclist injured in collision with railway train or railway vehicle : unspecified pedal cyclist injured in traffic accident
1188. V16 Pedal cyclist injured in collision with other nonmotor vehicle
1189. V16.0 Pedal cyclist injured in collision with other nonmotor vehicle : driver injured in nontraffic accident
1190. V16.1 Pedal cyclist injured in collision with other nonmotor vehicle : passenger injured in nontraffic accident
1191. V16.2 Pedal cyclist injured in collision with other nonmotor vehicle : unspecified pedal cyclist injured in nontraffic accident
1192. V16.3 Pedal cyclist injured in collision with other nonmotor vehicle : person injured while boarding or alighting
1193. V16.4 Pedal cyclist injured in collision with other nonmotor vehicle : driver injured in traffic accident
1194. V16.5 Pedal cyclist injured in collision with other nonmotor vehicle : passenger injured in traffic accident
1195. V16.9 Pedal cyclist injured in collision with other nonmotor vehicle : unspecified pedal cyclist injured in traffic accident
1196. V17 Pedal cyclist injured in collision with fixed or stationary object
1197. V17.0 Pedal cyclist injured in collision with fixed or stationary object : driver injured in nontraffic accident
1198. V17.1 Pedal cyclist injured in collision with fixed or stationary object : passenger injured in nontraffic accident
1199. V17.2 Pedal cyclist injured in collision with fixed or stationary object : unspecified pedal cyclist injured in nontraffic accident
1200. V17.3 Pedal cyclist injured in collision with fixed or stationary object : person injured while boarding or alighting
1201. V17.4 Pedal cyclist injured in collision with fixed or stationary object : driver injured in traffic accident
1202. V17.5 Pedal cyclist injured in collision with fixed or stationary object : passenger injured in traffic accident
1203. V17.9 Pedal cyclist injured in collision with fixed or stationary object : unspecified pedal cyclist injured in traffic accident
1204. V18 Pedal cyclist injured in noncollision transport accident
1205. V18.0 Pedal cyclist injured in noncollision transport accident : driver injured in nontraffic accident
1206. V18.1 Pedal cyclist injured in noncollision transport accident : passenger injured in nontraffic accident
1207. V18.2 Pedal cyclist injured in noncollision transport accident : unspecified pedal cyclist injured in nontraffic accident
1208. V18.3 Pedal cyclist injured in noncollision transport accident : person injured while boarding or alighting

1209. V18.4 Pedal cyclist injured in noncollision transport accident : driver injured in traffic accident
1210. V18.5 Pedal cyclist injured in noncollision transport accident : passenger injured in traffic accident
1211. V18.9 Pedal cyclist injured in noncollision transport accident : unspecified pedal cyclist injured in traffic accident
1212. V19 Pedal cyclist injured in other and unspecified transport accidents
1213. V19.0 Driver injured in collision with other and unspecified motor vehicles in nontraffic accident
1214. V19.1 Passenger injured in collision with other and unspecified motor vehicles in nontraffic accident
1215. V19.2 Unspecified pedal cyclist injured in collision with other and unspecified motor vehicles in nontraffic accident
1216. V19.3 Pedal cyclist [any] injured in unspecified nontraffic accident
1217. V19.4 Driver injured in collision with other and unspecified motor vehicles in traffic accident
1218. V19.5 Passenger injured in collision with other and unspecified motor vehicles in traffic accident
1219. V19.6 Unspecified pedal cyclist injured in collision with other and unspecified motor vehicles in traffic accident
1220. V19.8 Pedal cyclist [any] injured in other specified transport accidents
1221. V19.9 Pedal cyclist [any] injured in unspecified traffic accident
1222. V20 Motorcycle rider injured in collision with pedestrian or animal
1223. V20.0 Motorcycle rider injured in collision with pedestrian or animal : driver injured in nontraffic accident
1224. V20.1 Motorcycle rider injured in collision with pedestrian or animal : passenger injured in nontraffic accident
1225. V20.2 Motorcycle rider injured in collision with pedestrian or animal : unspecified motorcycle rider injured in nontraffic accident
1226. V20.3 Motorcycle rider injured in collision with pedestrian or animal : person injured while boarding or alighting
1227. V20.4 Motorcycle rider injured in collision with pedestrian or animal : driver injured in traffic accident
1228. V20.5 Motorcycle rider injured in collision with pedestrian or animal : passenger injured in traffic accident
1229. V20.9 Motorcycle rider injured in collision with pedestrian or animal : unspecified motorcycle rider injured in traffic accident
1230. V21 Motorcycle rider injured in collision with pedal cycle
1231. V21.0 Motorcycle rider injured in collision with pedal cycle : driver injured in nontraffic accident
1232. V21.1 Motorcycle rider injured in collision with pedal cycle : passenger injured in nontraffic accident
1233. V21.2 Motorcycle rider injured in collision with pedal cycle : unspecified motorcycle rider

injured in nontraffic accident
1234. V21.3 Motorcycle rider injured in collision with pedal cycle : person injured while boarding or alighting
1235. V21.4 Motorcycle rider injured in collision with pedal cycle : driver injured in traffic accident
1236. V21.5 Motorcycle rider injured in collision with pedal cycle : passenger injured in traffic accident
1237. V21.9 Motorcycle rider injured in collision with pedal cycle : unspecified motorcycle rider injured in traffic accident
1238. V22 Motorcycle rider injured in collision with two- or three-wheeled motor vehicle
1239. V22.0 Motorcycle rider injured in collision with two- or three-wheeled motor vehicle : driver injured in nontraffic accident
1240. V22.1 Motorcycle rider injured in collision with two- or three-wheeled motor vehicle : passenger injured in nontraffic accident
1241. V22.2 Motorcycle rider injured in collision with two- or three-wheeled motor vehicle : unspecified motorcycle rider injured in nontraffic accident
1242. V22.3 Motorcycle rider injured in collision with two- or three-wheeled motor vehicle : person injured while boarding or alighting
1243. V22.4 Motorcycle rider injured in collision with two- or three-wheeled motor vehicle : driver injured in traffic accident
1244. V22.5 Motorcycle rider injured in collision with two- or three-wheeled motor vehicle : passenger injured in traffic accident
1245. V22.9 Motorcycle rider injured in collision with two- or three-wheeled motor vehicle : unspecified motorcycle rider injured in traffic accident
1246. V23 Motorcycle rider injured in collision with car, pick-up truck or van
1247. V23.0 Motorcycle rider injured in collision with car, pick-up truck or van : driver injured in nontraffic accident
1248. V23.1 Motorcycle rider injured in collision with car, pick-up truck or van : passenger injured in nontraffic accident
1249. V23.2 Motorcycle rider injured in collision with car, pick-up truck or van : unspecified motorcycle rider injured in nontraffic accident
1250. V23.3 Motorcycle rider injured in collision with car, pick-up truck or van : person injured while boarding or alighting
1251. V23.4 Motorcycle rider injured in collision with car, pick-up truck or van : driver injured in traffic accident
1252. V23.5 Motorcycle rider injured in collision with car, pick-up truck or van : passenger injured in traffic accident
1253. V23.9 Motorcycle rider injured in collision with car, pick-up truck or van : unspecified motorcycle rider injured in traffic accident
1254. V24 Motorcycle rider injured in collision with heavy transport vehicle or bus
1255. V24.0 Motorcycle rider injured in collision with heavy transport vehicle or bus : driver injured in nontraffic accident
1256. V24.1 Motorcycle rider injured in collision with heavy transport vehicle or bus : passenger

injured in nontraffic accident
1257. V24.2 Motorcycle rider injured in collision with heavy transport vehicle or bus : unspecified motorcycle rider injured in nontraffic accident
1258. V24.3 Motorcycle rider injured in collision with heavy transport vehicle or bus : person injured while boarding or alighting
1259. V24.4 Motorcycle rider injured in collision with heavy transport vehicle or bus : driver injured in traffic accident
1260. V24.5 Motorcycle rider injured in collision with heavy transport vehicle or bus : passenger injured in traffic accident
1261. V24.9 Motorcycle rider injured in collision with heavy transport vehicle or bus : unspecified motorcycle rider injured in traffic accident
1262. V25 Motorcycle rider injured in collision with railway train or railway vehicle
1263. V25.0 Motorcycle rider injured in collision with railway train or railway vehicle : driver injured in nontraffic accident
1264. V25.1 Motorcycle rider injured in collision with railway train or railway vehicle : passenger injured in nontraffic accident
1265. V25.2 Motorcycle rider injured in collision with railway train or railway vehicle : unspecified motorcycle rider injured in nontraffic accident
1266. V25.3 Motorcycle rider injured in collision with railway train or railway vehicle : person injured while boarding or alighting
1267. V25.4 Motorcycle rider injured in collision with railway train or railway vehicle : driver injured in traffic accident
1268. V25.5 Motorcycle rider injured in collision with railway train or railway vehicle : passenger injured in traffic accident
1269. V25.9 Motorcycle rider injured in collision with railway train or railway vehicle : unspecified motorcycle rider injured in traffic accident
1270. V26 Motorcycle rider injured in collision with other nonmotor vehicle
1271. V26.0 Motorcycle rider injured in collision with other nonmotor vehicle : driver injured in nontraffic accident
1272. V26.1 Motorcycle rider injured in collision with other nonmotor vehicle : passenger injured in nontraffic accident
1273. V26.2 Motorcycle rider injured in collision with other nonmotor vehicle : unspecified motorcycle rider injured in nontraffic accident
1274. V26.3 Motorcycle rider injured in collision with other nonmotor vehicle : person injured while boarding or alighting
1275. V26.4 Motorcycle rider injured in collision with other nonmotor vehicle : driver injured in traffic accident
1276. V26.5 Motorcycle rider injured in collision with other nonmotor vehicle : passenger injured in traffic accident
1277. V26.9 Motorcycle rider injured in collision with other nonmotor vehicle : unspecified motorcycle rider injured in traffic accident
1278. V27 Motorcycle rider injured in collision with fixed or stationary object
1279. V27.0 Motorcycle rider injured in collision with fixed or stationary object : driver injured

in nontraffic accident
1280. V27.1 Motorcycle rider injured in collision with fixed or stationary object : passenger injured in nontraffic accident
1281. V27.2 Motorcycle rider injured in collision with fixed or stationary object : unspecified motorcycle rider injured in nontraffic accident
1282. V27.3 Motorcycle rider injured in collision with fixed or stationary object : person injured while boarding or alighting
1283. V27.4 Motorcycle rider injured in collision with fixed or stationary object : driver injured in traffic accident
1284. V27.5 Motorcycle rider injured in collision with fixed or stationary object : passenger injured in traffic accident
1285. V27.9 Motorcycle rider injured in collision with fixed or stationary object : unspecified motorcycle rider injured in traffic accident
1286. V28 Motorcycle rider injured in noncollision transport accident
1287. V28.0 Motorcycle rider injured in noncollision transport accident : driver injured in nontraffic accident
1288. V28.1 Motorcycle rider injured in noncollision transport accident : passenger injured in nontraffic accident
1289. V28.2 Motorcycle rider injured in noncollision transport accident : unspecified motorcycle rider injured in nontraffic accident
1290. V28.3 Motorcycle rider injured in noncollision transport accident : person injured while boarding or alighting
1291. V28.4 Motorcycle rider injured in noncollision transport accident : driver injured in traffic accident
1292. V28.5 Motorcycle rider injured in noncollision transport accident : passenger injured in traffic accident
1293. V28.9 Motorcycle rider injured in noncollision transport accident : unspecified motorcycle rider injured in traffic accident
1294. V29 Motorcycle rider injured in other and unspecified transport accidents
1295. V29.0 Driver injured in collision with other and unspecified motor vehicles in nontraffic accident
1296. V29.1 Passenger injured in collision with other and unspecified motor vehicles in nontraffic accident
1297. V29.2 Unspecified motorcycle rider injured in collision with other and unspecified motor vehicles in nontraffic accident
1298. V29.3 Motorcycle rider [any] injured in unspecified nontraffic accident
1299. V29.4 Driver injured in collision with other and unspecified motor vehicles in traffic accident
1300. V29.5 Passenger injured in collision with other and unspecified motor vehicles in traffic accident
1301. V29.6 Unspecified motorcycle rider injured in collision with other and unspecified motor vehicles in traffic accident
1302. V29.8 Motorcycle rider [any] injured in other specified transport accidents

1303. V29.9 Motorcycle rider [any] injured in unspecified traffic accident
1304. V30 Occupant of three-wheeled motor vehicle injured in collision with pedestrian or animal
1305. V30.0 Occupant of three-wheeled motor vehicle injured in collision with pedestrian or animal : driver injured in nontraffic accident
1306. V30.1 Occupant of three-wheeled motor vehicle injured in collision with pedestrian or animal : passenger injured in nontraffic accident
1307. V30.2 Occupant of three-wheeled motor vehicle injured in collision with pedestrian or animal : person on outside of vehicle injured in nontraffic accident
1308. V30.3 Occupant of three-wheeled motor vehicle injured in collision with pedestrian or animal : unspecified occupant of three-wheeled motor vehicle injured in nontraffic accident
1309. V30.4 Occupant of three-wheeled motor vehicle injured in collision with pedestrian or animal : person injured while boarding or alighting
1310. V30.5 Occupant of three-wheeled motor vehicle injured in collision with pedestrian or animal : driver injured in traffic accident
1311. V30.6 Occupant of three-wheeled motor vehicle injured in collision with pedestrian or animal : passenger injured in traffic accident
1312. V30.7 Occupant of three-wheeled motor vehicle injured in collision with pedestrian or animal : person on outside of vehicle injured in traffic accident
1313. V30.9 Occupant of three-wheeled motor vehicle injured in collision with pedestrian or animal : unspecified occupant of three-wheeled motor vehicle injured in traffic accident
1314. V31 Occupant of three-wheeled motor vehicle injured in collision with pedal cycle
1315. V31.0 Occupant of three-wheeled motor vehicle injured in collision with pedal cycle : driver injured in nontraffic accident
1316. V31.1 Occupant of three-wheeled motor vehicle injured in collision with pedal cycle : passenger injured in nontraffic accident
1317. V31.2 Occupant of three-wheeled motor vehicle injured in collision with pedal cycle : person on outside of vehicle injured in nontraffic accident
1318. V31.3 Occupant of three-wheeled motor vehicle injured in collision with pedal cycle : unspecified occupant of three-wheeled motor vehicle injured in nontraffic accident
1319. V31.4 Occupant of three-wheeled motor vehicle injured in collision with pedal cycle : person injured while boarding or alighting
1320. V31.5 Occupant of three-wheeled motor vehicle injured in collision with pedal cycle : driver injured in traffic accident
1321. V31.6 Occupant of three-wheeled motor vehicle injured in collision with pedal cycle : passenger injured in traffic accident
1322. V31.7 Occupant of three-wheeled motor vehicle injured in collision with pedal cycle : person on outside of vehicle injured in traffic accident
1323. V31.9 Occupant of three-wheeled motor vehicle injured in collision with pedal cycle : unspecified occupant of three-wheeled motor vehicle injured in traffic accident
1324. V32 Occupant of three-wheeled motor vehicle injured in collision with two- or three-wheeled motor vehicle
1325. V32.0 Occupant of three-wheeled motor vehicle injured in collision with two- or three-wheeled motor vehicle : driver injured in nontraffic accident

1326. V32.1 Occupant of three-wheeled motor vehicle injured in collision with two- or three-wheeled motor vehicle : passenger injured in nontraffic accident
1327. V32.2 Occupant of three-wheeled motor vehicle injured in collision with two- or three-wheeled motor vehicle : person on outside of vehicle injured in nontraffic accident
1328. V32.3 Occupant of three-wheeled motor vehicle injured in collision with two- or three-wheeled motor vehicle : unspecified occupant of three-wheeled motor vehicle injured in nontraffic accident
1329. V32.4 Occupant of three-wheeled motor vehicle injured in collision with two- or three-wheeled motor vehicle : person injured while boarding or alighting
1330. V32.5 Occupant of three-wheeled motor vehicle injured in collision with two- or three-wheeled motor vehicle : driver injured in traffic accident
1331. V32.6 Occupant of three-wheeled motor vehicle injured in collision with two- or three-wheeled motor vehicle : passenger injured in traffic accident
1332. V32.7 Occupant of three-wheeled motor vehicle injured in collision with two- or three-wheeled motor vehicle : person on outside of vehicle injured in traffic accident
1333. V32.9 Occupant of three-wheeled motor vehicle injured in collision with two- or three-wheeled motor vehicle : unspecified occupant of three-wheeled motor vehicle injured in traffic accident
1334. V33 Occupant of three-wheeled motor vehicle injured in collision with car, pick-up truck or van
1335. V33.0 Occupant of three-wheeled motor vehicle injured in collision with car, pick-up truck or van : driver injured in nontraffic accident
1336. V33.1 Occupant of three-wheeled motor vehicle injured in collision with car, pick-up truck or van : passenger injured in nontraffic accident
1337. V33.2 Occupant of three-wheeled motor vehicle injured in collision with car, pick-up truck or van : person on outside of vehicle injured in nontraffic accident
1338. V33.3 Occupant of three-wheeled motor vehicle injured in collision with car, pick-up truck or van : unspecified occupant of three-wheeled motor vehicle injured in nontraffic accident
1339. V33.4 Occupant of three-wheeled motor vehicle injured in collision with car, pick-up truck or van : person injured while boarding or alighting
1340. V33.5 Occupant of three-wheeled motor vehicle injured in collision with car, pick-up truck or van : driver injured in traffic accident
1341. V33.6 Occupant of three-wheeled motor vehicle injured in collision with car, pick-up truck or van : passenger injured in traffic accident
1342. V33.7 Occupant of three-wheeled motor vehicle injured in collision with car, pick-up truck or van : person on outside of vehicle injured in traffic accident
1343. V33.9 Occupant of three-wheeled motor vehicle injured in collision with car, pick-up truck or van : unspecified occupant of three-wheeled motor vehicle injured in traffic accident
1344. V34 Occupant of three-wheeled motor vehicle injured in collision with heavy transport vehicle or bus
1345. V34.0 Occupant of three-wheeled motor vehicle injured in collision with heavy transport vehicle or bus : driver injured in nontraffic accident
1346. V34.1 Occupant of three-wheeled motor vehicle injured in collision with heavy transport

vehicle or bus : passenger injured in nontraffic accident

1347. V34.2 Occupant of three-wheeled motor vehicle injured in collision with heavy transport vehicle or bus : person on outside of vehicle injured in nontraffic accident
1348. V34.3 Occupant of three-wheeled motor vehicle injured in collision with heavy transport vehicle or bus : unspecified occupant of three-wheeled motor vehicle injured in nontraffic accident
1349. V34.4 Occupant of three-wheeled motor vehicle injured in collision with heavy transport vehicle or bus : person injured while boarding or alighting
1350. V34.5 Occupant of three-wheeled motor vehicle injured in collision with heavy transport vehicle or bus : driver injured in traffic accident
1351. V34.6 Occupant of three-wheeled motor vehicle injured in collision with heavy transport vehicle or bus : passenger injured in traffic accident
1352. V34.7 Occupant of three-wheeled motor vehicle injured in collision with heavy transport vehicle or bus : person on outside of vehicle injured in traffic accident
1353. V34.9 Occupant of three-wheeled motor vehicle injured in collision with heavy transport vehicle or bus : unspecified occupant of three-wheeled motor vehicle injured in traffic accident
1354. V35 Occupant of three-wheeled motor vehicle injured in collision with railway train or railway vehicle
1355. V35.0 Occupant of three-wheeled motor vehicle injured in collision with railway train or railway vehicle : driver injured in nontraffic accident
1356. V35.1 Occupant of three-wheeled motor vehicle injured in collision with railway train or railway vehicle : passenger injured in nontraffic accident
1357. V35.2 Occupant of three-wheeled motor vehicle injured in collision with railway train or railway vehicle : person on outside of vehicle injured in nontraffic accident
1358. V35.3 Occupant of three-wheeled motor vehicle injured in collision with railway train or railway vehicle : unspecified occupant of three-wheeled motor vehicle injured in nontraffic accident
1359. V35.4 Occupant of three-wheeled motor vehicle injured in collision with railway train or railway vehicle : person injured while boarding or alighting
1360. V35.5 Occupant of three-wheeled motor vehicle injured in collision with railway train or railway vehicle : driver injured in traffic accident
1361. V35.6 Occupant of three-wheeled motor vehicle injured in collision with railway train or railway vehicle : passenger injured in traffic accident
1362. V35.7 Occupant of three-wheeled motor vehicle injured in collision with railway train or railway vehicle : person on outside of vehicle injured in traffic accident
1363. V35.9 Occupant of three-wheeled motor vehicle injured in collision with railway train or railway vehicle : unspecified occupant of three-wheeled motor vehicle injured in traffic accident
1364. V36 Occupant of three-wheeled motor vehicle injured in collision with other nonmotor vehicle
1365. V36.0 Occupant of three-wheeled motor vehicle injured in collision with other nonmotor vehicle : driver injured in nontraffic accident

1366. V36.1 Occupant of three-wheeled motor vehicle injured in collision with other nonmotor vehicle : passenger injured in nontraffic accident
1367. V36.2 Occupant of three-wheeled motor vehicle injured in collision with other nonmotor vehicle : person on outside of vehicle injured in nontraffic accident
1368. V36.3 Occupant of three-wheeled motor vehicle injured in collision with other nonmotor vehicle : unspecified occupant of three-wheeled motor vehicle injured in nontraffic accident
1369. V36.4 Occupant of three-wheeled motor vehicle injured in collision with other nonmotor vehicle : person injured while boarding or alighting
1370. V36.5 Occupant of three-wheeled motor vehicle injured in collision with other nonmotor vehicle : driver injured in traffic accident
1371. V36.6 Occupant of three-wheeled motor vehicle injured in collision with other nonmotor vehicle : passenger injured in traffic accident
1372. V36.7 Occupant of three-wheeled motor vehicle injured in collision with other nonmotor vehicle : person on outside of vehicle injured in traffic accident
1373. V36.9 Occupant of three-wheeled motor vehicle injured in collision with other nonmotor vehicle : unspecified occupant of three-wheeled motor vehicle injured in traffic accident
1374. V37 Occupant of three-wheeled motor vehicle injured in collision with fixed or stationary object
1375. V37.0 Occupant of three-wheeled motor vehicle injured in collision with fixed or stationary object : driver injured in nontraffic accident
1376. V37.1 Occupant of three-wheeled motor vehicle injured in collision with fixed or stationary object : passenger injured in nontraffic accident
1377. V37.2 Occupant of three-wheeled motor vehicle injured in collision with fixed or stationary object : person on outside of vehicle injured in nontraffic accident
1378. V37.3 Occupant of three-wheeled motor vehicle injured in collision with fixed or stationary object : unspecified occupant of three-wheeled motor vehicle injured in nontraffic accident
1379. V37.4 Occupant of three-wheeled motor vehicle injured in collision with fixed or stationary object : person injured while boarding or alighting
1380. V37.5 Occupant of three-wheeled motor vehicle injured in collision with fixed or stationary object : driver injured in traffic accident
1381. V37.6 Occupant of three-wheeled motor vehicle injured in collision with fixed or stationary object : passenger injured in traffic accident
1382. V37.7 Occupant of three-wheeled motor vehicle injured in collision with fixed or stationary object : person on outside of vehicle injured in traffic accident
1383. V37.9 Occupant of three-wheeled motor vehicle injured in collision with fixed or stationary object : unspecified occupant of three-wheeled motor vehicle injured in traffic accident
1384. V38 Occupant of three-wheeled motor vehicle injured in noncollision transport accident
1385. V38.0 Occupant of three-wheeled motor vehicle injured in noncollision transport accident : driver injured in nontraffic accident
1386. V38.1 Occupant of three-wheeled motor vehicle injured in noncollision transport accident : passenger injured in nontraffic accident
1387. V38.2 Occupant of three-wheeled motor vehicle injured in noncollision transport accident : person on outside of vehicle injured in nontraffic accident

1388. V38.3 Occupant of three-wheeled motor vehicle injured in noncollision transport accident : unspecified occupant of three-wheeled motor vehicle injured in nontraffic accident
1389. V38.4 Occupant of three-wheeled motor vehicle injured in noncollision transport accident : person injured while boarding or alighting
1390. V38.5 Occupant of three-wheeled motor vehicle injured in noncollision transport accident : driver injured in traffic accident
1391. V38.6 Occupant of three-wheeled motor vehicle injured in noncollision transport accident : passenger injured in traffic accident
1392. V38.7 Occupant of three-wheeled motor vehicle injured in noncollision transport accident : person on outside of vehicle injured in traffic accident
1393. V38.9 Occupant of three-wheeled motor vehicle injured in noncollision transport accident : unspecified occupant of three-wheeled motor vehicle injured in traffic accident
1394. V39 Occupant of three-wheeled motor vehicle injured in other and unspecified transport accidents
1395. V39.0 Driver injured in collision with other and unspecified motor vehicles in nontraffic accident
1396. V39.1 Passenger injured in collision with other and unspecified motor vehicles in nontraffic accident
1397. V39.2 Unspecified occupant of three-wheeled motor vehicle injured in collision with other and unspecified motor vehicles in nontraffic accident
1398. V39.3 Occupant [any] of three-wheeled motor vehicle injured in unspecified nontraffic accident
1399. V39.4 Driver injured in collision with other and unspecified motor vehicles in traffic accident
1400. V39.5 Passenger injured in collision with other and unspecified motor vehicles in traffic accident
1401. V39.6 Unspecified occupant of three-wheeled motor vehicle injured in collision with other and unspecified motor vehicles in traffic accident
1402. V39.8 Occupant [any] of three-wheeled motor vehicle injured in other specified transport accidents
1403. V39.9 Occupant [any] of three-wheeled motor vehicle injured in unspecified traffic accident
1404. V40 Car occupant injured in collision with pedestrian or animal
1405. V40.0 Car occupant injured in collision with pedestrian or animal : driver injured in nontraffic accident
1406. V40.1 Car occupant injured in collision with pedestrian or animal : passenger injured in nontraffic accident
1407. V40.2 Car occupant injured in collision with pedestrian or animal : person on outside of vehicle injured in nontraffic accident
1408. V40.3 Car occupant injured in collision with pedestrian or animal : unspecified car occupant injured in nontraffic accident
1409. V40.4 Car occupant injured in collision with pedestrian or animal : person injured while boarding or alighting
1410. V40.5 Car occupant injured in collision with pedestrian or animal : driver injured in traffic

accident

1411. V40.6 Car occupant injured in collision with pedestrian or animal : passenger injured in traffic accident
1412. V40.7 Car occupant injured in collision with pedestrian or animal : person on outside of vehicle injured in traffic accident
1413. V40.9 Car occupant injured in collision with pedestrian or animal : unspecified car occupant injured in traffic accident
1414. V41 Car occupant injured in collision with pedal cycle
1415. V41.0 Car occupant injured in collision with pedal cycle : driver injured in nontraffic accident
1416. V41.1 Car occupant injured in collision with pedal cycle : passenger injured in nontraffic accident
1417. V41.2 Car occupant injured in collision with pedal cycle : person on outside of vehicle injured in nontraffic accident
1418. V41.3 Car occupant injured in collision with pedal cycle : unspecified car occupant injured in nontraffic accident
1419. V41.4 Car occupant injured in collision with pedal cycle : person injured while boarding or alighting
1420. V41.5 Car occupant injured in collision with pedal cycle : driver injured in traffic accident
1421. V41.6 Car occupant injured in collision with pedal cycle : passenger injured in traffic accident
1422. V41.7 Car occupant injured in collision with pedal cycle : person on outside of vehicle injured in traffic accident
1423. V41.9 Car occupant injured in collision with pedal cycle : unspecified car occupant injured in traffic accident
1424. V42 Car occupant injured in collision with two- or three-wheeled motor vehicle
1425. V42.0 Car occupant injured in collision with two- or three-wheeled motor vehicle : driver injured in nontraffic accident
1426. V42.1 Car occupant injured in collision with two- or three-wheeled motor vehicle : passenger injured in nontraffic accident
1427. V42.2 Car occupant injured in collision with two- or three-wheeled motor vehicle : person on outside of vehicle injured in nontraffic accident
1428. V42.3 Car occupant injured in collision with two- or three-wheeled motor vehicle : unspecified car occupant injured in nontraffic accident
1429. V42.4 Car occupant injured in collision with two- or three-wheeled motor vehicle : person injured while boarding or alighting
1430. V42.5 Car occupant injured in collision with two- or three-wheeled motor vehicle : driver injured in traffic accident
1431. V42.6 Car occupant injured in collision with two- or three-wheeled motor vehicle : passenger injured in traffic accident
1432. V42.7 Car occupant injured in collision with two- or three-wheeled motor vehicle : person on outside of vehicle injured in traffic accident
1433. V42.9 Car occupant injured in collision with two- or three-wheeled motor vehicle : un-

specified car occupant injured in traffic accident
1434. V43 Car occupant injured in collision with car, pick-up truck or van
1435. V43.0 Car occupant injured in collision with car, pick-up truck or van : driver injured in nontraffic accident
1436. V43.1 Car occupant injured in collision with car, pick-up truck or van : passenger injured in nontraffic accident
1437. V43.2 Car occupant injured in collision with car, pick-up truck or van : person on outside of vehicle injured in nontraffic accident
1438. V43.3 Car occupant injured in collision with car, pick-up truck or van : unspecified car occupant injured in nontraffic accident
1439. V43.4 Car occupant injured in collision with car, pick-up truck or van : person injured while boarding or alighting
1440. V43.5 Car occupant injured in collision with car, pick-up truck or van : driver injured in traffic accident
1441. V43.6 Car occupant injured in collision with car, pick-up truck or van : passenger injured in traffic accident
1442. V43.7 Car occupant injured in collision with car, pick-up truck or van : person on outside of vehicle injured in traffic accident
1443. V43.9 Car occupant injured in collision with car, pick-up truck or van : unspecified car occupant injured in traffic accident
1444. V44 Car occupant injured in collision with heavy transport vehicle or bus
1445. V44.0 Car occupant injured in collision with heavy transport vehicle or bus : driver injured in nontraffic accident
1446. V44.1 Car occupant injured in collision with heavy transport vehicle or bus : passenger injured in nontraffic accident
1447. V44.2 Car occupant injured in collision with heavy transport vehicle or bus : person on outside of vehicle injured in nontraffic accident
1448. V44.3 Car occupant injured in collision with heavy transport vehicle or bus : unspecified car occupant injured in nontraffic accident
1449. V44.4 Car occupant injured in collision with heavy transport vehicle or bus : person injured while boarding or alighting
1450. V44.5 Car occupant injured in collision with heavy transport vehicle or bus : driver injured in traffic accident
1451. V44.6 Car occupant injured in collision with heavy transport vehicle or bus : passenger injured in traffic accident
1452. V44.7 Car occupant injured in collision with heavy transport vehicle or bus : person on outside of vehicle injured in traffic accident
1453. V44.9 Car occupant injured in collision with heavy transport vehicle or bus : unspecified car occupant injured in traffic accident
1454. V45 Car occupant injured in collision with railway train or railway vehicle
1455. V45.0 Car occupant injured in collision with railway train or railway vehicle : driver injured in nontraffic accident
1456. V45.1 Car occupant injured in collision with railway train or railway vehicle : passenger

injured in nontraffic accident
1457. V45.2 Car occupant injured in collision with railway train or railway vehicle : person on outside of vehicle injured in nontraffic accident
1458. V45.3 Car occupant injured in collision with railway train or railway vehicle : unspecified car occupant injured in nontraffic accident
1459. V45.4 Car occupant injured in collision with railway train or railway vehicle : person injured while boarding or alighting
1460. V45.5 Car occupant injured in collision with railway train or railway vehicle : driver injured in traffic accident
1461. V45.6 Car occupant injured in collision with railway train or railway vehicle : passenger injured in traffic accident
1462. V45.7 Car occupant injured in collision with railway train or railway vehicle : person on outside of vehicle injured in traffic accident
1463. V45.9 Car occupant injured in collision with railway train or railway vehicle : unspecified car occupant injured in traffic accident
1464. V46 Car occupant injured in collision with other nonmotor vehicle
1465. V46.0 Car occupant injured in collision with other nonmotor vehicle : driver injured in nontraffic accident
1466. V46.1 Car occupant injured in collision with other nonmotor vehicle : passenger injured in nontraffic accident
1467. V46.2 Car occupant injured in collision with other nonmotor vehicle : person on outside of vehicle injured in nontraffic accident
1468. V46.3 Car occupant injured in collision with other nonmotor vehicle : unspecified car occupant injured in nontraffic accident
1469. V46.4 Car occupant injured in collision with other nonmotor vehicle : person injured while boarding or alighting
1470. V46.5 Car occupant injured in collision with other nonmotor vehicle : driver injured in traffic accident
1471. V46.6 Car occupant injured in collision with other nonmotor vehicle : passenger injured in traffic accident
1472. V46.7 Car occupant injured in collision with other nonmotor vehicle : person on outside of vehicle injured in traffic accident
1473. V46.9 Car occupant injured in collision with other nonmotor vehicle : unspecified car occupant injured in traffic accident
1474. V47 Car occupant injured in collision with fixed or stationary object
1475. V47.0 Car occupant injured in collision with fixed or stationary object : driver injured in nontraffic accident
1476. V47.1 Car occupant injured in collision with fixed or stationary object : passenger injured in nontraffic accident
1477. V47.2 Car occupant injured in collision with fixed or stationary object : person on outside of vehicle injured in nontraffic accident
1478. V47.3 Car occupant injured in collision with fixed or stationary object : unspecified car occupant injured in nontraffic accident

1479. V47.4 Car occupant injured in collision with fixed or stationary object : person injured while boarding or alighting
1480. V47.5 Car occupant injured in collision with fixed or stationary object : driver injured in traffic accident
1481. V47.6 Car occupant injured in collision with fixed or stationary object : passenger injured in traffic accident
1482. V47.7 Car occupant injured in collision with fixed or stationary object : person on outside of vehicle injured in traffic accident
1483. V47.9 Car occupant injured in collision with fixed or stationary object : unspecified car occupant injured in traffic accident
1484. V48 Car occupant injured in noncollision transport accident
1485. V48.0 Car occupant injured in noncollision transport accident : driver injured in nontraffic accident
1486. V48.1 Car occupant injured in noncollision transport accident : passenger injured in nontraffic accident
1487. V48.2 Car occupant injured in noncollision transport accident : person on outside of vehicle injured in nontraffic accident
1488. V48.3 Car occupant injured in noncollision transport accident : unspecified car occupant injured in nontraffic accident
1489. V48.4 Car occupant injured in noncollision transport accident : person injured while boarding or alighting
1490. V48.5 Car occupant injured in noncollision transport accident : driver injured in traffic accident
1491. V48.6 Car occupant injured in noncollision transport accident : passenger injured in traffic accident
1492. V48.7 Car occupant injured in noncollision transport accident : person on outside of vehicle injured in traffic accident
1493. V48.9 Car occupant injured in noncollision transport accident : unspecified car occupant injured in traffic accident
1494. V49 Car occupant injured in other and unspecified transport accidents
1495. V49.0 Driver injured in collision with other and unspecified motor vehicles in nontraffic accident
1496. V49.1 Passenger injured in collision with other and unspecified motor vehicles in nontraffic accident
1497. V49.2 Unspecified car occupant injured in collision with other and unspecified motor vehicles in nontraffic accident
1498. V49.3 Car occupant [any] injured in unspecified nontraffic accident
1499. V49.4 Driver injured in collision with other and unspecified motor vehicles in traffic accident
1500. V49.5 Passenger injured in collision with other and unspecified motor vehicles in traffic accident
1501. V49.6 Unspecified car occupant injured in collision with other and unspecified motor vehicles in traffic accident

1502. V49.8 Car occupant [any] injured in other specified transport accidents
1503. V49.9 Car occupant [any] injured in unspecified traffic accident
1504. V50 Occupant of pick-up truck or van injured in collision with pedestrian or animal
1505. V50.0 Occupant of pick-up truck or van injured in collision with pedestrian or animal : driver injured in nontraffic accident
1506. V50.1 Occupant of pick-up truck or van injured in collision with pedestrian or animal : passenger injured in nontraffic accident
1507. V50.2 Occupant of pick-up truck or van injured in collision with pedestrian or animal : person on outside of vehicle injured in nontraffic accident
1508. V50.3 Occupant of pick-up truck or van injured in collision with pedestrian or animal : unspecified occupant of pick-up truck or van injured in nontraffic accident
1509. V50.4 Occupant of pick-up truck or van injured in collision with pedestrian or animal : person injured while boarding or alighting
1510. V50.5 Occupant of pick-up truck or van injured in collision with pedestrian or animal : driver injured in traffic accident
1511. V50.6 Occupant of pick-up truck or van injured in collision with pedestrian or animal : passenger injured in traffic accident
1512. V50.7 Occupant of pick-up truck or van injured in collision with pedestrian or animal : person on outside of vehicle injured in traffic accident
1513. V50.9 Occupant of pick-up truck or van injured in collision with pedestrian or animal : unspecified occupant of pick-up truck or van injured in traffic accident
1514. V51 Occupant of pick-up truck or van injured in collision with pedal cycle
1515. V51.0 Occupant of pick-up truck or van injured in collision with pedal cycle : driver injured in nontraffic accident
1516. V51.1 Occupant of pick-up truck or van injured in collision with pedal cycle : passenger injured in nontraffic accident
1517. V51.2 Occupant of pick-up truck or van injured in collision with pedal cycle : person on outside of vehicle injured in nontraffic accident
1518. V51.3 Occupant of pick-up truck or van injured in collision with pedal cycle : unspecified occupant of pick-up truck or van injured in nontraffic accident
1519. V51.4 Occupant of pick-up truck or van injured in collision with pedal cycle : person injured while boarding or alighting
1520. V51.5 Occupant of pick-up truck or van injured in collision with pedal cycle : driver injured in traffic accident
1521. V51.6 Occupant of pick-up truck or van injured in collision with pedal cycle : passenger injured in traffic accident
1522. V51.7 Occupant of pick-up truck or van injured in collision with pedal cycle : person on outside of vehicle injured in traffic accident
1523. V51.9 Occupant of pick-up truck or van injured in collision with pedal cycle : unspecified occupant of pick-up truck or van injured in traffic accident
1524. V52 Occupant of pick-up truck or van injured in collision with two- or three-wheeled motor vehicle
1525. V52.0 Occupant of pick-up truck or van injured in collision with two- or three-wheeled

motor vehicle : driver injured in nontraffic accident

1526. V52.1 Occupant of pick-up truck or van injured in collision with two- or three-wheeled motor vehicle : passenger injured in nontraffic accident
1527. V52.2 Occupant of pick-up truck or van injured in collision with two- or three-wheeled motor vehicle : person on outside of vehicle injured in nontraffic accident
1528. V52.3 Occupant of pick-up truck or van injured in collision with two- or three-wheeled motor vehicle : unspecified occupant of pick-up truck or van injured in nontraffic accident
1529. V52.4 Occupant of pick-up truck or van injured in collision with two- or three-wheeled motor vehicle : person injured while boarding or alighting
1530. V52.5 Occupant of pick-up truck or van injured in collision with two- or three-wheeled motor vehicle : driver injured in traffic accident
1531. V52.6 Occupant of pick-up truck or van injured in collision with two- or three-wheeled motor vehicle : passenger injured in traffic accident
1532. V52.7 Occupant of pick-up truck or van injured in collision with two- or three-wheeled motor vehicle : person on outside of vehicle injured in traffic accident
1533. V52.9 Occupant of pick-up truck or van injured in collision with two- or three-wheeled motor vehicle : unspecified occupant of pick-up truck or van injured in traffic accident
1534. V53 Occupant of pick-up truck or van injured in collision with car, pick-up truck or van
1535. V53.0 Occupant of pick-up truck or van injured in collision with car, pick-up truck or van : driver injured in nontraffic accident
1536. V53.1 Occupant of pick-up truck or van injured in collision with car, pick-up truck or van : passenger injured in nontraffic accident
1537. V53.2 Occupant of pick-up truck or van injured in collision with car, pick-up truck or van : person on outside of vehicle injured in nontraffic accident
1538. V53.3 Occupant of pick-up truck or van injured in collision with car, pick-up truck or van : unspecified occupant of pick-up truck or van injured in nontraffic accident
1539. V53.4 Occupant of pick-up truck or van injured in collision with car, pick-up truck or van : person injured while boarding or alighting
1540. V53.5 Occupant of pick-up truck or van injured in collision with car, pick-up truck or van : driver injured in traffic accident
1541. V53.6 Occupant of pick-up truck or van injured in collision with car, pick-up truck or van : passenger injured in traffic accident
1542. V53.7 Occupant of pick-up truck or van injured in collision with car, pick-up truck or van : person on outside of vehicle injured in traffic accident
1543. V53.9 Occupant of pick-up truck or van injured in collision with car, pick-up truck or van : unspecified occupant of pick-up truck or van injured in traffic accident
1544. V54 Occupant of pick-up truck or van injured in collision with heavy transport vehicle or bus
1545. V54.0 Occupant of pick-up truck or van injured in collision with heavy transport vehicle or bus : driver injured in nontraffic accident
1546. V54.1 Occupant of pick-up truck or van injured in collision with heavy transport vehicle or bus : passenger injured in nontraffic accident
1547. V54.2 Occupant of pick-up truck or van injured in collision with heavy transport vehicle

or bus : person on outside of vehicle injured in nontraffic accident
1548. V54.3 Occupant of pick-up truck or van injured in collision with heavy transport vehicle or bus : unspecified occupant of pick-up truck or van injured in nontraffic accident
1549. V54.4 Occupant of pick-up truck or van injured in collision with heavy transport vehicle or bus : person injured while boarding or alighting
1550. V54.5 Occupant of pick-up truck or van injured in collision with heavy transport vehicle or bus : driver injured in traffic accident
1551. V54.6 Occupant of pick-up truck or van injured in collision with heavy transport vehicle or bus : passenger injured in traffic accident
1552. V54.7 Occupant of pick-up truck or van injured in collision with heavy transport vehicle or bus : person on outside of vehicle injured in traffic accident
1553. V54.9 Occupant of pick-up truck or van injured in collision with heavy transport vehicle or bus : unspecified occupant of pick-up truck or van injured in traffic accident
1554. V55 Occupant of pick-up truck or van injured in collision with railway train or railway vehicle
1555. V55.0 Occupant of pick-up truck or van injured in collision with railway train or railway vehicle : driver injured in nontraffic accident
1556. V55.1 Occupant of pick-up truck or van injured in collision with railway train or railway vehicle : passenger injured in nontraffic accident
1557. V55.2 Occupant of pick-up truck or van injured in collision with railway train or railway vehicle : person on outside of vehicle injured in nontraffic accident
1558. V55.3 Occupant of pick-up truck or van injured in collision with railway train or railway vehicle : unspecified occupant of pick-up truck or van injured in nontraffic accident
1559. V55.4 Occupant of pick-up truck or van injured in collision with railway train or railway vehicle : person injured while boarding or alighting
1560. V55.5 Occupant of pick-up truck or van injured in collision with railway train or railway vehicle : driver injured in traffic accident
1561. V55.6 Occupant of pick-up truck or van injured in collision with railway train or railway vehicle : passenger injured in traffic accident
1562. V55.7 Occupant of pick-up truck or van injured in collision with railway train or railway vehicle : person on outside of vehicle injured in traffic accident
1563. V55.9 Occupant of pick-up truck or van injured in collision with railway train or railway vehicle : unspecified occupant of pick-up truck or van injured in traffic accident
1564. V56 Occupant of pick-up truck or van injured in collision with other nonmotor vehicle
1565. V56.0 Occupant of pick-up truck or van injured in collision with other nonmotor vehicle : driver injured in nontraffic accident
1566. V56.1 Occupant of pick-up truck or van injured in collision with other nonmotor vehicle : passenger injured in nontraffic accident
1567. V56.2 Occupant of pick-up truck or van injured in collision with other nonmotor vehicle : person on outside of vehicle injured in nontraffic accident
1568. V56.3 Occupant of pick-up truck or van injured in collision with other nonmotor vehicle : unspecified occupant of pick-up truck or van injured in nontraffic accident
1569. V56.4 Occupant of pick-up truck or van injured in collision with other nonmotor vehicle :

person injured while boarding or alighting
1570. V56.5 Occupant of pick-up truck or van injured in collision with other nonmotor vehicle : driver injured in traffic accident
1571. V56.6 Occupant of pick-up truck or van injured in collision with other nonmotor vehicle : passenger injured in traffic accident
1572. V56.7 Occupant of pick-up truck or van injured in collision with other nonmotor vehicle : person on outside of vehicle injured in traffic accident
1573. V56.9 Occupant of pick-up truck or van injured in collision with other nonmotor vehicle : unspecified occupant of pick-up truck or van injured in traffic accident
1574. V57 Occupant of pick-up truck or van injured in collision with fixed or stationary object
1575. V57.0 Occupant of pick-up truck or van injured in collision with fixed or stationary object : driver injured in nontraffic accident
1576. V57.1 Occupant of pick-up truck or van injured in collision with fixed or stationary object : passenger injured in nontraffic accident
1577. V57.2 Occupant of pick-up truck or van injured in collision with fixed or stationary object : person on outside of vehicle injured in nontraffic accident
1578. V57.3 Occupant of pick-up truck or van injured in collision with fixed or stationary object : unspecified occupant of pick-up truck or van injured in nontraffic accident
1579. V57.4 Occupant of pick-up truck or van injured in collision with fixed or stationary object : person injured while boarding or alighting
1580. V57.5 Occupant of pick-up truck or van injured in collision with fixed or stationary object : driver injured in traffic accident
1581. V57.6 Occupant of pick-up truck or van injured in collision with fixed or stationary object : passenger injured in traffic accident
1582. V57.7 Occupant of pick-up truck or van injured in collision with fixed or stationary object : person on outside of vehicle injured in traffic accident
1583. V57.9 Occupant of pick-up truck or van injured in collision with fixed or stationary object : unspecified occupant of pick-up truck or van injured in traffic accident
1584. V58 Occupant of pick-up truck or van injured in noncollision transport accident
1585. V58.0 Occupant of pick-up truck or van injured in noncollision transport accident : driver injured in nontraffic accident
1586. V58.1 Occupant of pick-up truck or van injured in noncollision transport accident : passenger injured in nontraffic accident
1587. V58.2 Occupant of pick-up truck or van injured in noncollision transport accident : person on outside of vehicle injured in nontraffic accident
1588. V58.3 Occupant of pick-up truck or van injured in noncollision transport accident : unspecified occupant of pick-up truck or van injured in nontraffic accident
1589. V58.4 Occupant of pick-up truck or van injured in noncollision transport accident : person injured while boarding or alighting
1590. V58.5 Occupant of pick-up truck or van injured in noncollision transport accident : driver injured in traffic accident
1591. V58.6 Occupant of pick-up truck or van injured in noncollision transport accident : passenger injured in traffic accident

1592. V58.7 Occupant of pick-up truck or van injured in noncollision transport accident : person on outside of vehicle injured in traffic accident
1593. V58.9 Occupant of pick-up truck or van injured in noncollision transport accident : unspecified occupant of pick-up truck or van injured in traffic accident
1594. V59 Occupant of pick-up truck or van injured in other and unspecified transport accidents
1595. V59.0 Driver injured in collision with other and unspecified motor vehicles in nontraffic accident
1596. V59.1 Passenger injured in collision with other and unspecified motor vehicles in nontraffic accident
1597. V59.2 Unspecified occupant of pick-up truck or van injured in collision with other and unspecified motor vehicles in nontraffic accident
1598. V59.3 Occupant [any] of pick-up truck or van injured in unspecified nontraffic accident
1599. V59.4 Driver injured in collision with other and unspecified motor vehicles in traffic accident
1600. V59.5 Passenger injured in collision with other and unspecified motor vehicles in traffic accident
1601. V59.6 Unspecified occupant of pick-up truck or van injured in collision with other and unspecified motor vehicles in traffic accident
1602. V59.8 Occupant [any] of pick-up truck or van injured in other specified transport accidents
1603. V59.9 Occupant [any] of pick-up truck or van injured in unspecified traffic accident
1604. V60 Occupant of heavy transport vehicle injured in collision with pedestrian or animal
1605. V60.0 Occupant of heavy transport vehicle injured in collision with pedestrian or animal : driver injured in nontraffic accident
1606. V60.1 Occupant of heavy transport vehicle injured in collision with pedestrian or animal : passenger injured in nontraffic accident
1607. V60.2 Occupant of heavy transport vehicle injured in collision with pedestrian or animal : person on outside of vehicle injured in nontraffic accident
1608. V60.3 Occupant of heavy transport vehicle injured in collision with pedestrian or animal : unspecified occupant of heavy transport vehicle injured in nontraffic accident
1609. V60.4 Occupant of heavy transport vehicle injured in collision with pedestrian or animal : person injured while boarding or alighting
1610. V60.5 Occupant of heavy transport vehicle injured in collision with pedestrian or animal : driver injured in traffic accident
1611. V60.6 Occupant of heavy transport vehicle injured in collision with pedestrian or animal : passenger injured in traffic accident
1612. V60.7 Occupant of heavy transport vehicle injured in collision with pedestrian or animal : person on outside of vehicle injured in traffic accident
1613. V60.9 Occupant of heavy transport vehicle injured in collision with pedestrian or animal : unspecified occupant of heavy transport vehicle injured in traffic accident
1614. V61 Occupant of heavy transport vehicle injured in collision with pedal cycle
1615. V61.0 Occupant of heavy transport vehicle injured in collision with pedal cycle : driver injured in nontraffic accident
1616. V61.1 Occupant of heavy transport vehicle injured in collision with pedal cycle : passenger

injured in nontraffic accident

1617. V61.2 Occupant of heavy transport vehicle injured in collision with pedal cycle : person on outside of vehicle injured in nontraffic accident
1618. V61.3 Occupant of heavy transport vehicle injured in collision with pedal cycle : unspecified occupant of heavy transport vehicle injured in nontraffic accident
1619. V61.4 Occupant of heavy transport vehicle injured in collision with pedal cycle : person injured while boarding or alighting
1620. V61.5 Occupant of heavy transport vehicle injured in collision with pedal cycle : driver injured in traffic accident
1621. V61.6 Occupant of heavy transport vehicle injured in collision with pedal cycle : passenger injured in traffic accident
1622. V61.7 Occupant of heavy transport vehicle injured in collision with pedal cycle : person on outside of vehicle injured in traffic accident
1623. V61.9 Occupant of heavy transport vehicle injured in collision with pedal cycle : unspecified occupant of heavy transport vehicle injured in traffic accident
1624. V62 Occupant of heavy transport vehicle injured in collision with two- or three-wheeled motor vehicle
1625. V62.0 Occupant of heavy transport vehicle injured in collision with two- or three-wheeled motor vehicle : driver injured in nontraffic accident
1626. V62.1 Occupant of heavy transport vehicle injured in collision with two- or three-wheeled motor vehicle : passenger injured in nontraffic accident
1627. V62.2 Occupant of heavy transport vehicle injured in collision with two- or three-wheeled motor vehicle : person on outside of vehicle injured in nontraffic accident
1628. V62.3 Occupant of heavy transport vehicle injured in collision with two- or three-wheeled motor vehicle : unspecified occupant of heavy transport vehicle injured in nontraffic accident
1629. V62.4 Occupant of heavy transport vehicle injured in collision with two- or three-wheeled motor vehicle : person injured while boarding or alighting
1630. V62.5 Occupant of heavy transport vehicle injured in collision with two- or three-wheeled motor vehicle : driver injured in traffic accident
1631. V62.6 Occupant of heavy transport vehicle injured in collision with two- or three-wheeled motor vehicle : passenger injured in traffic accident
1632. V62.7 Occupant of heavy transport vehicle injured in collision with two- or three-wheeled motor vehicle : person on outside of vehicle injured in traffic accident
1633. V62.9 Occupant of heavy transport vehicle injured in collision with two- or three-wheeled motor vehicle : unspecified occupant of heavy transport vehicle injured in traffic accident
1634. V63 Occupant of heavy transport vehicle injured in collision with car, pick-up truck or van
1635. V63.0 Occupant of heavy transport vehicle injured in collision with car, pick-up truck or van : driver injured in nontraffic accident
1636. V63.1 Occupant of heavy transport vehicle injured in collision with car, pick-up truck or van : passenger injured in nontraffic accident
1637. V63.2 Occupant of heavy transport vehicle injured in collision with car, pick-up truck or van : person on outside of vehicle injured in nontraffic accident

1638. V63.3 Occupant of heavy transport vehicle injured in collision with car, pick-up truck or van : unspecified occupant of heavy transport vehicle injured in nontraffic accident
1639. V63.4 Occupant of heavy transport vehicle injured in collision with car, pick-up truck or van : person injured while boarding or alighting
1640. V63.5 Occupant of heavy transport vehicle injured in collision with car, pick-up truck or van : driver injured in traffic accident
1641. V63.6 Occupant of heavy transport vehicle injured in collision with car, pick-up truck or van : passenger injured in traffic accident
1642. V63.7 Occupant of heavy transport vehicle injured in collision with car, pick-up truck or van : person on outside of vehicle injured in traffic accident
1643. V63.9 Occupant of heavy transport vehicle injured in collision with car, pick-up truck or van : unspecified occupant of heavy transport vehicle injured in traffic accident
1644. V64 Occupant of heavy transport vehicle injured in collision with heavy transport vehicle or bus
1645. V64.0 Occupant of heavy transport vehicle injured in collision with heavy transport vehicle or bus : driver injured in nontraffic accident
1646. V64.1 Occupant of heavy transport vehicle injured in collision with heavy transport vehicle or bus : passenger injured in nontraffic accident
1647. V64.2 Occupant of heavy transport vehicle injured in collision with heavy transport vehicle or bus : person on outside of vehicle injured in nontraffic accident
1648. V64.3 Occupant of heavy transport vehicle injured in collision with heavy transport vehicle or bus : unspecified occupant of heavy transport vehicle injured in nontraffic accident
1649. V64.4 Occupant of heavy transport vehicle injured in collision with heavy transport vehicle or bus : person injured while boarding or alighting
1650. V64.5 Occupant of heavy transport vehicle injured in collision with heavy transport vehicle or bus : driver injured in traffic accident
1651. V64.6 Occupant of heavy transport vehicle injured in collision with heavy transport vehicle or bus : passenger injured in traffic accident
1652. V64.7 Occupant of heavy transport vehicle injured in collision with heavy transport vehicle or bus : person on outside of vehicle injured in traffic accident
1653. V64.9 Occupant of heavy transport vehicle injured in collision with heavy transport vehicle or bus : unspecified occupant of heavy transport vehicle injured in traffic accident
1654. V65 Occupant of heavy transport vehicle injured in collision with railway train or railway vehicle
1655. V65.0 Occupant of heavy transport vehicle injured in collision with railway train or railway vehicle : driver injured in nontraffic accident
1656. V65.1 Occupant of heavy transport vehicle injured in collision with railway train or railway vehicle : passenger injured in nontraffic accident
1657. V65.2 Occupant of heavy transport vehicle injured in collision with railway train or railway vehicle : person on outside of vehicle injured in nontraffic accident
1658. V65.3 Occupant of heavy transport vehicle injured in collision with railway train or railway vehicle : unspecified occupant of heavy transport vehicle injured in nontraffic accident
1659. V65.4 Occupant of heavy transport vehicle injured in collision with railway train or railway

vehicle : person injured while boarding or alighting
1660. V65.5 Occupant of heavy transport vehicle injured in collision with railway train or railway vehicle : driver injured in traffic accident
1661. V65.6 Occupant of heavy transport vehicle injured in collision with railway train or railway vehicle : passenger injured in traffic accident
1662. V65.7 Occupant of heavy transport vehicle injured in collision with railway train or railway vehicle : person on outside of vehicle injured in traffic accident
1663. V65.9 Occupant of heavy transport vehicle injured in collision with railway train or railway vehicle : unspecified occupant of heavy transport vehicle injured in traffic accident
1664. V66 Occupant of heavy transport vehicle injured in collision with other nonmotor vehicle
1665. V66.0 Occupant of heavy transport vehicle injured in collision with other nonmotor vehicle : driver injured in nontraffic accident
1666. V66.1 Occupant of heavy transport vehicle injured in collision with other nonmotor vehicle : passenger injured in nontraffic accident
1667. V66.2 Occupant of heavy transport vehicle injured in collision with other nonmotor vehicle : person on outside of vehicle injured in nontraffic accident
1668. V66.3 Occupant of heavy transport vehicle injured in collision with other nonmotor vehicle : unspecified occupant of heavy transport vehicle injured in nontraffic accident
1669. V66.4 Occupant of heavy transport vehicle injured in collision with other nonmotor vehicle : person injured while boarding or alighting
1670. V66.5 Occupant of heavy transport vehicle injured in collision with other nonmotor vehicle : driver injured in traffic accident
1671. V66.6 Occupant of heavy transport vehicle injured in collision with other nonmotor vehicle : passenger injured in traffic accident
1672. V66.7 Occupant of heavy transport vehicle injured in collision with other nonmotor vehicle : person on outside of vehicle injured in traffic accident
1673. V66.9 Occupant of heavy transport vehicle injured in collision with other nonmotor vehicle : unspecified occupant of heavy transport vehicle injured in traffic accident
1674. V67 Occupant of heavy transport vehicle injured in collision with fixed or stationary object
1675. V67.0 Occupant of heavy transport vehicle injured in collision with fixed or stationary object : driver injured in nontraffic accident
1676. V67.1 Occupant of heavy transport vehicle injured in collision with fixed or stationary object : passenger injured in nontraffic accident
1677. V67.2 Occupant of heavy transport vehicle injured in collision with fixed or stationary object : person on outside of vehicle injured in nontraffic accident
1678. V67.3 Occupant of heavy transport vehicle injured in collision with fixed or stationary object : unspecified occupant of heavy transport vehicle injured in nontraffic accident
1679. V67.4 Occupant of heavy transport vehicle injured in collision with fixed or stationary object : person injured while boarding or alighting
1680. V67.5 Occupant of heavy transport vehicle injured in collision with fixed or stationary object : driver injured in traffic accident
1681. V67.6 Occupant of heavy transport vehicle injured in collision with fixed or stationary object : passenger injured in traffic accident

1682. V67.7 Occupant of heavy transport vehicle injured in collision with fixed or stationary object : person on outside of vehicle injured in traffic accident
1683. V67.9 Occupant of heavy transport vehicle injured in collision with fixed or stationary object : unspecified occupant of heavy transport vehicle injured in traffic accident
1684. V68 Occupant of heavy transport vehicle injured in noncollision transport accident
1685. V68.0 Occupant of heavy transport vehicle injured in noncollision transport accident : driver injured in nontraffic accident
1686. V68.1 Occupant of heavy transport vehicle injured in noncollision transport accident : passenger injured in nontraffic accident
1687. V68.2 Occupant of heavy transport vehicle injured in noncollision transport accident : person on outside of vehicle injured in nontraffic accident
1688. V68.3 Occupant of heavy transport vehicle injured in noncollision transport accident : unspecified occupant of heavy transport vehicle injured in nontraffic accident
1689. V68.4 Occupant of heavy transport vehicle injured in noncollision transport accident : person injured while boarding or alighting
1690. V68.5 Occupant of heavy transport vehicle injured in noncollision transport accident : driver injured in traffic accident
1691. V68.6 Occupant of heavy transport vehicle injured in noncollision transport accident : passenger injured in traffic accident
1692. V68.7 Occupant of heavy transport vehicle injured in noncollision transport accident : person on outside of vehicle injured in traffic accident
1693. V68.9 Occupant of heavy transport vehicle injured in noncollision transport accident : unspecified occupant of heavy transport vehicle injured in traffic accident
1694. V69 Occupant of heavy transport vehicle injured in other and unspecified transport accidents
1695. V69.0 Driver injured in collision with other and unspecified motor vehicles in nontraffic accident
1696. V69.1 Passenger injured in collision with other and unspecified motor vehicles in nontraffic accident
1697. V69.2 Unspecified occupant of heavy transport vehicle injured in collision with other and unspecified motor vehicles in nontraffic accident
1698. V69.3 Occupant [any] of heavy transport vehicle injured in unspecified nontraffic accident
1699. V69.4 Driver injured in collision with other and unspecified motor vehicles in traffic accident
1700. V69.5 Passenger injured in collision with other and unspecified motor vehicles in traffic accident
1701. V69.6 Unspecified occupant of heavy transport vehicle injured in collision with other and unspecified motor vehicles in traffic accident
1702. V69.8 Occupant [any] of heavy transport vehicle injured in other specified transport accidents
1703. V69.9 Occupant [any] of heavy transport vehicle injured in unspecified traffic accident
1704. V70 Bus occupant injured in collision with pedestrian or animal
1705. V70.0 Bus occupant injured in collision with pedestrian or animal : driver injured in

nontraffic accident

1706. V70.1 Bus occupant injured in collision with pedestrian or animal : passenger injured in nontraffic accident
1707. V70.2 Bus occupant injured in collision with pedestrian or animal : person on outside of vehicle injured in nontraffic accident
1708. V70.3 Bus occupant injured in collision with pedestrian or animal : unspecified bus occupant injured in nontraffic accident
1709. V70.4 Bus occupant injured in collision with pedestrian or animal : person injured while boarding or alighting
1710. V70.5 Bus occupant injured in collision with pedestrian or animal : driver injured in traffic accident
1711. V70.6 Bus occupant injured in collision with pedestrian or animal : passenger injured in traffic accident
1712. V70.7 Bus occupant injured in collision with pedestrian or animal : person on outside of vehicle injured in traffic accident
1713. V70.9 Bus occupant injured in collision with pedestrian or animal : unspecified bus occupant injured in traffic accident
1714. V71 Bus occupant injured in collision with pedal cycle
1715. V71.0 Bus occupant injured in collision with pedal cycle : driver injured in nontraffic accident
1716. V71.1 Bus occupant injured in collision with pedal cycle : passenger injured in nontraffic accident
1717. V71.2 Bus occupant injured in collision with pedal cycle : person on outside of vehicle injured in nontraffic accident
1718. V71.3 Bus occupant injured in collision with pedal cycle : unspecified bus occupant injured in nontraffic accident
1719. V71.4 Bus occupant injured in collision with pedal cycle : person injured while boarding or alighting
1720. V71.5 Bus occupant injured in collision with pedal cycle : driver injured in traffic accident
1721. V71.6 Bus occupant injured in collision with pedal cycle : passenger injured in traffic accident
1722. V71.7 Bus occupant injured in collision with pedal cycle : person on outside of vehicle injured in traffic accident
1723. V71.9 Bus occupant injured in collision with pedal cycle : unspecified bus occupant injured in traffic accident
1724. V72 Bus occupant injured in collision with two- or three-wheeled motor vehicle
1725. V72.0 Bus occupant injured in collision with two- or three-wheeled motor vehicle : driver injured in nontraffic accident
1726. V72.1 Bus occupant injured in collision with two- or three-wheeled motor vehicle : passenger injured in nontraffic accident
1727. V72.2 Bus occupant injured in collision with two- or three-wheeled motor vehicle : person on outside of vehicle injured in nontraffic accident
1728. V72.3 Bus occupant injured in collision with two- or three-wheeled motor vehicle : un-

specified bus occupant injured in nontraffic accident
1729. V72.4 Bus occupant injured in collision with two- or three-wheeled motor vehicle : person injured while boarding or alighting
1730. V72.5 Bus occupant injured in collision with two- or three-wheeled motor vehicle : driver injured in traffic accident
1731. V72.6 Bus occupant injured in collision with two- or three-wheeled motor vehicle : passenger injured in traffic accident
1732. V72.7 Bus occupant injured in collision with two- or three-wheeled motor vehicle : person on outside of vehicle injured in traffic accident
1733. V72.9 Bus occupant injured in collision with two- or three-wheeled motor vehicle : unspecified bus occupant injured in traffic accident
1734. V73 Bus occupant injured in collision with car, pick-up truck or van
1735. V73.0 Bus occupant injured in collision with car, pick-up truck or van : driver injured in nontraffic accident
1736. V73.1 Bus occupant injured in collision with car, pick-up truck or van : passenger injured in nontraffic accident
1737. V73.2 Bus occupant injured in collision with car, pick-up truck or van : person on outside of vehicle injured in nontraffic accident
1738. V73.3 Bus occupant injured in collision with car, pick-up truck or van : unspecified bus occupant injured in nontraffic accident
1739. V73.4 Bus occupant injured in collision with car, pick-up truck or van : person injured while boarding or alighting
1740. V73.5 Bus occupant injured in collision with car, pick-up truck or van : driver injured in traffic accident
1741. V73.6 Bus occupant injured in collision with car, pick-up truck or van : passenger injured in traffic accident
1742. V73.7 Bus occupant injured in collision with car, pick-up truck or van : person on outside of vehicle injured in traffic accident
1743. V73.9 Bus occupant injured in collision with car, pick-up truck or van : unspecified bus occupant injured in traffic accident
1744. V74 Bus occupant injured in collision with heavy transport vehicle or bus
1745. V74.0 Bus occupant injured in collision with heavy transport vehicle or bus : driver injured in nontraffic accident
1746. V74.1 Bus occupant injured in collision with heavy transport vehicle or bus : passenger injured in nontraffic accident
1747. V74.2 Bus occupant injured in collision with heavy transport vehicle or bus : person on outside of vehicle injured in nontraffic accident
1748. V74.3 Bus occupant injured in collision with heavy transport vehicle or bus : unspecified bus occupant injured in nontraffic accident
1749. V74.4 Bus occupant injured in collision with heavy transport vehicle or bus : person injured while boarding or alighting
1750. V74.5 Bus occupant injured in collision with heavy transport vehicle or bus : driver injured in traffic accident

1751. V74.6 Bus occupant injured in collision with heavy transport vehicle or bus : passenger injured in traffic accident
1752. V74.7 Bus occupant injured in collision with heavy transport vehicle or bus : person on outside of vehicle injured in traffic accident
1753. V74.9 Bus occupant injured in collision with heavy transport vehicle or bus : unspecified bus occupant injured in traffic accident
1754. V75 Bus occupant injured in collision with railway train or railway vehicle
1755. V75.0 Bus occupant injured in collision with railway train or railway vehicle : driver injured in nontraffic accident
1756. V75.1 Bus occupant injured in collision with railway train or railway vehicle : passenger injured in nontraffic accident
1757. V75.2 Bus occupant injured in collision with railway train or railway vehicle : person on outside of vehicle injured in nontraffic accident
1758. V75.3 Bus occupant injured in collision with railway train or railway vehicle : unspecified bus occupant injured in nontraffic accident
1759. V75.4 Bus occupant injured in collision with railway train or railway vehicle : person injured while boarding or alighting
1760. V75.5 Bus occupant injured in collision with railway train or railway vehicle : driver injured in traffic accident
1761. V75.6 Bus occupant injured in collision with railway train or railway vehicle : passenger injured in traffic accident
1762. V75.7 Bus occupant injured in collision with railway train or railway vehicle : person on outside of vehicle injured in traffic accident
1763. V75.9 Bus occupant injured in collision with railway train or railway vehicle : unspecified bus occupant injured in traffic accident
1764. V76 Bus occupant injured in collision with other nonmotor vehicle
1765. V76.0 Bus occupant injured in collision with other nonmotor vehicle : driver injured in nontraffic accident
1766. V76.1 Bus occupant injured in collision with other nonmotor vehicle : passenger injured in nontraffic accident
1767. V76.2 Bus occupant injured in collision with other nonmotor vehicle : person on outside of vehicle injured in nontraffic accident
1768. V76.3 Bus occupant injured in collision with other nonmotor vehicle : unspecified bus occupant injured in nontraffic accident
1769. V76.4 Bus occupant injured in collision with other nonmotor vehicle : person injured while boarding or alighting
1770. V76.5 Bus occupant injured in collision with other nonmotor vehicle : driver injured in traffic accident
1771. V76.6 Bus occupant injured in collision with other nonmotor vehicle : passenger injured in traffic accident
1772. V76.7 Bus occupant injured in collision with other nonmotor vehicle : person on outside of vehicle injured in traffic accident
1773. V76.9 Bus occupant injured in collision with other nonmotor vehicle : unspecified bus

occupant injured in traffic accident
1774. V77 Bus occupant injured in collision with fixed or stationary object
1775. V77.0 Bus occupant injured in collision with fixed or stationary object : driver injured in nontraffic accident
1776. V77.1 Bus occupant injured in collision with fixed or stationary object : passenger injured in nontraffic accident
1777. V77.2 Bus occupant injured in collision with fixed or stationary object : person on outside of vehicle injured in nontraffic accident
1778. V77.3 Bus occupant injured in collision with fixed or stationary object : unspecified bus occupant injured in nontraffic accident
1779. V77.4 Bus occupant injured in collision with fixed or stationary object : person injured while boarding or alighting
1780. V77.5 Bus occupant injured in collision with fixed or stationary object : driver injured in traffic accident
1781. V77.6 Bus occupant injured in collision with fixed or stationary object : passenger injured in traffic accident
1782. V77.7 Bus occupant injured in collision with fixed or stationary object : person on outside of vehicle injured in traffic accident
1783. V77.9 Bus occupant injured in collision with fixed or stationary object : unspecified bus occupant injured in traffic accident
1784. V78 Bus occupant injured in noncollision transport accident
1785. V78.0 Bus occupant injured in noncollision transport accident : driver injured in nontraffic accident
1786. V78.1 Bus occupant injured in noncollision transport accident : passenger injured in nontraffic accident
1787. V78.2 Bus occupant injured in noncollision transport accident : person on outside of vehicle injured in nontraffic accident
1788. V78.3 Bus occupant injured in noncollision transport accident : unspecified bus occupant injured in nontraffic accident
1789. V78.4 Bus occupant injured in noncollision transport accident : person injured while boarding or alighting
1790. V78.5 Bus occupant injured in noncollision transport accident : driver injured in traffic accident
1791. V78.6 Bus occupant injured in noncollision transport accident : passenger injured in traffic accident
1792. V78.7 Bus occupant injured in noncollision transport accident : person on outside of vehicle injured in traffic accident
1793. V78.9 Bus occupant injured in noncollision transport accident : unspecified bus occupant injured in traffic accident
1794. V79 Bus occupant injured in other and unspecified transport accidents
1795. V79.0 Driver injured in collision with other and unspecified motor vehicles in nontraffic accident
1796. V79.1 Passenger injured in collision with other and unspecified motor vehicles in nontraffic

accident

1797. V79.2 Unspecified bus occupant injured in collision with other and unspecified motor vehicles in nontraffic accident
1798. V79.3 Bus occupant [any] injured in unspecified nontraffic accident
1799. V79.4 Driver injured in collision with other and unspecified motor vehicles in traffic accident
1800. V79.5 Passenger injured in collision with other and unspecified motor vehicles in traffic accident
1801. V79.6 Unspecified bus occupant injured in collision with other and unspecified motor vehicles in traffic accident
1802. V79.8 Bus occupant [any] injured in other specified transport accidents
1803. V79.9 Bus occupant [any] injured in unspecified traffic accident
1804. V80 Animal-rider or occupant of animal-drawn vehicle injured in transport accident
1805. V80.0 Rider or occupant injured by fall from or being thrown from animal or animal-drawn vehicle in noncollision accident
1806. V80.1 Rider or occupant injured in collision with pedestrian or animal
1807. V80.2 Rider or occupant injured in collision with pedal cycle
1808. V80.3 Rider or occupant injured in collision with two- or three-wheeled motor vehicle
1809. V80.4 Rider or occupant injured in collision with car, pick-up truck, van, heavy transport vehicle or bus
1810. V80.5 Rider or occupant injured in collision with other specified motor vehicle
1811. V80.6 Rider or occupant injured in collision with railway train or railway vehicle
1812. V80.7 Rider or occupant injured in collision with other nonmotor vehicle
1813. V80.8 Rider or occupant injured in collision with fixed or stationary object
1814. V80.9 Rider or occupant injured in other and unspecified transport accidents
1815. V81 Occupant of railway train or railway vehicle injured in transport accident
1816. V81.0 Occupant of railway train or railway vehicle injured in collision with motor vehicle in nontraffic accident
1817. V81.1 Occupant of railway train or railway vehicle injured in collision with motor vehicle in traffic accident
1818. V81.2 Occupant of railway train or railway vehicle injured in collision with or hit by rolling stock
1819. V81.3 Occupant of railway train or railway vehicle injured in collision with other object
1820. V81.4 Person injured while boarding or alighting from railway train or railway vehicle
1821. V81.5 Occupant of railway train or railway vehicle injured by fall in railway train or railway vehicle
1822. V81.6 Occupant of railway train or railway vehicle injured by fall from railway train or railway vehicle
1823. V81.7 Occupant of railway train or railway vehicle injured in derailment without antecedent collision
1824. V81.8 Occupant of railway train or railway vehicle injured in other specified railway accidents
1825. V81.9 Occupant of railway train or railway vehicle injured in unspecified railway accident

1826. V82 Occupant of streetcar injured in transport accident
1827. V82.0 Occupant of streetcar injured in collision with motor vehicle in nontraffic accident
1828. V82.1 Occupant of streetcar injured in collision with motor vehicle in traffic accident
1829. V82.2 Occupant of streetcar injured in collision with or hit by rolling stock
1830. V82.3 Occupant of streetcar injured in collision with other object
1831. V82.4 Person injured while boarding or alighting from streetcar
1832. V82.5 Occupant of streetcar injured by fall in streetcar
1833. V82.6 Occupant of streetcar injured by fall from streetcar
1834. V82.7 Occupant of streetcar injured in derailment without antecedent collision
1835. V82.8 Occupant of streetcar injured in other specified transport accidents
1836. V82.9 Occupant of streetcar injured in unspecified traffic accident
1837. V83 Occupant of special vehicle mainly used on industrial premises injured in transport accident
1838. V83.0 Driver of special industrial vehicle injured in traffic accident
1839. V83.1 Passenger of special industrial vehicle injured in traffic accident
1840. V83.2 Person on outside of special industrial vehicle injured in traffic accident
1841. V83.3 Unspecified occupant of special industrial vehicle injured in traffic accident
1842. V83.4 Person injured while boarding or alighting from special industrial vehicle
1843. V83.5 Driver of special industrial vehicle injured in nontraffic accident
1844. V83.6 Passenger of special industrial vehicle injured in nontraffic accident
1845. V83.7 Person on outside of special industrial vehicle injured in nontraffic accident
1846. V83.9 Unspecified occupant of special industrial vehicle injured in nontraffic accident
1847. V84 Occupant of special vehicle mainly used in agriculture injured in transport accident
1848. V84.0 Driver of special agricultural vehicle injured in traffic accident
1849. V84.1 Passenger of special agricultural vehicle injured in traffic accident
1850. V84.2 Person on outside of special agricultural vehicle injured in traffic accident
1851. V84.3 Unspecified occupant of special agricultural vehicle injured in traffic accident
1852. V84.4 Person injured while boarding or alighting from special agricultural vehicle
1853. V84.5 Driver of special agricultural vehicle injured in nontraffic accident
1854. V84.6 Passenger of special agricultural vehicle injured in nontraffic accident
1855. V84.7 Person on outside of special agricultural vehicle injured in nontraffic accident
1856. V84.9 Unspecified occupant of special agricultural vehicle injured in nontraffic accident
1857. V85 Occupant of special construction vehicle injured in transport accident
1858. V85.0 Driver of special construction vehicle injured in traffic accident
1859. V85.1 Passenger of special construction vehicle injured in traffic accident
1860. V85.2 Person on outside of special construction vehicle injured in traffic accident
1861. V85.3 Unspecified occupant of special construction vehicle injured in traffic accident
1862. V85.4 Person injured while boarding or alighting from special construction vehicle
1863. V85.5 Driver of special construction vehicle injured in nontraffic accident
1864. V85.6 Passenger of special construction vehicle injured in nontraffic accident
1865. V85.7 Person on outside of special construction vehicle injured in nontraffic accident
1866. V85.9 Unspecified occupant of special construction vehicle injured in nontraffic accident
1867. V86 Occupant of special all-terrain or other motor vehicle designed primarily for off-road

use, injured in transport accident
1868. V86.0 Driver of all-terrain or other off-road motor vehicle injured in traffic accident
1869. V86.1 Passenger of all-terrain or other off-road motor vehicle injured in traffic accident
1870. V86.2 Person on outside of all-terrain or other off-road motor vehicle injured in traffic accident
1871. V86.3 Unspecified occupant of all-terrain or other off-road motor vehicle injured in traffic accident
1872. V86.4 Person injured while boarding or alighting from all-terrain or other off-road motor vehicle
1873. V86.5 Driver of all-terrain or other off-road motor vehicle injured in nontraffic accident
1874. V86.6 Passenger of all-terrain or other off-road motor vehicle injured in nontraffic accident
1875. V86.7 Person on outside of all-terrain or other off-road motor vehicle injured in nontraffic accident
1876. V86.9 Unspecified occupant of all-terrain or other off-road motor vehicle injured in nontraffic accident
1877. V87 Traffic accident of specified type but victim's mode of transport unknown
1878. V87.0 Person injured in collision between car and two- or three-wheeled motor vehicle (traffic)
1879. V87.1 Person injured in collision between other motor vehicle and two- or three-wheeled motor vehicle (traffic)
1880. V87.2 Person injured in collision between car and pick-up truck or van (traffic)
1881. V87.3 Person injured in collision between car and bus (traffic)
1882. V87.4 Person injured in collision between car and heavy transport vehicle (traffic)
1883. V87.5 Person injured in collision between heavy transport vehicle and bus (traffic)
1884. V87.6 Person injured in collision between railway train or railway vehicle and car (traffic)
1885. V87.7 Person injured in collision between other specified motor vehicles (traffic)
1886. V87.8 Person injured in other specified noncollision transport accidents involving motor vehicle (traffic)
1887. V87.9 Person injured in other specified (collision)(noncollision) transport accidents involving nonmotor vehicle (traffic)
1888. V88 Nontraffic accident of specified type but victim's mode of transport unknown
1889. V88.0 Person injured in collision between car and two- or three-wheeled motor vehicle, nontraffic
1890. V88.1 Person injured in collision between other motor vehicle and two- or three-wheeled motor vehicle, nontraffic
1891. V88.2 Person injured in collision between car and pick-up truck or van, nontraffic
1892. V88.3 Person injured in collision between car and bus, nontraffic
1893. V88.4 Person injured in collision between car and heavy transport vehicle, nontraffic
1894. V88.5 Person injured in collision between heavy transport vehicle and bus, nontraffic
1895. V88.6 Person injured in collision between railway train or railway vehicle and car, nontraffic
1896. V88.7 Person injured in collision between other specified motor vehicles, nontraffic
1897. V88.8 Person injured in other specified noncollision transport accidents involving motor vehicle, nontraffic

1898. V88.9 Person injured in other specified (collision)(noncollision) transport accidents involving nonmotor vehicle, nontraffic
1899. V89 Motor- or nonmotor-vehicle accident, type of vehicle unspecified
1900. V89.0 Person injured in unspecified motor-vehicle accident, nontraffic
1901. V89.1 Person injured in unspecified nonmotor-vehicle accident, nontraffic
1902. V89.2 Person injured in unspecified motor-vehicle accident, traffic
1903. V89.3 Person injured in unspecified nonmotor-vehicle accident, traffic
1904. V89.9 Person injured in unspecified vehicle accident
1905. V90 Accident to watercraft causing drowning and submersion
1906. V90.0 Accident to watercraft causing drowning and submersion : merchant ship
1907. V90.1 Accident to watercraft causing drowning and submersion : passenger ship
1908. V90.2 Accident to watercraft causing drowning and submersion : fishing boat
1909. V90.3 Accident to watercraft causing drowning and submersion : other powered watercraft
1910. V90.4 Accident to watercraft causing drowning and submersion : sailboat
1911. V90.5 Accident to watercraft causing drowning and submersion : canoe or kayak
1912. V90.6 Accident to watercraft causing drowning and submersion : inflatable craft (nonpowered)
1913. V90.7 Accident to watercraft causing drowning and submersion : water-skis
1914. V90.8 Accident to watercraft causing drowning and submersion : other unpowered watercraft
1915. V90.9 Accident to watercraft causing drowning and submersion : unspecified watercraft
1916. V91 Accident to watercraft causing other injury
1917. V91.0 Accident to watercraft causing other injury : merchant ship
1918. V91.1 Accident to watercraft causing other injury : passenger ship
1919. V91.2 Accident to watercraft causing other injury : fishing boat
1920. V91.3 Accident to watercraft causing other injury : other powered watercraft
1921. V91.4 Accident to watercraft causing other injury : sailboat
1922. V91.5 Accident to watercraft causing other injury : canoe or kayak
1923. V91.6 Accident to watercraft causing other injury : inflatable craft (nonpowered)
1924. V91.7 Accident to watercraft causing other injury : water-skis
1925. V91.8 Accident to watercraft causing other injury : other unpowered watercraft
1926. V91.9 Accident to watercraft causing other injury : unspecified watercraft
1927. V92 Water-transport-related drowning and submersion without accident to watercraft
1928. V92.0 Water-transport-related drowning and submersion without accident to watercraft : merchant ship
1929. V92.1 Water-transport-related drowning and submersion without accident to watercraft : passenger ship
1930. V92.2 Water-transport-related drowning and submersion without accident to watercraft : fishing boat
1931. V92.3 Water-transport-related drowning and submersion without accident to watercraft : other powered watercraft
1932. V92.4 Water-transport-related drowning and submersion without accident to watercraft : sailboat

1933. V92.5 Water-transport-related drowning and submersion without accident to watercraft : canoe or kayak
1934. V92.6 Water-transport-related drowning and submersion without accident to watercraft : inflatable craft (nonpowered)
1935. V92.7 Water-transport-related drowning and submersion without accident to watercraft : water-skis
1936. V92.8 Water-transport-related drowning and submersion without accident to watercraft : other unpowered watercraft
1937. V92.9 Water-transport-related drowning and submersion without accident to watercraft : unspecified watercraft
1938. V93 Accident on board watercraft without accident to watercraft, not causing drowning and submersion
1939. V93.0 Accident on board watercraft without accident to watercraft, not causing drowning and submersion : merchant ship
1940. V93.1 Accident on board watercraft without accident to watercraft, not causing drowning and submersion : passenger ship
1941. V93.2 Accident on board watercraft without accident to watercraft, not causing drowning and submersion : fishing boat
1942. V93.3 Accident on board watercraft without accident to watercraft, not causing drowning and submersion : other powered watercraft
1943. V93.4 Accident on board watercraft without accident to watercraft, not causing drowning and submersion : sailboat
1944. V93.5 Accident on board watercraft without accident to watercraft, not causing drowning and submersion : canoe or kayak
1945. V93.6 Accident on board watercraft without accident to watercraft, not causing drowning and submersion : inflatable craft (nonpowered)
1946. V93.7 Accident on board watercraft without accident to watercraft, not causing drowning and submersion : water-skis
1947. V93.8 Accident on board watercraft without accident to watercraft, not causing drowning and submersion : other unpowered watercraft
1948. V93.9 Accident on board watercraft without accident to watercraft, not causing drowning and submersion : unspecified watercraft
1949. V94 Other and unspecified water transport accidents
1950. V94.0 Other and unspecified water transport accidents : merchant ship
1951. V94.1 Other and unspecified water transport accidents : passenger ship
1952. V94.2 Other and unspecified water transport accidents : fishing boat
1953. V94.3 Other and unspecified water transport accidents : other powered watercraft
1954. V94.4 Other and unspecified water transport accidents : sailboat
1955. V94.5 Other and unspecified water transport accidents : canoe or kayak
1956. V94.6 Other and unspecified water transport accidents : inflatable craft (nonpowered)
1957. V94.7 Other and unspecified water transport accidents : water-skis
1958. V94.8 Other and unspecified water transport accidents : other unpowered watercraft
1959. V94.9 Other and unspecified water transport accidents : unspecified watercraft

1960. V95 Accident to powered aircraft causing injury to occupant
1961. V95.0 Helicopter accident injuring occupant
1962. V95.1 Ultralight, microlight or powered-glider accident injuring occupant
1963. V95.2 Accident to other private fixed-wing aircraft, injuring occupant
1964. V95.3 Accident to commercial fixed-wing aircraft, injuring occupant
1965. V95.4 Spacecraft accident injuring occupant
1966. V95.8 Other aircraft accidents injuring occupant
1967. V95.9 Unspecified aircraft accident injuring occupant
1968. V96 Accident to nonpowered aircraft causing injury to occupant
1969. V96.0 Balloon accident injuring occupant
1970. V96.1 Hang-glider accident injuring occupant
1971. V96.2 Glider (nonpowered) accident injuring occupant
1972. V96.8 Other nonpowered-aircraft accidents injuring occupant
1973. V96.9 Unspecified nonpowered-aircraft accident injuring occupant
1974. V97 Other specified air transport accidents
1975. V97.0 Occupant of aircraft injured in other specified air transport accidents
1976. V97.1 Person injured while boarding or alighting from aircraft
1977. V97.2 Parachutist injured in air transport accident
1978. V97.3 Person on ground injured in air transport accident
1979. V97.8 Other air transport accidents, not elsewhere classified
1980. V98 Other specified transport accidents
1981. V99 Unspecified transport accident
1982. W00 Fall on same level involving ice and snow
1983. W01 Fall on same level from slipping, tripping and stumbling
1984. W02 Fall involving ice-skates, skis, roller-skates or skateboards
1985. W03 Other fall on same level due to collision with, or pushing by, another person
1986. W04 Fall while being carried or supported by other persons
1987. W05 Fall involving wheelchair
1988. W06 Fall involving bed
1989. W07 Fall involving chair
1990. W08 Fall involving other furniture
1991. W09 Fall involving playground equipment
1992. W10 Fall on and from stairs and steps
1993. W11 Fall on and from ladder
1994. W12 Fall on and from scaffolding
1995. W13 Fall from, out of or through building or structure
1996. W14 Fall from tree
1997. W15 Fall from cliff
1998. W16 Diving or jumping into water causing injury other than drowning or submersion
1999. W17 Other fall from one level to another
2000. W18 Other fall on same level
2001. W19 Unspecified fall
2002. W20 Struck by thrown, projected or falling object(s)

2003. W21 Striking against or struck by sports equipment
2004. W22 Striking against or struck by other object(s)
2005. W23 Caught, crushed, jammed or pinched in or between objects
2006. W24 Contact with lifting and transmission device(s), not elsewhere classified
2007. W25 Contact with sharp glass
2008. W26 Contact with other sharp object(s)
2009. W27 Contact with nonpowered hand tool
2010. W28 Contact with powered lawnmower
2011. W29 Contact with other powered hand tools and household machinery
2012. W30 Contact with agricultural machinery
2013. W31 Contact with other and unspecified machinery
2014. W32 Handgun discharge
2015. W33 Rifle, shotgun and larger firearm discharge
2016. W34 Discharge from other and unspecified firearms
2017. W35 Explosion and rupture of boiler
2018. W36 Explosion and rupture of gas cylinder
2019. W37 Explosion and rupture of pressurized tyre, pipe or hose
2020. W38 Explosion and rupture of other specified pressurized devices
2021. W39 Discharge of firework
2022. W40 Explosion of other materials
2023. W41 Exposure to high-pressure jet
2024. W42 Exposure to noise
2025. W43 Exposure to vibration
2026. W44 Foreign body entering into or through eye or natural orifice
2027. W45 Foreign body or object entering through skin
2028. W46 Contact with hypodermic needle
2029. W49 Exposure to other and unspecified inanimate mechanical forces
2030. W50 Hit, struck, kicked, twisted, bitten or scratched by another person
2031. W51 Striking against or bumped into by another person
2032. W52 Crushed, pushed or stepped on by crowd or human stampede
2033. W53 Bitten by rat
2034. W54 Bitten or struck by dog
2035. W55 Bitten or struck by other mammals
2036. W56 Contact with marine animal
2037. W57 Bitten or stung by nonvenomous insect and other nonvenomous arthropods
2038. W58 Bitten or struck by crocodile or alligator
2039. W59 Bitten or crushed by other reptiles
2040. W60 Contact with plant thorns and spines and sharp leaves
2041. W64 Exposure to other and unspecified animate mechanical forces
2042. W65 Drowning and submersion while in bath-tub
2043. W66 Drowning and submersion following fall into bath-tub
2044. W67 Drowning and submersion while in swimming-pool
2045. W68 Drowning and submersion following fall into swimming-pool

2046. W69 Drowning and submersion while in natural water
2047. W70 Drowning and submersion following fall into natural water
2048. W73 Other specified drowning and submersion
2049. W74 Unspecified drowning and submersion
2050. W75 Accidental suffocation and strangulation in bed
2051. W76 Other accidental hanging and strangulation
2052. W77 Threat to breathing due to cave-in, falling earth and other substances
2053. W78 Inhalation of gastric contents
2054. W79 Inhalation and ingestion of food causing obstruction of respiratory tract
2055. W80 Inhalation and ingestion of other objects causing obstruction of respiratory tract
2056. W81 Confined to or trapped in a low-oxygen environment
2057. W83 Other specified threats to breathing
2058. W84 Unspecified threat to breathing
2059. W85 Exposure to electric transmission lines
2060. W86 Exposure to other specified electric current
2061. W87 Exposure to unspecified electric current
2062. W88 Exposure to ionizing radiation
2063. W89 Exposure to man-made visible and ultraviolet light
2064. W90 Exposure to other nonionizing radiation
2065. W91 Exposure to unspecified type of radiation
2066. W92 Exposure to excessive heat of man-made origin
2067. W93 Exposure to excessive cold of man-made origin
2068. W94 Exposure to high and low air pressure and changes in air pressure
2069. W99 Exposure to other and unspecified man-made environmental factors
2070. X00 Exposure to uncontrolled fire in building or structure
2071. X01 Exposure to uncontrolled fire, not in building or structure
2072. X02 Exposure to controlled fire in building or structure
2073. X03 Exposure to controlled fire, not in building or structure
2074. X04 Exposure to ignition of highly flammable material
2075. X05 Exposure to ignition or melting of nightwear
2076. X06 Exposure to ignition or melting of other clothing and apparel
2077. X08 Exposure to other specified smoke, fire and flames
2078. X09 Exposure to unspecified smoke, fire and flames
2079. X10 Contact with hot drinks, food, fats and cooking oils
2080. X11 Contact with hot tap-water
2081. X12 Contact with other hot fluids
2082. X13 Contact with steam and hot vapours
2083. X14 Contact with hot air and gases
2084. X15 Contact with hot household appliances
2085. X16 Contact with hot heating appliances, radiators and pipes
2086. X17 Contact with hot engines, machinery and tools
2087. X18 Contact with other hot metals
2088. X19 Contact with other and unspecified heat and hot substances

2089. X20 Contact with venomous snakes and lizards
2090. X21 Contact with venomous spiders
2091. X22 Contact with scorpions
2092. X23 Contact with hornets, wasps and bees
2093. X24 Contact with centipedes and venomous millipedes (tropical)
2094. X25 Contact with other venomous arthropods
2095. X26 Contact with venomous marine animals and plants
2096. X27 Contact with other specified venomous animals
2097. X28 Contact with other specified venomous plants
2098. X29 Contact with unspecified venomous animal or plant
2099. X30 Exposure to excessive natural heat
2100. X31 Exposure to excessive natural cold
2101. X32 Exposure to sunlight
2102. X33 Victim of lightning
2103. X34 Victim of earthquake
2104. X35 Victim of volcanic eruption
2105. X36 Victim of avalanche, landslide and other earth movements
2106. X37 Victim of cataclysmic storm
2107. X38 Victim of flood
2108. X39 Exposure to other and unspecified forces of nature
2109. X60 Intentional self-poisoning by and exposure to nonopioid analgesics, antipyretics and antirheumatics
2110. X61 Intentional self-poisoning by and exposure to antiepileptic, sedative-hypnotic, antiparkinsonism and psychotropic drugs, not elsewhere classified
2111. X62 Intentional self-poisoning by and exposure to narcotics and psychodysleptics [hallucinogens], not elsewhere classified
2112. X63 Intentional self-poisoning by and exposure to other drugs acting on the autonomic nervous system
2113. X64 Intentional self-poisoning by and exposure to other and unspecified drugs, medicaments and biological substances
2114. X65 Intentional self-poisoning by and exposure to alcohol
2115. X66 Intentional self-poisoning by and exposure to organic solvents and halogenated hydrocarbons and their vapours
2116. X67 Intentional self-poisoning by and exposure to carbon monoxide and other gases and vapours
2117. X67.0 Intentional self-poisoning by and exposure to carbon monoxide from combustion engine exhaust
2118. X67.1 Intentional self-poisoning by and exposure to carbon monoxide from utility gas
2119. X67.2 Intentional self-poisoning by and exposure to carbon monoxide from other domestic fuels
2120. X67.3 Intentional self-poisoning by and exposure to carbon monoxide from other sources
2121. X67.4 Intentional self-poisoning by and exposure to carbon monoxide from unspecified sources

2122. X67.8 Intentional self-poisoning by and exposure to other specified gases and vapours
2123. X67.9 Intentional self-poisoning by and exposure to unspecified gases and vapours
2124. X68 Intentional self-poisoning by and exposure to pesticides
2125. X69 Intentional self-poisoning by and exposure to other and unspecified chemicals and noxious substances
2126. X70 Intentional self-harm by hanging, strangulation and suffocation
2127. X71 Intentional self-harm by drowning and submersion
2128. X72 Intentional self-harm by handgun discharge
2129. X73 Intentional self-harm by rifle, shotgun and larger firearm discharge
2130. X74 Intentional self-harm by other and unspecified firearm discharge
2131. X75 Intentional self-harm by explosive material
2132. X76 Intentional self-harm by smoke, fire and flames
2133. X77 Intentional self-harm by steam, hot vapours and hot objects
2134. X78 Intentional self-harm by sharp object
2135. X79 Intentional self-harm by blunt object
2136. X80 Intentional self-harm by jumping from a high place
2137. X81 Intentional self-harm by jumping or lying before moving object
2138. X82 Intentional self-harm by crashing of motor vehicle
2139. X83 Intentional self-harm by other specified means
2140. X84 Intentional self-harm by unspecified means
2141. X85 Assault by drugs, medicaments and biological substances
2142. X86 Assault by corrosive substance
2143. X87 Assault by pesticides
2144. X88 Assault by gases and vapours
2145. X88.0 Assault by carbon monoxide from combustion engine exhaust
2146. X88.1 Assault by carbon monoxide from utility gas
2147. X88.2 Assault by carbon monoxide from other domestic fuels
2148. X88.3 Assault by carbon monoxide from other sources
2149. X88.4 Assault by carbon monoxide from unspecified sources
2150. X88.8 Assault by other specified gases and vapours
2151. X88.9 Assault by unspecified gases and vapours
2152. X89 Assault by other specified chemicals and noxious substances
2153. X90 Assault by unspecified chemical or noxious substance
2154. X91 Assault by hanging, strangulation and suffocation
2155. X92 Assault by drowning and submersion
2156. X93 Assault by handgun discharge
2157. X94 Assault by rifle, shotgun and larger firearm discharge
2158. X95 Assault by other and unspecified firearm discharge
2159. X96 Assault by explosive material
2160. X97 Assault by smoke, fire and flames
2161. X98 Assault by steam, hot vapours and hot objects
2162. X99 Assault by sharp object
2163. Y00 Assault by blunt object

2164. Y01 Assault by pushing from high place
2165. Y02 Assault by pushing or placing victim before moving object
2166. Y03 Assault by crashing of motor vehicle
2167. Y04 Assault by bodily force
2168. Y05 Sexual assault by bodily force
2169. Y06 Neglect and abandonment
2170. Y06.0 By spouse or partner
2171. Y06.1 By parent
2172. Y06.2 By acquaintance or friend
2173. Y06.8 By other specified persons
2174. Y06.9 By unspecified person
2175. Y07 Other maltreatment
2176. Y07.0 By spouse or partner
2177. Y07.1 By parent
2178. Y07.2 By acquaintance or friend
2179. Y07.3 By official authorities
2180. Y07.8 By other specified persons
2181. Y07.9 By unspecified person
2182. Y08 Assault by other specified means
2183. Y09 Assault by unspecified means
2184. Y20 Hanging, strangulation and suffocation, undetermined intent
2185. Y21 Drowning and submersion, undetermined intent
2186. Y22 Handgun discharge, undetermined intent
2187. Y23 Rifle, shotgun and larger firearm discharge, undetermined intent
2188. Y24 Other and unspecified firearm discharge, undetermined intent
2189. Y25 Contact with explosive material, undetermined intent
2190. Y26 Exposure to smoke, fire and flames, undetermined intent
2191. Y27 Contact with steam, hot vapours and hot objects, undetermined intent
2192. Y28 Contact with sharp object, undetermined intent
2193. Y29 Contact with blunt object, undetermined intent
2194. Y30 Falling, jumping or pushed from a high place, undetermined intent
2195. Y31 Falling, lying or running before or into moving object, undetermined intent
2196. Y32 Crashing of motor vehicle, undetermined intent
2197. Y33 Other specified events, undetermined intent
2198. Y34 Unspecified event, undetermined intent
2199. Y35 Legal intervention
2200. Y35.0 Legal intervention involving firearm discharge
2201. Y35.1 Legal intervention involving explosives
2202. Y35.2 Legal intervention involving gas
2203. Y35.3 Legal intervention involving blunt objects
2204. Y35.4 Legal intervention involving sharp objects
2205. Y35.5 Legal execution
2206. Y35.6 Legal intervention involving other specified means

2207. Y35.7 Legal intervention, means unspecified
2208. Y36 Operations of war
2209. Y36.0 War operations involving explosion of marine weapons
2210. Y36.1 War operations involving destruction of aircraft
2211. Y36.2 War operations involving other explosions and fragments
2212. Y36.3 War operations involving fires, conflagrations and hot substances
2213. Y36.4 War operations involving firearm discharge and other forms of conventional warfare
2214. Y36.5 War operations involving nuclear weapons
2215. Y36.6 War operations involving biological weapons
2216. Y36.7 War operations involving chemical weapons and other forms of unconventional warfare
2217. Y36.8 War operations occurring after cessation of hostilities
2218. Y36.9 War operations, unspecified

- **11. Clinical Frailty Scale Very Fit - People who are robust, active, energetic and motivated. These people commonly exercise regularly. They are among the fittest for their age. Fit - People who have no active disease symptoms but are less fit than category 1. Often, they exercise or are very active occasionally, e.g. seasonally. Managing well - People whose medical problems are well controlled, but are not regularly active beyond routine walking. Living with very mild frailty - While not dependent on others for daily help, often symptoms limit activities. A common complaint is being "slowed up", and/or being tired during the day. Living with mild frailty - These people often have more evident slowing, and need help in high order IADLs (finances, transportation, heavy housework, medications). Typically, mild frailty progressively impairs shopping and walking outside alone, meal preparation and housework. Living with moderate frailty - People need help with all outside activities and with keeping house. Inside, they often have problems with stairs and need help with bathing and might need minimal assistance (cuing, standby) with dressing. Living with severe frailty - Completely dependent for personal care, from whatever cause (physical or cognitive). Even so, they seem stable and not at high risk of dying (within ~ 6 months). Living with very severe frailty - Completely dependent, approaching the end of life. Typically, they could not recover even from a minor illness. Terminally ill - Approaching the end of life. This category applies to people with a life expectancy** Source: Medical record or treating physician

    1. 1. Very Fit
    2. 2. Fit
    3. 3. Managing well
    4. 4. Living with very mild frailty
    5. 5. Living with mild frailty
    6. 6. Living with moderate frailty
    7. 7. Living with severe frailty
    8. 8. Living with very severe frailty
    9. 9. Terminally ill
    10. Not known

- **12. Comorbidities (Charlson Comorbidity Index) Myocardial infarction - History of definite or probable MI (EKG changes and/or enzyme changes) Congestive heart failure - Exertional or paroxysmal nocturnal dyspnea and has responded to digitalis, diuretics, or afterload reducing agents Peripheral vascular disease - Intermittent claudication or past bypass for chronic arterial insufficiency, history of gangrene or acute arterial insufficiency, or untreated thoracic or abdominal aneurysm (≥6 cm) Cerebrovascular disease - History of a cerebrovascular accident with minor or no residua and transient ischemic attacks Dementia - Chronic cognitive deficit Chronic pulmonary disease - Asthma, chronic bronchitis, emphysema, and other lung disease who have ongoing symptoms such as dyspnea or cough, with mild or moderate activity. Peptic ulcer disease - Any history of treatment for ulcer disease or history of ulcer bleeding Liver disease - Severe = cirrhosis and portal hypertension with variceal bleeding history, moderate = cirrhosis and portal hypertension but no variceal bleeding history, mild = chronic hepatitis (or cirrhosis without portal hypertension) Renal disease - Severe = on dialysis, status post kidney transplant, uremia, moderate = creatinine >3 mg/dL (0.27 mmol/L)** Source: Medical record, treating physician or interview

    1. Myocardial infarction
    2. Congestive heart failure
    3. Peripheral vascular disease
    4. Cerebrovascular disease
    5. Dementia
    6. Chronic pulmonary disease
    7. Rheumatologic disease
    8. Peptic ulcer disease
    9. Liver disease
    10. Diabetes
    11. Hemiplegia or paraplegia
    12. Renal disease
    13. Malignancy
    14. Leukemia
    15. Lymphoma
    16. AIDS
    17. Not known
    18. None
    19. Other

- **12.1 Please specify**

- **13. Severity of liver disease** Source: Medical record, treating physician or interview

    1. Mild
    2. Moderate or severe
    3. Not known

- **14. Severity of diabetes** Source: Medical record, treating physician or interview

    1. Controlled
    2. Uncontrolled
    3. Not known

- **15. Severity of malignancy** Source: Medical record, treating physician or interview

    1. Localized
    2. Metastatic tumor
    3. Not known

- **Any other comments?**

18.2.5 Prehospital

- **1. Date of injury** Source: Medical record of interview
- **1.1 Time of injury**
- **2. Mode of transport to the participating hospital** Source: Medical record of interview

    1. Ambulance
    2. Police
    3. Private vehicle
    4. Walking
    5. Others
    6. Not known

- **2.1 If other, please specify** Source: Medical record of interview
- **3. Referred or transferred to the participating hospital from another hospital** Source: Medical record of interview

    1. Yes
    2. No
    3. Not known

- **4. Date of referring or transferring**
- **5. Time of referring or transferring**
- **Any other comments?**

18.2.6 ATLS adherence

- **1. Airway patency checked** Source: Observation
    1. Yes
    2. No
- **1. Chest wall palpated** Source: Observation
    1. Yes
    2. No
- **2. Breath sounds checked** Source: Observation
    1. Yes
    2. No
- **3. Respiratory rate measured** Source: Observation
    1. Yes
    2. No
- **4. Saturation (SpO2) measured** Source: Observation
    1. Yes
    2. No
- **1. Heart rate measured** Source: Observation
    1. Yes
    2. No
- **2. Blood pressure measured** Source: Observation
    1. Yes
    2. No
- **3. Abdomen palpated** Source: Observation
    1. Yes
    2. No
- **4. Thighs palpated** Source: Observation
    1. Yes
    2. No
- **5. IV access obtained** Source: Observation

1. Yes
2. No

- **1. GCS checked** Source: Observation

1. Yes
2. No

- **2. Pupils checked** Source: Observation

1. Yes
2. No

- **1. Patients exposed for assessment**

1. Yes
2. No

- **2. Temperature measured** Source: Observation

1. Yes
2. No

- **1. Which airway interventions were performed?** Source: Observation

1. None
2. Manual airway procedure such as chin lift or jaw thrust
3. Nasopharyngeal or Oropharyngeal airway inserted
4. Supraglottic airway device
5. Tracheal intubation
6. Surgical airway
7. Other
8. Not known

- **2. If other airway interventions given, specify**
- **3. Were airway interventions performed while minimising c-spine movement?** Source: Observation

1. Yes
2. No
3. Not known

- **1. Which breathing interventions were performed?** Source: Observation

1. None
2. Oxygen applied
3. Intracostal drain placement

4. Other
5. Not known

- **2. If other breathing Interventions done, specify**
- **1. Which circulation interventions and adjuncts were performed?** Source: Observation

1. None
2. Control of external bleeding
3. Fluid bolus
4. Blood transfusion
5. eFast
6. Pelvic binder applied
7. Reduction of highly displaced fracture
8. Other
9. Not known

- **2. If other circulation Interventions done, specify**
- **1. Which disability intervention was performed?** Source: Observation

1. None
2. Placement of definitive airway if the patient had a GCS of 8 or less
3. Log Rolling
4. Spine board during transportation
5. Other
6. Not known

- **2. If other disability interventions done, specify**
- **1. Which exposure intervention was performed?** Source: Observation

1. None
2. Covered with warmer or blanket
3. Warm fluids administered
4. Other
5. Not known

- **2. If other exposure interventions done, specify**
- **Any other comments?**

18.2.7 Emergency department

- **1. Date of arrival to the emergency department at the participating hospital.** Source: Medical record of interview
- **2 Time of arrival to the emergency department at the participating hospital**
- **3. First recorded systolic blood pressure (mmHg)** Source: Medical record
- **4. First recorded diastolic blood pressure (mmHg)** Source: Medical record
- **5. First recorded heart rate (beats per minute)** Source: Medical record
- **6. First recorded respiratory rate (breaths per minute)** Source: Medical record
- **7. First recorded Glasgow Coma Scale** Source: Medical record
- **8. First recorded body temperature (°F)** Source: Medical record
- **9. First recorded oxygen saturation (%)** Source: Medical record
- **10. Emergency department disposition** Source: Medical record

    1. Admitted
    2. Referred or transferred for admission
    3. Dead
    4. Others
    5. Not known

- **10.1 Date of death**
- **10.2 Time of death**
- **11. If other, please specify** Source: Medical record
- **12. Date and time of referral or transfer for admission** Source: Medical record
- **Any other comments?**

18.2.8 Hospital

- **1. Date of admission to the participating hospital** Source: Medical record
- **2 Time of admission to the participating hospital** Source: Medical record
- **3. Type of admitting ward** Source: Medical record

    1. General surgery
    2. Orthopaedics
    3. Neurosurgery
    4. Intensive care unit
    5. High dependency unit

6. Medicine
7. Trauma ward
8. Not known

- **4. Ward name or number** Source: Medical record
- **5. Admitted to intensive care unit during admission** Source: Medical record

1. Yes
2. No
3. Not known

- **6. Date of first intensive care unit admission** Source: Medical record
- **7. Time of first intensive care unit admission** Source: Medical record
- **8. Was the participant discharged from intensive care unit?**

1. Yes
2. No

- **9. Date of first intensive care unit discharge** Source: Medical record
- **10. Time of first intensive care unit discharge** Source: Medical record
- **11. Hospital disposition** Source: Medical record

1. Alive
2. Dead
3. Transferred for admission
4. Not known

- **12. Was the patient transferred to another hospital for admission?** Source: Medical record

1. Yes
2. No
3. Not known

- **13. Date of discharge or transfer from participating hospital** Source: Medical record
- **14. Time of discharge or transfer from participating hospital** Source: Medical record
- **Any other comments?**

18.2.9 Surgery

- **1. Surgery done?**

    1. Yes
    2. No

- **2. Date of surgical procedure** A surgical procedure is defined as any procedure performed in the operating room, interventional dropdownlogy suite, or at the bedside, requiring general or regional anesthesia. Source: Medical record
- **3. Time of surgical procedure** A surgical procedure is defined as any procedure performed in the operating room, interventional dropdownlogy suite, or at the bedside, requiring general or regional anesthesia. Source: Medical record
- **4. Preoperative ASA score** Source: Medical record or treating physician

    1. 1. A normal healthy patient
    2. 2. A patient with mild systemic disease
    3. 3. A patient with severe systemic disease
    4. 4. A patient with severe systemic disease that is a constant threat to life
    5. 5. A moribund patient who is not expected to survive without the operation
    6. 6. A declared brain-dead patient whose organs are being removed for donor purposes
    7. 999. Not known

- **5. Description of procedure** Source: Medical record
- **6. Procedure coded according to SNOMED CT** Source: Medical record
- **Any other comments?**

18.2.10 Imaging

- **1. Date and time of imaging** Source: Medical record
- **2 Time of imaging** Source: Medical record
- **3. Type of imaging** Source: Medical record

    1. Ultrasound
    2. X-ray
    3. Computed Tomography (CT)
    4. Magnetic Resonance Imaging (MRI)

- **Any other comments?**

18.2.11 Transfusion

- **1. Transfusion done?**

    1. Yes
    2. No

- **2. Date of transfusion** Source: Medical record
- **3. Time of transfusion** Source: Medical record
- **4. Type of blood product** Source: Medical record

    1. Packed red blood cells
    2. Platelets
    3. Fresh frozen plasma
    4. Whole blood
    5. Other

- **5 Other specify**
- **6. Number of units transfused** Source: Medical record
- **Any other comments?**

18.2.12 Injury

- **1. Injury description** Source: Medical record
- **2. Injury ICD 10 code** Coded using ICD 10. Source: Medical record

    1. S00 Superficial injury of head
    2. S00.0 Superficial injury of scalp
    3. S00.1 Contusion of eyelid and periocular area
    4. S00.2 Other superficial injuries of eyelid and periocular area
    5. S00.3 Superficial injury of nose
    6. S00.4 Superficial injury of ear
    7. S00.5 Superficial injury of lip and oral cavity
    8. S00.7 Multiple superficial injuries of head
    9. S00.8 Superficial injury of other parts of head
    10. S00.9 Superficial injury of head, part unspecified
    11. S01 Open wound of head
    12. S01.0 Open wound of scalp
    13. S01.1 Open wound of eyelid and periocular area
    14. S01.2 Open wound of nose
    15. S01.3 Open wound of ear
    16. S01.4 Open wound of cheek and temporomandibular area
    17. S01.5 Open wound of lip and oral cavity

18. S01.7 Multiple open wounds of head
19. S01.8 Open wound of other parts of head
20. S01.9 Open wound of head, part unspecified
21. S02 Fracture of skull and facial bones
22. S02.0 Fracture of vault of skull
23. S02.1 Fracture of base of skull
24. S02.2 Fracture of nasal bones
25. S02.3 Fracture of orbital floor
26. S02.4 Fracture of malar and maxillary bones
27. S02.5 Fracture of tooth
28. S02.6 Fracture of mandible
29. S02.7 Multiple fractures involving skull and facial bones
30. S02.8 Fractures of other skull and facial bones
31. S02.9 Fracture of skull and facial bones, part unspecified
32. S03 Dislocation, sprain and strain of joints and ligaments of head
33. S03.0 Dislocation of jaw
34. S03.1 Dislocation of septal cartilage of nose
35. S03.2 Dislocation of tooth
36. S03.3 Dislocation of other and unspecified parts of head
37. S03.4 Sprain and strain of jaw
38. S03.5 Sprain and strain of joints and ligaments of other and unspecified parts of head
39. S04 Injury of cranial nerves
40. S04.0 Injury of optic nerve and pathways
41. S04.1 Injury of oculomotor nerve
42. S04.2 Injury of trochlear nerve
43. S04.3 Injury of trigeminal nerve
44. S04.4 Injury of abducent nerve
45. S04.5 Injury of facial nerve
46. S04.6 Injury of acoustic nerve
47. S04.7 Injury of accessory nerve
48. S04.8 Injury of other cranial nerves
49. S04.9 Injury of unspecified cranial nerve
50. S05 Injury of eye and orbit
51. S05.0 Injury of conjunctiva and corneal abrasion without mention of foreign body
52. S05.1 Contusion of eyeball and orbital tissues
53. S05.2 Ocular laceration and rupture with prolapse or loss of intraocular tissue
54. S05.3 Ocular laceration without prolapse or loss of intraocular tissue
55. S05.4 Penetrating wound of orbit with or without foreign body
56. S05.5 Penetrating wound of eyeball with foreign body
57. S05.6 Penetrating wound of eyeball without foreign body
58. S05.7 Avulsion of eye
59. S05.8 Other injuries of eye and orbit
60. S05.9 Injury of eye and orbit, unspecified

61. S06 Intracranial injury
62. S06.0 Concussion
63. S06.1 Traumatic cerebral oedema
64. S06.2 Diffuse brain injury
65. S06.3 Focal brain injury
66. S06.4 Epidural haemorrhage
67. S06.5 Traumatic subdural haemorrhage
68. S06.6 Traumatic subarachnoid haemorrhage
69. S06.7 Intracranial injury with prolonged coma
70. S06.8 Other intracranial injuries
71. S06.9 Intracranial injury, unspecified
72. S07 Crushing injury of head
73. S07.0 Crushing injury of face
74. S07.1 Crushing injury of skull
75. S07.8 Crushing injury of other parts of head
76. S07.9 Crushing injury of head, part unspecified
77. S08 Traumatic amputation of part of head
78. S08.0 Avulsion of scalp
79. S08.1 Traumatic amputation of ear
80. S08.8 Traumatic amputation of other parts of head
81. S08.9 Traumatic amputation of unspecified part of head
82. S09 Other and unspecified injuries of head
83. S09.0 Injury of blood vessels of head, not elsewhere classified
84. S09.1 Injury of muscle and tendon of head
85. S09.2 Traumatic rupture of ear drum
86. S09.7 Multiple injuries of head
87. S09.8 Other specified injuries of head
88. S09.9 Unspecified injury of head
89. S10 Superficial injury of neck
90. S10.0 Contusion of throat
91. S10.1 Other and unspecified superficial injuries of throat
92. S10.7 Multiple superficial injuries of neck
93. S10.8 Superficial injury of other parts of neck
94. S10.9 Superficial injury of neck, part unspecified
95. S11 Open wound of neck
96. S11.0 Open wound involving larynx and trachea
97. S11.1 Open wound involving thyroid gland
98. S11.2 Open wound involving pharynx and cervical oesophagus
99. S11.7 Multiple open wounds of neck
100. S11.8 Open wound of other parts of neck
101. S11.9 Open wound of neck, part unspecified
102. S12 Fracture of neck
103. S12.0 Fracture of first cervical vertebra

104. S12.1 Fracture of second cervical vertebra
105. S12.2 Fracture of other specified cervical vertebra
106. S12.7 Multiple fractures of cervical spine
107. S12.8 Fracture of other parts of neck
108. S12.9 Fracture of neck, part unspecified
109. S13 Dislocation, sprain and strain of joints and ligaments at neck level
110. S13.0 Traumatic rupture of cervical intervertebral disc
111. S13.1 Dislocation of cervical vertebra
112. S13.2 Dislocation of other and unspecified parts of neck
113. S13.3 Multiple dislocations of neck
114. S13.4 Sprain and strain of cervical spine
115. S13.5 Sprain and strain of thyroid region
116. S13.6 Sprain and strain of joints and ligaments of other and unspecified parts of neck
117. S14 Injury of nerves and spinal cord at neck level
118. S14.0 Concussion and oedema of cervical spinal cord
119. S14.1 Other and unspecified injuries of cervical spinal cord
120. S14.2 Injury of nerve root of cervical spine
121. S14.3 Injury of brachial plexus
122. S14.4 Injury of peripheral nerves of neck
123. S14.5 Injury of cervical sympathetic nerves
124. S14.6 Injury of other and unspecified nerves of neck
125. S15 Injury of blood vessels at neck level
126. S15.0 Injury of carotid artery
127. S15.1 Injury of vertebral artery
128. S15.2 Injury of external jugular vein
129. S15.3 Injury of internal jugular vein
130. S15.7 Injury of multiple blood vessels at neck level
131. S15.8 Injury of other blood vessels at neck level
132. S15.9 Injury of unspecified blood vessel at neck level
133. S16 Injury of muscle and tendon at neck level
134. S17 Crushing injury of neck
135. S17.0 Crushing injury of larynx and trachea
136. S17.8 Crushing injury of other parts of neck
137. S17.9 Crushing injury of neck, part unspecified
138. S18 Traumatic amputation at neck level
139. S19 Other and unspecified injuries of neck
140. S19.7 Multiple injuries of neck
141. S19.8 Other specified injuries of neck
142. S19.9 Unspecified injury of neck
143. S20 Superficial injury of thorax
144. S20.0 Contusion of breast
145. S20.1 Other and unspecified superficial injuries of breast
146. S20.2 Contusion of thorax

147. S20.3 Other superficial injuries of front wall of thorax
148. S20.4 Other superficial injuries of back wall of thorax
149. S20.7 Multiple superficial injuries of thorax
150. S20.8 Superficial injury of other and unspecified parts of thorax
151. S21 Open wound of thorax
152. S21.0 Open wound of breast
153. S21.1 Open wound of front wall of thorax
154. S21.2 Open wound of back wall of thorax
155. S21.7 Multiple open wounds of thoracic wall
156. S21.8 Open wound of other parts of thorax
157. S21.9 Open wound of thorax, part unspecified
158. S22 Fracture of rib(s), sternum and thoracic spine
159. S22.0 Fracture of thoracic vertebra
160. S22.1 Multiple fractures of thoracic spine
161. S22.2 Fracture of sternum
162. S22.3 Fracture of rib
163. S22.4 Multiple fractures of ribs
164. S22.5 Flail chest
165. S22.8 Fracture of other parts of bony thorax
166. S22.9 Fracture of bony thorax, part unspecified
167. S23 Dislocation, sprain and strain of joints and ligaments of thorax
168. S23.0 Traumatic rupture of thoracic intervertebral disc
169. S23.1 Dislocation of thoracic vertebra
170. S23.2 Dislocation of other and unspecified parts of thorax
171. S23.3 Sprain and strain of thoracic spine
172. S23.4 Sprain and strain of ribs and sternum
173. S23.5 Sprain and strain of other and unspecified parts of thorax
174. S24 Injury of nerves and spinal cord at thorax level
175. S24.0 Concussion and oedema of thoracic spinal cord
176. S24.1 Other and unspecified injuries of thoracic spinal cord
177. S24.2 Injury of nerve root of thoracic spine
178. S24.3 Injury of peripheral nerves of thorax
179. S24.4 Injury of thoracic sympathetic nerves
180. S24.5 Injury of other nerves of thorax
181. S24.6 Injury of unspecified nerve of thorax
182. S25 Injury of blood vessels of thorax
183. S25.0 Injury of thoracic aorta
184. S25.1 Injury of innominate or subclavian artery
185. S25.2 Injury of superior vena cava
186. S25.3 Injury of innominate or subclavian vein
187. S25.4 Injury of pulmonary blood vessels
188. S25.5 Injury of intercostal blood vessels
189. S25.7 Injury of multiple blood vessels of thorax

190. S25.8 Injury of other blood vessels of thorax
191. S25.9 Injury of unspecified blood vessel of thorax
192. S26 Injury of heart
193. S26.0 Injury of heart with haemopericardium
194. S26.8 Other injuries of heart
195. S26.9 Injury of heart, unspecified
196. S27 Injury of other and unspecified intrathoracic organs
197. S27.0 Traumatic pneumothorax
198. S27.1 Traumatic haemothorax
199. S27.2 Traumatic haemopneumothorax
200. S27.3 Other injuries of lung
201. S27.4 Injury of bronchus
202. S27.5 Injury of thoracic trachea
203. S27.6 Injury of pleura
204. S27.7 Multiple injuries of intrathoracic organs
205. S27.8 Injury of other specified intrathoracic organs
206. S27.9 Injury of unspecified intrathoracic organ
207. S28 Crushing injury of thorax and traumatic amputation of part of thorax
208. S28.0 Crushed chest
209. S28.1 Traumatic amputation of part of thorax
210. S29 Other and unspecified injuries of thorax
211. S29.0 Injury of muscle and tendon at thorax level
212. S29.7 Multiple injuries of thorax
213. S29.8 Other specified injuries of thorax
214. S29.9 Unspecified injury of thorax
215. S30 Superficial injury of abdomen, lower back and pelvis
216. S30.0 Contusion of lower back and pelvis
217. S30.1 Contusion of abdominal wall
218. S30.2 Contusion of external genital organs
219. S30.7 Multiple superficial injuries of abdomen, lower back and pelvis
220. S30.8 Other superficial injuries of abdomen, lower back and pelvis
221. S30.9 Superficial injury of abdomen, lower back and pelvis, part unspecified
222. S31 Open wound of abdomen, lower back and pelvis
223. S31.0 Open wound of lower back and pelvis
224. S31.1 Open wound of abdominal wall
225. S31.2 Open wound of penis
226. S31.3 Open wound of scrotum and testes
227. S31.4 Open wound of vagina and vulva
228. S31.5 Open wound of other and unspecified external genital organs
229. S31.7 Multiple open wounds of abdomen, lower back and pelvis
230. S31.8 Open wound of other and unspecified parts of abdomen
231. S32 Fracture of lumbar spine and pelvis
232. S32.0 Fracture of lumbar vertebra

233. S32.1 Fracture of sacrum
234. S32.2 Fracture of coccyx
235. S32.3 Fracture of ilium
236. S32.4 Fracture of acetabulum
237. S32.5 Fracture of pubis
238. S32.7 Multiple fractures of lumbar spine and pelvis
239. S32.8 Fracture of other and unspecified parts of lumbar spine and pelvis
240. S33 Dislocation, sprain and strain of joints and ligaments of lumbar spine and pelvis
241. S33.0 Traumatic rupture of lumbar intervertebral disc
242. S33.1 Dislocation of lumbar vertebra
243. S33.2 Dislocation of sacroiliac and sacrococcygeal joint
244. S33.3 Dislocation of other and unspecified parts of lumbar spine and pelvis
245. S33.4 Traumatic rupture of symphysis pubis
246. S33.5 Sprain and strain of lumbar spine
247. S33.6 Sprain and strain of sacroiliac joint
248. S33.7 Sprain and strain of other and unspecified parts of lumbar spine and pelvis
249. S34 Injury of nerves and lumbar spinal cord at abdomen, lower back and pelvis level
250. S34.0 Concussion and oedema of lumbar spinal cord
251. S34.1 Other injury of lumbar spinal cord
252. S34.2 Injury of nerve root of lumbar and sacral spine
253. S34.3 Injury of cauda equina
254. S34.4 Injury of lumbosacral plexus
255. S34.5 Injury of lumbar, sacral and pelvic sympathetic nerves
256. S34.6 Injury of peripheral nerve(s) of abdomen, lower back and pelvis
257. S34.8 Injury of other and unspecified nerves at abdomen, lower back and pelvis level
258. S35 Injury of blood vessels at abdomen, lower back and pelvis level
259. S35.0 Injury of abdominal aorta
260. S35.1 Injury of inferior vena cava
261. S35.2 Injury of coeliac or mesenteric artery
262. S35.3 Injury of portal or splenic vein
263. S35.4 Injury of renal blood vessels
264. S35.5 Injury of iliac blood vessels
265. S35.7 Injury of multiple blood vessels at abdomen, lower back and pelvis level
266. S35.8 Injury of other blood vessels at abdomen, lower back and pelvis level
267. S35.9 Injury of unspecified blood vessel at abdomen, lower back and pelvis level
268. S36 Injury of intra-abdominal organs
269. S36.0 Injury of spleen
270. S36.1 Injury of liver or gallbladder
271. S36.2 Injury of pancreas
272. S36.3 Injury of stomach
273. S36.4 Injury of small intestine
274. S36.5 Injury of colon
275. S36.6 Injury of rectum

276. S36.7 Injury of multiple intra-abdominal organs
277. S36.8 Injury of other intra-abdominal organs
278. S36.9 Injury of unspecified intra-abdominal organ
279. S37 Injury of urinary and pelvic organs
280. S37.0 Injury of kidney
281. S37.1 Injury of ureter
282. S37.2 Injury of bladder
283. S37.3 Injury of urethra
284. S37.4 Injury of ovary
285. S37.5 Injury of fallopian tube
286. S37.6 Injury of uterus
287. S37.7 Injury of multiple pelvic organs
288. S37.8 Injury of other pelvic organs
289. S37.9 Injury of unspecified pelvic organ
290. S38 Crushing injury and traumatic amputation of part of abdomen, lower back and pelvis
291. S38.0 Crushing injury of external genital organs
292. S38.1 Crushing injury of other and unspecified parts of abdomen, lower back and pelvis
293. S38.2 Traumatic amputation of external genital organs
294. S38.3 Traumatic amputation of other and unspecified parts of abdomen, lower back and pelvis
295. S39 Other and unspecified injuries of abdomen, lower back and pelvis
296. S39.0 Injury of muscle and tendon of abdomen, lower back and pelvis
297. S39.6 Injury of intra-abdominal organ(s) with pelvic organ(s)
298. S39.7 Other multiple injuries of abdomen, lower back and pelvis
299. S39.8 Other specified injuries of abdomen, lower back and pelvis
300. S39.9 Unspecified injury of abdomen, lower back and pelvis
301. S40 Superficial injury of shoulder and upper arm
302. S40.0 Contusion of shoulder and upper arm
303. S40.7 Multiple superficial injuries of shoulder and upper arm
304. S40.8 Other superficial injuries of shoulder and upper arm
305. S40.9 Superficial injury of shoulder and upper arm, unspecified
306. S41 Open wound of shoulder and upper arm
307. S41.0 Open wound of shoulder
308. S41.1 Open wound of upper arm
309. S41.7 Multiple open wounds of shoulder and upper arm
310. S41.8 Open wound of other and unspecified parts of shoulder girdle
311. S42 Fracture of shoulder and upper arm
312. S42.0 Fracture of clavicle
313. S42.1 Fracture of scapula
314. S42.2 Fracture of upper end of humerus
315. S42.3 Fracture of shaft of humerus
316. S42.4 Fracture of lower end of humerus
317. S42.7 Multiple fractures of clavicle, scapula and humerus

318. S42.8 Fracture of other parts of shoulder and upper arm
319. S42.9 Fracture of shoulder girdle, part unspecified
320. S43 Dislocation, sprain and strain of joints and ligaments of shoulder girdle
321. S43.0 Dislocation of shoulder joint
322. S43.1 Dislocation of acromioclavicular joint
323. S43.2 Dislocation of sternoclavicular joint
324. S43.3 Dislocation of other and unspecified parts of shoulder girdle
325. S43.4 Sprain and strain of shoulder joint
326. S43.5 Sprain and strain of acromioclavicular joint
327. S43.6 Sprain and strain of sternoclavicular joint
328. S43.7 Sprain and strain of other and unspecified parts of shoulder girdle
329. S44 Injury of nerves at shoulder and upper arm level
330. S44.0 Injury of ulnar nerve at upper arm level
331. S44.1 Injury of median nerve at upper arm level
332. S44.2 Injury of radial nerve at upper arm level
333. S44.3 Injury of axillary nerve
334. S44.4 Injury of musculocutaneous nerve
335. S44.5 Injury of cutaneous sensory nerve at shoulder and upper arm level
336. S44.7 Injury of multiple nerves at shoulder and upper arm level
337. S44.8 Injury of other nerves at shoulder and upper arm level
338. S44.9 Injury of unspecified nerve at shoulder and upper arm level
339. S45 Injury of blood vessels at shoulder and upper arm level
340. S45.0 Injury of axillary artery
341. S45.1 Injury of brachial artery
342. S45.2 Injury of axillary or brachial vein
343. S45.3 Injury of superficial vein at shoulder and upper arm level
344. S45.7 Injury of multiple blood vessels at shoulder and upper arm level
345. S45.8 Injury of other blood vessels at shoulder and upper arm level
346. S45.9 Injury of unspecified blood vessel at shoulder and upper arm level
347. S46 Injury of muscle and tendon at shoulder and upper arm level
348. S46.0 Injury of muscle(s) and tendon(s) of the rotator cuff of shoulder
349. S46.1 Injury of muscle and tendon of long head of biceps
350. S46.2 Injury of muscle and tendon of other parts of biceps
351. S46.3 Injury of muscle and tendon of triceps
352. S46.7 Injury of multiple muscles and tendons at shoulder and upper arm level
353. S46.8 Injury of other muscles and tendons at shoulder and upper arm level
354. S46.9 Injury of unspecified muscle and tendon at shoulder and upper arm level
355. S47 Crushing injury of shoulder and upper arm
356. S48 Traumatic amputation of shoulder and upper arm
357. S48.0 Traumatic amputation at shoulder joint
358. S48.1 Traumatic amputation at level between shoulder and elbow
359. S48.9 Traumatic amputation of shoulder and upper arm, level unspecified
360. S49 Other and unspecified injuries of shoulder and upper arm

361. S49.7 Multiple injuries of shoulder and upper arm
362. S49.8 Other specified injuries of shoulder and upper arm
363. S49.9 Unspecified injury of shoulder and upper arm
364. S50 Superficial injury of forearm
365. S50.0 Contusion of elbow
366. S50.1 Contusion of other and unspecified parts of forearm
367. S50.7 Multiple superficial injuries of forearm
368. S50.8 Other superficial injuries of forearm
369. S50.9 Superficial injury of forearm, unspecified
370. S51 Open wound of forearm
371. S51.0 Open wound of elbow
372. S51.7 Multiple open wounds of forearm
373. S51.8 Open wound of other parts of forearm
374. S51.9 Open wound of forearm, part unspecified
375. S52 Fracture of forearm
376. S52.0 Fracture of upper end of ulna
377. S52.1 Fracture of upper end of radius
378. S52.2 Fracture of shaft of ulna
379. S52.3 Fracture of shaft of radius
380. S52.4 Fracture of shafts of both ulna and radius
381. S52.5 Fracture of lower end of radius
382. S52.6 Fracture of lower end of both ulna and radius
383. S52.7 Multiple fractures of forearm
384. S52.8 Fracture of other parts of forearm
385. S52.9 Fracture of forearm, part unspecified
386. S53 Dislocation, sprain and strain of joints and ligaments of elbow
387. S53.0 Dislocation of radial head
388. S53.1 Dislocation of elbow, unspecified
389. S53.2 Traumatic rupture of radial collateral ligament
390. S53.3 Traumatic rupture of ulnar collateral ligament
391. S53.4 Sprain and strain of elbow
392. S54 Injury of nerves at forearm level
393. S54.0 Injury of ulnar nerve at forearm level
394. S54.1 Injury of median nerve at forearm level
395. S54.2 Injury of radial nerve at forearm level
396. S54.3 Injury of cutaneous sensory nerve at forearm level
397. S54.7 Injury of multiple nerves at forearm level
398. S54.8 Injury of other nerves at forearm level
399. S54.9 Injury of unspecified nerve at forearm level
400. S55 Injury of blood vessels at forearm level
401. S55.0 Injury of ulnar artery at forearm level
402. S55.1 Injury of radial artery at forearm level
403. S55.2 Injury of vein at forearm level

404. S55.7 Injury of multiple blood vessels at forearm level
405. S55.8 Injury of other blood vessels at forearm level
406. S55.9 Injury of unspecified blood vessel at forearm level
407. S56 Injury of muscle and tendon at forearm level
408. S56.0 Injury of flexor muscle and tendon of thumb at forearm level
409. S56.1 Injury of long flexor muscle and tendon of other finger(s) at forearm level
410. S56.2 Injury of other flexor muscle and tendon at forearm level
411. S56.3 Injury of extensor or abductor muscles and tendons of thumb at forearm level
412. S56.4 Injury of extensor muscle and tendon of other finger(s) at forearm level
413. S56.5 Injury of other extensor muscle and tendon at forearm level
414. S56.7 Injury of multiple muscles and tendons at forearm level
415. S56.8 Injury of other and unspecified muscles and tendons at forearm level
416. S57 Crushing injury of forearm
417. S57.0 Crushing injury of elbow
418. S57.8 Crushing injury of other parts of forearm
419. S57.9 Crushing injury of forearm, part unspecified
420. S58 Traumatic amputation of forearm
421. S58.0 Traumatic amputation at elbow level
422. S58.1 Traumatic amputation at level between elbow and wrist
423. S58.9 Traumatic amputation of forearm, level unspecified
424. S59 Other and unspecified injuries of forearm
425. S59.7 Multiple injuries of forearm
426. S59.8 Other specified injuries of forearm
427. S59.9 Unspecified injury of forearm
428. S60 Superficial injury of wrist and hand
429. S60.0 Contusion of finger(s) without damage to nail
430. S60.1 Contusion of finger(s) with damage to nail
431. S60.2 Contusion of other parts of wrist and hand
432. S60.7 Multiple superficial injuries of wrist and hand
433. S60.8 Other superficial injuries of wrist and hand
434. S60.9 Superficial injury of wrist and hand, unspecified
435. S61 Open wound of wrist and hand
436. S61.0 Open wound of finger(s) without damage to nail
437. S61.1 Open wound of finger(s) with damage to nail
438. S61.7 Multiple open wounds of wrist and hand
439. S61.8 Open wound of other parts of wrist and hand
440. S61.9 Open wound of wrist and hand, part unspecified
441. S62 Fracture at wrist and hand level
442. S62.0 Fracture of navicular [scaphoid] bone of hand
443. S62.1 Fracture of other carpal bone(s)
444. S62.2 Fracture of first metacarpal bone
445. S62.3 Fracture of other metacarpal bone
446. S62.4 Multiple fractures of metacarpal bones

447. S62.5 Fracture of thumb
448. S62.6 Fracture of other finger
449. S62.7 Multiple fractures of fingers
450. S62.8 Fracture of other and unspecified parts of wrist and hand
451. S63 Dislocation, sprain and strain of joints and ligaments at wrist and hand level
452. S63.0 Dislocation of wrist
453. S63.1 Dislocation of finger
454. S63.2 Multiple dislocations of fingers
455. S63.3 Traumatic rupture of ligament of wrist and carpus
456. S63.4 Traumatic rupture of ligament of finger at metacarpophalangeal and interphalangeal joint(s)
457. S63.5 Sprain and strain of wrist
458. S63.6 Sprain and strain of finger(s)
459. S63.7 Sprain and strain of other and unspecified parts of hand
460. S64 Injury of nerves at wrist and hand level
461. S64.0 Injury of ulnar nerve at wrist and hand level
462. S64.1 Injury of median nerve at wrist and hand level
463. S64.2 Injury of radial nerve at wrist and hand level
464. S64.3 Injury of digital nerve of thumb
465. S64.4 Injury of digital nerve of other finger
466. S64.7 Injury of multiple nerves at wrist and hand level
467. S64.8 Injury of other nerves at wrist and hand level
468. S64.9 Injury of unspecified nerve at wrist and hand level
469. S65 Injury of blood vessels at wrist and hand level
470. S65.0 Injury of ulnar artery at wrist and hand level
471. S65.1 Injury of radial artery at wrist and hand level
472. S65.2 Injury of superficial palmar arch
473. S65.3 Injury of deep palmar arch
474. S65.4 Injury of blood vessel(s) of thumb
475. S65.5 Injury of blood vessel(s) of other finger
476. S65.7 Injury of multiple blood vessels at wrist and hand level
477. S65.8 Injury of other blood vessels at wrist and hand level
478. S65.9 Injury of unspecified blood vessel at wrist and hand level
479. S66 Injury of muscle and tendon at wrist and hand level
480. S66.0 Injury of long flexor muscle and tendon of thumb at wrist and hand level
481. S66.1 Injury of flexor muscle and tendon of other finger at wrist and hand level
482. S66.2 Injury of extensor muscle and tendon of thumb at wrist and hand level
483. S66.3 Injury of extensor muscle and tendon of other finger at wrist and hand level
484. S66.4 Injury of intrinsic muscle and tendon of thumb at wrist and hand level
485. S66.5 Injury of intrinsic muscle and tendon of other finger at wrist and hand level
486. S66.6 Injury of multiple flexor muscles and tendons at wrist and hand level
487. S66.7 Injury of multiple extensor muscles and tendons at wrist and hand level
488. S66.8 Injury of other muscles and tendons at wrist and hand level

489. S66.9 Injury of unspecified muscle and tendon at wrist and hand level
490. S67 Crushing injury of wrist and hand
491. S67.0 Crushing injury of thumb and other finger(s)
492. S67.8 Crushing injury of other and unspecified parts of wrist and hand
493. S68 Traumatic amputation of wrist and hand
494. S68.0 Traumatic amputation of thumb (complete)(partial)
495. S68.1 Traumatic amputation of other single finger (complete)(partial)
496. S68.2 Traumatic amputation of two or more fingers alone (complete)(partial)
497. S68.3 Combined traumatic amputation of (part of) finger(s) with other parts of wrist and hand
498. S68.4 Traumatic amputation of hand at wrist level
499. S68.8 Traumatic amputation of other parts of wrist and hand
500. S68.9 Traumatic amputation of wrist and hand, level unspecified
501. S69 Other and unspecified injuries of wrist and hand
502. S69.7 Multiple injuries of wrist and hand
503. S69.8 Other specified injuries of wrist and hand
504. S69.9 Unspecified injury of wrist and hand
505. S70 Superficial injury of hip and thigh
506. S70.0 Contusion of hip
507. S70.1 Contusion of thigh
508. S70.7 Multiple superficial injuries of hip and thigh
509. S70.8 Other superficial injuries of hip and thigh
510. S70.9 Superficial injury of hip and thigh, unspecified
511. S71 Open wound of hip and thigh
512. S71.0 Open wound of hip
513. S71.1 Open wound of thigh
514. S71.7 Multiple open wounds of hip and thigh
515. S71.8 Open wound of other and unspecified parts of pelvic girdle
516. S72 Fracture of femur
517. S72.0 Fracture of neck of femur
518. S72.1 Pertrochanteric fracture
519. S72.2 Subtrochanteric fracture
520. S72.3 Fracture of shaft of femur
521. S72.4 Fracture of lower end of femur
522. S72.7 Multiple fractures of femur
523. S72.8 Fractures of other parts of femur
524. S72.9 Fracture of femur, part unspecified
525. S73 Dislocation, sprain and strain of joint and ligaments of hip
526. S73.0 Dislocation of hip
527. S73.1 Sprain and strain of hip
528. S74 Injury of nerves at hip and thigh level
529. S74.0 Injury of sciatic nerve at hip and thigh level
530. S74.1 Injury of femoral nerve at hip and thigh level

531. S74.2 Injury of cutaneous sensory nerve at hip and thigh level
532. S74.7 Injury of multiple nerves at hip and thigh level
533. S74.8 Injury of other nerves at hip and thigh level
534. S74.9 Injury of unspecified nerve at hip and thigh level
535. S75 Injury of blood vessels at hip and thigh level
536. S75.0 Injury of femoral artery
537. S75.1 Injury of femoral vein at hip and thigh level
538. S75.2 Injury of greater saphenous vein at hip and thigh level
539. S75.7 Injury of multiple blood vessels at hip and thigh level
540. S75.8 Injury of other blood vessels at hip and thigh level
541. S75.9 Injury of unspecified blood vessel at hip and thigh level
542. S76 Injury of muscle and tendon at hip and thigh level
543. S76.0 Injury of muscle and tendon of hip
544. S76.1 Injury of quadriceps muscle and tendon
545. S76.2 Injury of adductor muscle and tendon of thigh
546. S76.3 Injury of muscle and tendon of the posterior muscle group at thigh level
547. S76.4 Injury of other and unspecified muscles and tendons at thigh level
548. S76.7 Injury of multiple muscles and tendons at hip and thigh level
549. S77 Crushing injury of hip and thigh
550. S77.0 Crushing injury of hip
551. S77.1 Crushing injury of thigh
552. S77.2 Crushing injury of hip with thigh
553. S78 Traumatic amputation of hip and thigh
554. S78.0 Traumatic amputation at hip joint
555. S78.1 Traumatic amputation at level between hip and knee
556. S78.9 Traumatic amputation of hip and thigh, level unspecified
557. S79 Other and unspecified injuries of hip and thigh
558. S79.7 Multiple injuries of hip and thigh
559. S79.8 Other specified injuries of hip and thigh
560. S79.9 Unspecified injury of hip and thigh
561. S80 Superficial injury of lower leg
562. S80.0 Contusion of knee
563. S80.1 Contusion of other and unspecified parts of lower leg
564. S80.7 Multiple superficial injuries of lower leg
565. S80.8 Other superficial injuries of lower leg
566. S80.9 Superficial injury of lower leg, unspecified
567. S81 Open wound of lower leg
568. S81.0 Open wound of knee
569. S81.7 Multiple open wounds of lower leg
570. S81.8 Open wound of other parts of lower leg
571. S81.9 Open wound of lower leg, part unspecified
572. S82 Fracture of lower leg, including ankle
573. S82.0 Fracture of patella

574. S82.1 Fracture of upper end of tibia
575. S82.2 Fracture of shaft of tibia
576. S82.3 Fracture of lower end of tibia
577. S82.4 Fracture of fibula alone
578. S82.5 Fracture of medial malleolus
579. S82.6 Fracture of lateral malleolus
580. S82.7 Multiple fractures of lower leg
581. S82.8 Fractures of other parts of lower leg
582. S82.9 Fracture of lower leg, part unspecified
583. S83 Dislocation, sprain and strain of joints and ligaments of knee
584. S83.0 Dislocation of patella
585. S83.1 Dislocation of knee
586. S83.2 Tear of meniscus, current
587. S83.3 Tear of articular cartilage of knee, current
588. S83.4 Sprain and strain involving (fibular)(tibial) collateral ligament of knee
589. S83.5 Sprain and strain involving (anterior)(posterior) cruciate ligament of knee
590. S83.6 Sprain and strain of other and unspecified parts of knee
591. S83.7 Injury to multiple structures of knee
592. S84 Injury of nerves at lower leg level
593. S84.0 Injury of tibial nerve at lower leg level
594. S84.1 Injury of peroneal nerve at lower leg level
595. S84.2 Injury of cutaneous sensory nerve at lower leg level
596. S84.7 Injury of multiple nerves at lower leg level
597. S84.8 Injury of other nerves at lower leg level
598. S84.9 Injury of unspecified nerve at lower leg level
599. S85 Injury of blood vessels at lower leg level
600. S85.0 Injury of popliteal artery
601. S85.1 Injury of (anterior)(posterior) tibial artery
602. S85.2 Injury of peroneal artery
603. S85.3 Injury of greater saphenous vein at lower leg level
604. S85.4 Injury of lesser saphenous vein at lower leg level
605. S85.5 Injury of popliteal vein
606. S85.7 Injury of multiple blood vessels at lower leg level
607. S85.8 Injury of other blood vessels at lower leg level
608. S85.9 Injury of unspecified blood vessel at lower leg level
609. S86 Injury of muscle and tendon at lower leg level
610. S86.0 Injury of Achilles tendon
611. S86.1 Injury of other muscle(s) and tendon(s) of posterior muscle group at lower leg level
612. S86.2 Injury of muscle(s) and tendon(s) of anterior muscle group at lower leg level
613. S86.3 Injury of muscle(s) and tendon(s) of peroneal muscle group at lower leg level
614. S86.7 Injury of multiple muscles and tendons at lower leg level
615. S86.8 Injury of other muscles and tendons at lower leg level
616. S86.9 Injury of unspecified muscle and tendon at lower leg level

617. S87 Crushing injury of lower leg
618. S87.0 Crushing injury of knee
619. S87.8 Crushing injury of other and unspecified parts of lower leg
620. S88 Traumatic amputation of lower leg
621. S88.0 Traumatic amputation at knee level
622. S88.1 Traumatic amputation at level between knee and ankle
623. S88.9 Traumatic amputation of lower leg, level unspecified
624. S89 Other and unspecified injuries of lower leg
625. S89.7 Multiple injuries of lower leg
626. S89.8 Other specified injuries of lower leg
627. S89.9 Unspecified injury of lower leg
628. S90 Superficial injury of ankle and foot
629. S90.0 Contusion of ankle
630. S90.1 Contusion of toe(s) without damage to nail
631. S90.2 Contusion of toe(s) with damage to nail
632. S90.3 Contusion of other and unspecified parts of foot
633. S90.7 Multiple superficial injuries of ankle and foot
634. S90.8 Other superficial injuries of ankle and foot
635. S90.9 Superficial injury of ankle and foot, unspecified
636. S91 Open wound of ankle and foot
637. S91.0 Open wound of ankle
638. S91.1 Open wound of toe(s) without damage to nail
639. S91.2 Open wound of toe(s) with damage to nail
640. S91.3 Open wound of other parts of foot
641. S91.7 Multiple open wounds of ankle and foot
642. S92 Fracture of foot, except ankle
643. S92.0 Fracture of calcaneus
644. S92.1 Fracture of talus
645. S92.2 Fracture of other tarsal bone(s)
646. S92.3 Fracture of metatarsal bone
647. S92.4 Fracture of great toe
648. S92.5 Fracture of other toe
649. S92.7 Multiple fractures of foot
650. S92.9 Fracture of foot, unspecified
651. S93 Dislocation, sprain and strain of joints and ligaments at ankle and foot level
652. S93.0 Dislocation of ankle joint
653. S93.1 Dislocation of toe(s)
654. S93.2 Rupture of ligaments at ankle and foot level
655. S93.3 Dislocation of other and unspecified parts of foot
656. S93.4 Sprain and strain of ankle
657. S93.5 Sprain and strain of toe(s)
658. S93.6 Sprain and strain of other and unspecified parts of foot
659. S94 Injury of nerves at ankle and foot level

660. S94.0 Injury of lateral plantar nerve
661. S94.1 Injury of medial plantar nerve
662. S94.2 Injury of deep peroneal nerve at ankle and foot level
663. S94.3 Injury of cutaneous sensory nerve at ankle and foot level
664. S94.7 Injury of multiple nerves at ankle and foot level
665. S94.8 Injury of other nerves at ankle and foot level
666. S94.9 Injury of unspecified nerve at ankle and foot level
667. S95 Injury of blood vessels at ankle and foot level
668. S95.0 Injury of dorsal artery of foot
669. S95.1 Injury of plantar artery of foot
670. S95.2 Injury of dorsal vein of foot
671. S95.7 Injury of multiple blood vessels at ankle and foot level
672. S95.8 Injury of other blood vessels at ankle and foot level
673. S95.9 Injury of unspecified blood vessel at ankle and foot level
674. S96 Injury of muscle and tendon at ankle and foot level
675. S96.0 Injury of muscle and tendon of long flexor muscle of toe at ankle and foot level
676. S96.1 Injury of muscle and tendon of long extensor muscle of toe at ankle and foot level
677. S96.2 Injury of intrinsic muscle and tendon at ankle and foot level
678. S96.7 Injury of multiple muscles and tendons at ankle and foot level
679. S96.8 Injury of other muscles and tendons at ankle and foot level
680. S96.9 Injury of unspecified muscle and tendon at ankle and foot level
681. S97 Crushing injury of ankle and foot
682. S97.0 Crushing injury of ankle
683. S97.1 Crushing injury of toe(s)
684. S97.8 Crushing injury of other parts of ankle and foot
685. S98 Traumatic amputation of ankle and foot
686. S98.0 Traumatic amputation of foot at ankle level
687. S98.1 Traumatic amputation of one toe
688. S98.2 Traumatic amputation of two or more toes
689. S98.3 Traumatic amputation of other parts of foot
690. S98.4 Traumatic amputation of foot, level unspecified
691. S99 Other and unspecified injuries of ankle and foot
692. S99.7 Multiple injuries of ankle and foot
693. S99.8 Other specified injuries of ankle and foot
694. S99.9 Unspecified injury of ankle and foot
695. T00 Superficial injuries involving multiple body regions
696. T00.0 Superficial injuries involving head with neck
697. T00.1 Superficial injuries involving thorax with abdomen, lower back and pelvis
698. T00.2 Superficial injuries involving multiple regions of upper limb(s)
699. T00.3 Superficial injuries involving multiple regions of lower limb(s)
700. T00.6 Superficial injuries involving multiple regions of upper limb(s) with lower limb(s)
701. T00.8 Superficial injuries involving other combinations of body regions
702. T00.9 Multiple superficial injuries, unspecified

703. T01 Open wounds involving multiple body regions
704. T01.0 Open wounds involving head with neck
705. T01.1 Open wounds involving thorax with abdomen, lower back and pelvis
706. T01.2 Open wounds involving multiple regions of upper limb(s)
707. T01.3 Open wounds involving multiple regions of lower limb(s)
708. T01.6 Open wounds involving multiple regions of upper limb(s) with lower limb(s)
709. T01.8 Open wounds involving other combinations of body regions
710. T01.9 Multiple open wounds, unspecified
711. T02 Fractures involving multiple body regions
712. T02.0 Fractures involving head with neck
713. T02.1 Fractures involving thorax with lower back and pelvis
714. T02.2 Fractures involving multiple regions of one upper limb
715. T02.3 Fractures involving multiple regions of one lower limb
716. T02.4 Fractures involving multiple regions of both upper limbs
717. T02.5 Fractures involving multiple regions of both lower limbs
718. T02.6 Fractures involving multiple regions of upper limb(s) with lower limb(s)
719. T02.7 Fractures involving thorax with lower back and pelvis with limb(s)
720. T02.8 Fractures involving other combinations of body regions
721. T02.9 Multiple fractures, unspecified
722. T03 Dislocations, sprains and strains involving multiple body regions
723. T03.0 Dislocations, sprains and strains involving head with neck
724. T03.1 Dislocations, sprains and strains involving thorax with lower back and pelvis
725. T03.2 Dislocations, sprains and strains involving multiple regions of upper limb(s)
726. T03.3 Dislocations, sprains and strains involving multiple regions of lower limb(s)
727. T03.4 Dislocations, sprains and strains involving multiple regions of upper limb(s) with lower limb(s)
728. T03.8 Dislocations, sprains and strains involving other combinations of body regions
729. T03.9 Multiple dislocations, sprains and strains, unspecified
730. T04 Crushing injuries involving multiple body regions
731. T04.0 Crushing injuries involving head with neck
732. T04.1 Crushing injuries involving thorax with abdomen, lower back and pelvis
733. T04.2 Crushing injuries involving multiple regions of upper limb(s)
734. T04.3 Crushing injuries involving multiple regions of lower limb(s)
735. T04.4 Crushing injuries involving multiple regions of upper limb(s) with lower limb(s)
736. T04.7 Crushing injuries of thorax with abdomen, lower back and pelvis with limb(s)
737. T04.8 Crushing injuries involving other combinations of body regions
738. T04.9 Multiple crushing injuries, unspecified
739. T05 Traumatic amputations involving multiple body regions
740. T05.0 Traumatic amputation of both hands
741. T05.1 Traumatic amputation of one hand and other arm [any level, except hand]
742. T05.2 Traumatic amputation of both arms [any level]
743. T05.3 Traumatic amputation of both feet
744. T05.4 Traumatic amputation of one foot and other leg [any level, except foot]

745. T05.5 Traumatic amputation of both legs [any level]
746. T05.6 Traumatic amputation of upper and lower limbs, any combination [any level]
747. T05.8 Traumatic amputations involving other combinations of body regions
748. T05.9 Multiple traumatic amputations, unspecified
749. T06 Other injuries involving multiple body regions, not elsewhere classified
750. T06.0 Injuries of brain and cranial nerves with injuries of nerves and spinal cord at neck level
751. T06.1 Injuries of nerves and spinal cord involving other multiple body regions
752. T06.2 Injuries of nerves involving multiple body regions
753. T06.3 Injuries of blood vessels involving multiple body regions
754. T06.4 Injuries of muscles and tendons involving multiple body regions
755. T06.5 Injuries of intrathoracic organs with intra-abdominal and pelvic organs
756. T06.8 Other specified injuries involving multiple body regions
757. T07 Unspecified multiple injuries
758. T08 Fracture of spine, level unspecified
759. T09 Other injuries of spine and trunk, level unspecified
760. T09.0 Superficial injury of trunk, level unspecified
761. T09.1 Open wound of trunk, level unspecified
762. T09.2 Dislocation, sprain and strain of unspecified joint and ligament of trunk
763. T09.3 Injury of spinal cord, level unspecified
764. T09.4 Injury of unspecified nerve, spinal nerve root and plexus of trunk
765. T09.5 Injury of unspecified muscle and tendon of trunk
766. T09.6 Traumatic amputation of trunk, level unspecified
767. T09.8 Other specified injuries of trunk, level unspecified
768. T09.9 Unspecified injury of trunk, level unspecified
769. T10 Fracture of upper limb, level unspecified
770. T11 Other injuries of upper limb, level unspecified
771. T11.0 Superficial injury of upper limb, level unspecified
772. T11.1 Open wound of upper limb, level unspecified
773. T11.2 Dislocation, sprain and strain of unspecified joint and ligament of upper limb, level unspecified
774. T11.3 Injury of unspecified nerve of upper limb, level unspecified
775. T11.4 Injury of unspecified blood vessel of upper limb, level unspecified
776. T11.5 Injury of unspecified muscle and tendon of upper limb, level unspecified
777. T11.6 Traumatic amputation of upper limb, level unspecified
778. T11.8 Other specified injuries of upper limb, level unspecified
779. T11.9 Unspecified injury of upper limb, level unspecified
780. T12 Fracture of lower limb, level unspecified
781. T13 Other injuries of lower limb, level unspecified
782. T13.0 Superficial injury of lower limb, level unspecified
783. T13.1 Open wound of lower limb, level unspecified
784. T13.2 Dislocation, sprain and strain of unspecified joint and ligament of lower limb, level unspecified

785. T13.3 Injury of unspecified nerve of lower limb, level unspecified
786. T13.4 Injury of unspecified blood vessel of lower limb, level unspecified
787. T13.5 Injury of unspecified muscle and tendon of lower limb, level unspecified
788. T13.6 Traumatic amputation of lower limb, level unspecified
789. T13.8 Other specified injuries of lower limb, level unspecified
790. T13.9 Unspecified injury of lower limb, level unspecified
791. T14 Injury of unspecified body region
792. T14.0 Superficial injury of unspecified body region
793. T14.1 Open wound of unspecified body region
794. T14.2 Fracture of unspecified body region
795. T14.3 Dislocation, sprain and strain of unspecified body region
796. T14.4 Injury of nerve(s) of unspecified body region
797. T14.5 Injury of blood vessel(s) of unspecified body region
798. T14.6 Injury of muscles and tendons of unspecified body region
799. T14.7 Crushing injury and traumatic amputation of unspecified body region
800. T14.8 Other injuries of unspecified body region
801. T14.9 Injury, unspecified
802. T15 Foreign body on external eye
803. T15.0 Foreign body in cornea
804. T15.1 Foreign body in conjunctival sac
805. T15.8 Foreign body in other and multiple parts of external eye
806. T15.9 Foreign body on external eye, part unspecified
807. T16 Foreign body in ear
808. T17 Foreign body in respiratory tract
809. T17.0 Foreign body in nasal sinus
810. T17.1 Foreign body in nostril
811. T17.2 Foreign body in pharynx
812. T17.3 Foreign body in larynx
813. T17.4 Foreign body in trachea
814. T17.5 Foreign body in bronchus
815. T17.8 Foreign body in other and multiple parts of respiratory tract
816. T17.9 Foreign body in respiratory tract, part unspecified
817. T18 Foreign body in alimentary tract
818. T18.0 Foreign body in mouth
819. T18.1 Foreign body in oesophagus
820. T18.2 Foreign body in stomach
821. T18.3 Foreign body in small intestine
822. T18.4 Foreign body in colon
823. T18.5 Foreign body in anus and rectum
824. T18.8 Foreign body in other and multiple parts of alimentary tract
825. T18.9 Foreign body in alimentary tract, part unspecified
826. T19 Foreign body in genitourinary tract
827. T19.0 Foreign body in urethra

828. T19.1 Foreign body in bladder
829. T19.2 Foreign body in vulva and vagina
830. T19.3 Foreign body in uterus [any part]
831. T19.8 Foreign body in other and multiple parts of genitourinary tract
832. T19.9 Foreign body in genitourinary tract, part unspecified
833. T20 Burn and corrosion of head and neck
834. T20.0 Burn of unspecified degree of head and neck
835. T20.1 Burn of first degree of head and neck
836. T20.2 Burn of second degree of head and neck
837. T20.3 Burn of third degree of head and neck
838. T20.4 Corrosion of unspecified degree of head and neck
839. T20.5 Corrosion of first degree of head and neck
840. T20.6 Corrosion of second degree of head and neck
841. T20.7 Corrosion of third degree of head and neck
842. T21 Burn and corrosion of trunk
843. T21.0 Burn of unspecified degree of trunk
844. T21.1 Burn of first degree of trunk
845. T21.2 Burn of second degree of trunk
846. T21.3 Burn of third degree of trunk
847. T21.4 Corrosion of unspecified degree of trunk
848. T21.5 Corrosion of first degree of trunk
849. T21.6 Corrosion of second degree of trunk
850. T21.7 Corrosion of third degree of trunk
851. T22 Burn and corrosion of shoulder and upper limb, except wrist and hand
852. T22.0 Burn of unspecified degree of shoulder and upper limb, except wrist and hand
853. T22.1 Burn of first degree of shoulder and upper limb, except wrist and hand
854. T22.2 Burn of second degree of shoulder and upper limb, except wrist and hand
855. T22.3 Burn of third degree of shoulder and upper limb, except wrist and hand
856. T22.4 Corrosion of unspecified degree of shoulder and upper limb, except wrist and hand
857. T22.5 Corrosion of first degree of shoulder and upper limb, except wrist and hand
858. T22.6 Corrosion of second degree of shoulder and upper limb, except wrist and hand
859. T22.7 Corrosion of third degree of shoulder and upper limb, except wrist and hand
860. T23 Burn and corrosion of wrist and hand
861. T23.0 Burn of unspecified degree of wrist and hand
862. T23.1 Burn of first degree of wrist and hand
863. T23.2 Burn of second degree of wrist and hand
864. T23.3 Burn of third degree of wrist and hand
865. T23.4 Corrosion of unspecified degree of wrist and hand
866. T23.5 Corrosion of first degree of wrist and hand
867. T23.6 Corrosion of second degree of wrist and hand
868. T23.7 Corrosion of third degree of wrist and hand
869. T24 Burn and corrosion of hip and lower limb, except ankle and foot
870. T24.0 Burn of unspecified degree of hip and lower limb, except ankle and foot

871. T24.1 Burn of first degree of hip and lower limb, except ankle and foot
872. T24.2 Burn of second degree of hip and lower limb, except ankle and foot
873. T24.3 Burn of third degree of hip and lower limb, except ankle and foot
874. T24.4 Corrosion of unspecified degree of hip and lower limb, except ankle and foot
875. T24.5 Corrosion of first degree of hip and lower limb, except ankle and foot
876. T24.6 Corrosion of second degree of hip and lower limb, except ankle and foot
877. T24.7 Corrosion of third degree of hip and lower limb, except ankle and foot
878. T25 Burn and corrosion of ankle and foot
879. T25.0 Burn of unspecified degree of ankle and foot
880. T25.1 Burn of first degree of ankle and foot
881. T25.2 Burn of second degree of ankle and foot
882. T25.3 Burn of third degree of ankle and foot
883. T25.4 Corrosion of unspecified degree of ankle and foot
884. T25.5 Corrosion of first degree of ankle and foot
885. T25.6 Corrosion of second degree of ankle and foot
886. T25.7 Corrosion of third degree of ankle and foot
887. T26 Burn and corrosion confined to eye and adnexa
888. T26.0 Burn of eyelid and periocular area
889. T26.1 Burn of cornea and conjunctival sac
890. T26.2 Burn with resulting rupture and destruction of eyeball
891. T26.3 Burn of other parts of eye and adnexa
892. T26.4 Burn of eye and adnexa, part unspecified
893. T26.5 Corrosion of eyelid and periocular area
894. T26.6 Corrosion of cornea and conjunctival sac
895. T26.7 Corrosion with resulting rupture and destruction of eyeball
896. T26.8 Corrosion of other parts of eye and adnexa
897. T26.9 Corrosion of eye and adnexa, part unspecified
898. T27 Burn and corrosion of respiratory tract
899. T27.0 Burn of larynx and trachea
900. T27.1 Burn involving larynx and trachea with lung
901. T27.2 Burn of other parts of respiratory tract
902. T27.3 Burn of respiratory tract, part unspecified
903. T27.4 Corrosion of larynx and trachea
904. T27.5 Corrosion involving larynx and trachea with lung
905. T27.6 Corrosion of other parts of respiratory tract
906. T27.7 Corrosion of respiratory tract, part unspecified
907. T28 Burn and corrosion of other internal organs
908. T28.0 Burn of mouth and pharynx
909. T28.1 Burn of oesophagus
910. T28.2 Burn of other parts of alimentary tract
911. T28.3 Burn of internal genitourinary organs
912. T28.4 Burn of other and unspecified internal organs
913. T28.5 Corrosion of mouth and pharynx

914. T28.6 Corrosion of oesophagus
915. T28.7 Corrosion of other parts of alimentary tract
916. T28.8 Corrosion of internal genitourinary organs
917. T28.9 Corrosion of other and unspecified internal organs
918. T29 Burns and corrosions of multiple body regions
919. T29.0 Burns of multiple regions, unspecified degree
920. T29.1 Burns of multiple regions, no more than first-degree burns mentioned
921. T29.2 Burns of multiple regions, no more than second-degree burns mentioned
922. T29.3 Burns of multiple regions, at least one burn of third degree mentioned
923. T29.4 Corrosions of multiple regions, unspecified degree
924. T29.5 Corrosions of multiple regions, no more than first-degree corrosions mentioned
925. T29.6 Corrosions of multiple regions, no more than second-degree corrosions mentioned
926. T29.7 Corrosions of multiple regions, at least one corrosion of third degree mentioned
927. T30 Burn and corrosion, body region unspecified
928. T30.0 Burn of unspecified body region, unspecified degree
929. T30.1 Burn of first degree, body region unspecified
930. T30.2 Burn of second degree, body region unspecified
931. T30.3 Burn of third degree, body region unspecified
932. T30.4 Corrosion of unspecified body region, unspecified degree
933. T30.5 Corrosion of first degree, body region unspecified
934. T30.6 Corrosion of second degree, body region unspecified
935. T30.7 Corrosion of third degree, body region unspecified
936. T31 Burns classified according to extent of body surface involved
937. T31.0 Burns involving less than 10% of body surface
938. T31.1 Burns involving 10-19% of body surface
939. T31.2 Burns involving 20-29% of body surface
940. T31.3 Burns involving 30-39% of body surface
941. T31.4 Burns involving 40-49% of body surface
942. T31.5 Burns involving 50-59% of body surface
943. T31.6 Burns involving 60-69% of body surface
944. T31.7 Burns involving 70-79% of body surface
945. T31.8 Burns involving 80-89% of body surface
946. T31.9 Burns involving 90% or more of body surface
947. T32 Corrosions classified according to extent of body surface involved
948. T32.0 Corrosions involving less than 10% of body surface
949. T32.1 Corrosions involving 10-19% of body surface
950. T32.2 Corrosions involving 20-29% of body surface
951. T32.3 Corrosions involving 30-39% of body surface
952. T32.4 Corrosions involving 40-49% of body surface
953. T32.5 Corrosions involving 50-59% of body surface
954. T32.6 Corrosions involving 60-69% of body surface
955. T32.7 Corrosions involving 70-79% of body surface
956. T32.8 Corrosions involving 80-89% of body surface

957. T32.9 Corrosions involving 90% or more of body surface
958. T33 Superficial frostbite
959. T33.0 Superficial frostbite of head
960. T33.1 Superficial frostbite of neck
961. T33.2 Superficial frostbite of thorax
962. T33.3 Superficial frostbite of abdominal wall, lower back and pelvis
963. T33.4 Superficial frostbite of arm
964. T33.5 Superficial frostbite of wrist and hand
965. T33.6 Superficial frostbite of hip and thigh
966. T33.7 Superficial frostbite of knee and lower leg
967. T33.8 Superficial frostbite of ankle and foot
968. T33.9 Superficial frostbite of other and unspecified sites
969. T34 Frostbite with tissue necrosis
970. T34.0 Frostbite with tissue necrosis of head
971. T34.1 Frostbite with tissue necrosis of neck
972. T34.2 Frostbite with tissue necrosis of thorax
973. T34.3 Frostbite with tissue necrosis of abdominal wall, lower back and pelvis
974. T34.4 Frostbite with tissue necrosis of arm
975. T34.5 Frostbite with tissue necrosis of wrist and hand
976. T34.6 Frostbite with tissue necrosis of hip and thigh
977. T34.7 Frostbite with tissue necrosis of knee and lower leg
978. T34.8 Frostbite with tissue necrosis of ankle and foot
979. T34.9 Frostbite with tissue necrosis of other and unspecified sites
980. T35 Frostbite involving multiple body regions and unspecified frostbite
981. T35.0 Superficial frostbite involving multiple body regions
982. T35.1 Frostbite with tissue necrosis involving multiple body regions
983. T35.2 Unspecified frostbite of head and neck
984. T35.3 Unspecified frostbite of thorax, abdomen, lower back and pelvis
985. T35.4 Unspecified frostbite of upper limb
986. T35.5 Unspecified frostbite of lower limb
987. T35.6 Unspecified frostbite involving multiple body regions
988. T35.7 Unspecified frostbite of unspecified site
989. T74 Maltreatment syndromes
990. T74.0 Neglect or abandonment
991. T74.1 Physical abuse
992. T74.2 Sexual abuse
993. T74.3 Psychological abuse
994. T74.8 Other maltreatment syndromes
995. T74.9 Maltreatment syndrome, unspecified

- **3. Injury source data** Source: Medical record

    1. Medical record
    2. X-ray report

3. CT-report
4. Surgical notes

- **4. AIS code**
- **Any other comments?**

18.2.13 Individual mortality status

- **1. Is the patient dead?** Source: Medical record or interview

1. Yes
2. No

- **2. Date of death** Source: Medical record or interview
- **3. Time of death**
- **Any other comments?**

18.2.14 Quality of life (EQ5D5L)

- **1. Date of filling this form**
- **2. First, I would like to ask you about MOBILITY. Would you say that:** Requires opt-in consent, not routinely recorded. Source: Interview

1. You have no problems in walking about?
2. You have slight problems in walking about?
3. You have moderate problems in walking about?
4. You have severe problems in walking about?
5. You are unable to walk about?

- **3. Next, I would like to ask you about SELF-CARE. Would you say that:** Requires opt-in consent, not routinely recorded. Source: Interview

1. You have no problems washing or dressing yourself?
2. You have slight problems washing or dressing yourself?
3. You have moderate problems washing or dressing yourself?
4. You have severe problems washing or dressing yourself?
5. You are unable to wash or dress yourself?

- **4. Next, I would like to ask you about USUAL ACTIVITIES, for example, work, study, housework, family or leisure activities. Would you say that:** Requires opt-in consent, not routinely recorded. Source: Interview

1. You have no problems doing your usual activities?
2. You have slight problems doing your usual activities?
3. You have moderate problems doing your usual activities?

4. You have severe problems doing your usual activities?
5. You are unable to do your usual activities?

- **5. Next, I would like to ask you about PAIN OR DISCOMFORT. Would you say that:** Requires opt-in consent, not routinely recorded. Source: Interview

1. You have no pain or discomfort?
2. You have slight pain or discomfort?
3. You have moderate pain or discomfort?
4. You have severe pain or discomfort?
5. You have extreme pain or discomfort?

- **6. Finally, I would like to ask you about ANXIETY OR DEPRESSION. Would you say that:** Requires opt-in consent, not routinely recorded. Source: Interview

1. You are not anxious or depressed?
2. You are slightly anxious or depressed?
3. You are moderately anxious or depressed?
4. You are severely anxious or depressed?
5. You are extremely anxious or depressed?

- **I would now like you to tell me the point on this line where you would put your health TODAY.(Note to interviewer: mark the line at the point indicating the respondent's health today.)** Requires opt-in consent, not routinely recorded. Source: Interview
- **Any other comments?**

18.2.15 Disability (WHODAS 2.0)

- **Date of form filling**
- **1. Who are you interviewing?** Requires opt-in consent, not routinely recorded. Source: Interview

1. Patient participant
2. Patient representative

- **2. What is the relationship between the representative and the participant?** Requires opt-in consent, not routinely recorded. Source: Interview

1. Husband or wife
2. Parent
3. Son or daughter
4. Brother or sister
5. Other relative
6. Friend
7. Professional carer

8. Other (specify)

- **3. If other, please specify**
- **1. Standing for long periods such as 30 minutes?** Requires opt-in consent, not routinely recorded. Source: Interview

1. None
2. Mild
3. Moderate
4. Severe
5. Extreme or cannot do
6. None
7. Mild
8. Moderate
9. Severe
10. Extreme or cannot do

- **2. Taking care of your household responsibilities?** Requires opt-in consent, not routinely recorded. Source: Interview

1. None
2. Mild
3. Moderate
4. Severe
5. Extreme or cannot do

- **3. Learning a new task, for example, learning how to get to a new place?** Requires opt-in consent, not routinely recorded. Source: Interview

1. None
2. Mild
3. Moderate
4. Severe
5. Extreme or cannot do
6. None
7. Mild
8. Moderate
9. Severe
10. Extreme or cannot do

- **4. How much of a problem did you have joining in community activities (for example, festivities, religious or other activities) in the same way as anyone else can?** Requires opt-in consent, not routinely recorded. Source: Interview

1. None

2. Mild
3. Moderate
4. Severe
5. Extreme or cannot do

- **5. How much have you been emotionally affected by your health problems?** Requires opt-in consent, not routinely recorded. Source: Interview

1. None
2. Mild
3. Moderate
4. Severe
5. Extreme or cannot do

- **1. Concentrating on doing something for ten minutes?** Requires opt-in consent, not routinely recorded. Source: Interview

1. None
2. Mild
3. Moderate
4. Severe
5. Extreme or cannot do
6. None
7. Mild
8. Moderate
9. Severe
10. Extreme or cannot do

- **2. Walking a long distance such as a kilometre [or equivalent]?** Requires opt-in consent, not routinely recorded. Source: Interview

1. None
2. Mild
3. Moderate
4. Severe
5. Extreme or cannot do
6. None
7. Mild
8. Moderate
9. Severe
10. Extreme or cannot do

- **3. Washing your whole body?** Requires opt-in consent, not routinely recorded. Source: Interview

1. None

2. Mild
3. Moderate
4. Severe
5. Extreme or cannot do

- **4. Getting dressed?** Requires opt-in consent, not routinely recorded. Source: Interview

1. None
2. Mild
3. Moderate
4. Severe
5. Extreme or cannot do
6. None
7. Mild
8. Moderate
9. Severe
10. Extreme or cannot do

- **5. Dealing with people you do not know?** Requires opt-in consent, not routinely recorded. Source: Interview

1. None
2. Mild
3. Moderate
4. Severe
5. Extreme or cannot do

- **6. Maintaining a friendship?** Requires opt-in consent, not routinely recorded. Source: Interview

1. None
2. Mild
3. Moderate
4. Severe
5. Extreme or cannot do
6. None
7. Mild
8. Moderate
9. Severe
10. Extreme or cannot do

- **7. Your day-to-day work/school?** Requires opt-in consent, not routinely recorded. Source: Interview

1. None
2. Mild

3. Moderate
4. Severe
5. Extreme or cannot do

- **2. Taking care of his or her household responsibilities?** Requires opt-in consent, not routinely recorded. Source: Interview

1. None
2. Mild
3. Moderate
4. Severe
5. Extreme or cannot do

- **4. How much of a problem did he or she have joining in community activities (for example, festivities, religious or other activities) in the same way as anyone else can?** Requires opt-in consent, not routinely recorded. Source: Interview

1. None
2. Mild
3. Moderate
4. Severe
5. Extreme or cannot do

- **5. How much has your relative been emotionally affected by his or her health condition?** Requires opt-in consent, not routinely recorded. Source: Interview

1. None
2. Mild
3. Moderate
4. Severe
5. Extreme or cannot do

- **3. Washing his or her whole body?** Requires opt-in consent, not routinely recorded. Source: Interview

1. None
2. Mild
3. Moderate
4. Severe
5. Extreme or cannot do

- **5. Dealing with people he or she does not know?** Requires opt-in consent, not routinely recorded. Source: Interview

1. None
2. Mild

3. Moderate
4. Severe
5. Extreme or cannot do

- **7. His or her day-to-day work/school?** Requires opt-in consent, not routinely recorded. Source: Interview

1. None
2. Mild
3. Moderate
4. Severe
5. Extreme or cannot do

- **1. Overall, in the past 30 days, how many days were these difficulties present?** Requires opt-in consent, not routinely recorded. Source: Interview
- **2. In the past 30 days, for how many days were you totally unable to carry out your usual activities or work because of any health condition?** Requires opt-in consent, not routinely recorded. Source: Interview
- **3. In the past 30 days, not counting the days that you were totally unable, for how many days did you cut back or reduce your usual activities or work because of any health condition?** Requires opt-in consent, not routinely recorded. Source: Interview
- **Any other comments?**

18.2.16 Return to work

- **1. Date of form filling**
- **2. Did the participant return to work?**

1. Yes
2. No

- **3. Date and time of return to work** Requires opt-in consent, not routinely recorded. Source: Interview
- **4. Work status** Requires opt-in consent, not routinely recorded. Source: Interview

1. Paid work
2. Self-employed, such as own your business or farming
3. Non-paid work, such as volunteer or charity
4. Student
5. Keeping house/homemaker
6. Not known

- **Any other comments?**

18.2.17 Safety events

- **1. Is any safety event reported?**

    1. Yes
    2. No

- **2. Date reported to trial management team of safety event**
- **3. Type of safety event** Source: Medical record or treating physician

    1. Prolonged mechanical ventilation (> 7 days)
    2. Initiation of renal replacement therapy
    3. Prolonged (> 2 days) use of vasopressors such as norepinephrine or vasopressin
    4. Renewed (restart after at least 2 days without) use of vasopressors such as norepinephrine or vasopressin
    5. Other

- **4. Elaborate on other safety event** Source: Medical record or treating physician
- **5. Investigator assessment of safety event** Source: Investigator
- **Any other comments?**

18.2.18 End of study

- **1. What is the reason for the end of study?**

    1. Completed follow up
    2. Lost to follow up
    3. Death
    4. Discharge and no consent for follow up
    5. Opt-out from routinely recorded (in-hospital) data collection and no consent for follow-up
    6. Opt-out from routinely recorded (in-hospital) data collection and withdrawn consent for follow-up

- **2. Date of end of study**
- **Any other comments?**

18.2.19 Scheduling

- **Day 7**
- **Day 30**
- **Day 90**

18.2.20 Review

- **Status**
    1. Review pending
    2. Replies to queries pending
    3. Review done
- **Reviewed by**
    1. Prashant
    2. Debojit
    3. Sara
    4. Martin
- **Review comments** For example, comments about main quality issues, if any.

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49. ICMJE | Recommendations | Defining the Role of Authors and Contributors.

20 Appendices

20.1 Initial hospital screening instrument

Screening call

This form is for screening potentially eligible clusters for the ATLS vs standard care trial. Please fill it in while talking to the hospital representative. Thank you so much for helping with this task!

Please complete the questions below.

Synopsis Title Effects of Advanced Trauma Life Support® Training Compared to Standard Care on Adult Trauma Patient Outcomes: A Cluster Randomised Trial

Rationale Trauma is a massive global health issue. Many training programmes have been developed to help physicians in the initial management of trauma patients. Advanced Trauma Life Support® (ATLS®) is the most popular of these programmes and have been used to train over one million physicians worldwide. Despite its widespread use, there are no controlled trials showing that ATLS® improves patient outcomes. Multiple systematic reviews emphasise the need for such trials.

Aim To compare the effects of ATLS® training with standard care on outcomes in adult trauma patients.

Primary Outcome In-hospital mortality within 30 days of arrival at the emergency department.

Trial Design Batched stepped-wedge cluster randomised trial in India.

Trial Population Adult trauma patients presenting to the emergency department of a participating hospital.

Sample Size 30 clusters and 4320 patients.

Eligibility Criteria

Cluster will be hospitals with a baseline admission rate of at least 400 patients with trauma per year or 35 patients with trauma per month for at least the last six months, that provide emergency surgical and orthopaedic services around the clock, and where no more than 25% of initial trauma care providers trained in a formalised trauma life support training programme.

Patients will be at least 15 years old, who present to the emergency department of participating hospitals with a history of trauma occuring less than 48 hours before arrival, and who are admitted or die between and admission, or who are transferred from the emergency department of a participating hospital to another hospital for admission.

Intervention The intervention will be ATLS® training, a proprietary 2.5 day course teaching a standardised approach to trauma patient care using the concepts of a primary and secondary survey. Physicians will be trained in an accredited ATLS® training facility in India.

Ethical Considerations In-hospital data collection will be conducted under a waiver of informed consent. Patients will be informed about the trial and their right to opt out of data collection. Patients will be informed that they can withdraw their data from the trial at any time.

Trial Period 2024-10-01 to 2029-10-01

Hospital details

Hospital name and address

Eligibility assessment	
Does the hospital admit trauma patients?	◯ Yes ◯ No
Is the hospital representative interested in potentially participating?	◯ Yes ◯ No (Comment: _____)
If the hospital representative is not interested in participating then you can go ahead and submit the form.	
How many trauma patients aged 15 years or older, excluding patients with isolated limb injuries, are admitted each month?	◯ < 30 ◯ 30-60 ◯ > 60 ◯ Not sure (Comment: _____)
Does the hospital provide emergency surgery and orthopaedic services around the clock?	◯ Yes ◯ No (Comment: _____)
Out of the physicians involved in the initial resuscitation of trauma patients, are less than 25% trained in a formalised trauma life support training programme like ATLS or Primary Trauma Care?	◯ Yes ◯ No (Comment (like name of other training programme): _____)
Unfortunately, the hospital does not fulfil the eligibility criteria and you may go ahead and submit the form.	
The hospital fulfills the cluster eligibility criteria. Please enter the hospital representative's contact details: Name E-mail Phone number _____ _____ _____ Before you can proceed to fill in descriptive information about the potential cluster, please confirm that the contact details are complete: _____	

Cluster descriptive information	
Out of the patients with trauma who present to the emergency department, what percentage are referrals or transfers from other hospitals?	◯ < 20% ◯ 20-50% ◯ 51-80% ◯ > 80% ◯ Not sure (Comment: _____)
Out of the patients with trauma who are admitted, what percentage are transferred to other hospitals?	◯ < 10% ◯ 10-20% ◯ 21-30% ◯ >30% ◯ Not sure (Comment: _____)

Who performs the initial resuscitation of trauma patients as they arrive to the hospital?	☐ Casualty medical officers ☐ Emergency medicine residents ☐ Surgical residents ☐ Not sure (Comment: _____)
How many CMOs work in the emergency department? This question is here to help us estimate the number of people we will need to train.	◯ < 10 ◯ 10-20 ◯ 21-30 ◯ > 30 ◯ Not sure (Comment: _____)
How many emergency medicine/general surgery residents are admitted each year? This question is here to help us estimate the number of people we will need to train.	◯ < 10 ◯ 10-20 ◯ 21-30 ◯ > 30 ◯ Not sure (Comment: _____)
What specialities are available around the clock to care for trauma patients?	☐ General surgery ☐ Orthopaedics ☐ Neurosurgery ☐ Vascular surgery ☐ Interventional radiology ☐ Emergency medicine ☐ Not sure (Comment: _____)
What facilities are available around the clock?	☐ X-ray ☐ Ultrasound/FAST ☐ CT ☐ MRI ☐ Blood bank ☐ Not sure (Comment: _____)
How many beds does the hospital have?	◯ < 250 ◯ 250-500 ◯ 501-750 ◯ 751-1000 ◯ >1000 ◯ Not sure (Comment: _____)
How many ICU beds does the hospital have?	◯ No ICU beds ◯ 1-10 ◯ 11-20 ◯ 21-30 ◯ > 30 ◯ Not sure (Comment: _____)
How many dedicated trauma beds does the hospital have?	◯ No dedicated trauma beds ◯ 1-10 ◯ 11-20 ◯ 21-30 ◯ > 30 ◯ Not sure (Comment: _____)

Please submit the form once you have completed all questions above.

20.2 In-depth hospital screening interview instrument

Hospital screening interview for the ATLS vs standard care trial

This is the screening interview form for the Advanced Trauma Life Support® vs Standard Care trial planned by Karolinska Institutet along with The George Institute. You have expressed preliminary interest inparticipating in this trial. We are undertaking this hospital screening interview inorder to assess whether the study could be conducted at your hospital. We appreciate your efforts to answer as many of these questions as possible and we will follow up on your responses and any questions you may have in a separate call. Thank you!

Synopsis Effects of Advanced Trauma Life Support® Training Compared to Standard Care on Adult Trauma Patient Outcomes: A Cluster Randomised Trial Rationale Trauma is a massive global health issue. Many training programmes have been developed to help physicians in the initial management of trauma patients. Among these programmes, Advanced Trauma Life Support® (ATLS®) is the most popular, having trained over one million physicians worldwide. Despite its widespread use, there are no controlled trials showing that ATLS® improves patient outcomes. Multiple systematic reviews emphasise the need for such trials.

Aim To compare the effects of ATLS® training with standard care on outcomes in adult trauma patients.

Primary Outcome All-cause mortality within 30 days of arrival at the emergency department.

Trial Design Batched stepped-wedge cluster randomised trial in India.

Trial Population Adult trauma patients presenting to the emergency department of a participating hospital.

Sample Size 30 clusters and 4320 patients.

Eligibility Criteria

Hospitals are secondary or tertiary hospitals in India that admit or refer/transfer for admission at least 400 patients with trauma per year.

Clusters are one or more units of physicians providing initial trauma care in the emergency department of tertiary hospitals in India.

Patients participants are adult trauma patients who presents to the emergency department of participating hospitals and are admitted or transferred for admission.

Intervention The intervention will be ATLS® training, a proprietary 2.5 day course teaching a standardised approach to trauma patient care using the concepts of a primary and secondary survey. Physicians will be trained in an accredited ATLS® training facility in India.

Ethical Considerations We will use an opt-out consent approach, in which consent is presumed unless actively declined. Note that consent here refers to consent to data collection, as it will not be possible for patients to opt out from being subjected to the intervention. This approach is justified because the trial can be considered to involve only minimal risk and the data

collection is non-invasive and mostly involve extracting routinely collected data from medical records. Patient participants will be informed about the study and their right to opt out once they are admitted or telephonically if they are transferred. Patients will be informed that they can withdraw their data from the trial at any time before final analysis of the data.

Trial Period October 1, 2024, to September 30, 2029

Hospital details

[hospital_address]

Contact person details

Name [contact_name]
E-mail [contact_email]
Phone number [contact_phone_number]

Will you [contact_name] also be the site investigator?	◯ Yes ◯ No

Investigator details

Please enter the name and contact details of the site investigator

Name _____
E-mail _____
Phone number _____

Please enter these additional investigator details

Designation _____
Specialization _____
State Medical Council registration number _____

Is the investigator trained in International Council for Harmonisation, Guideline for Good Clinical Practice (ICH GCP)?	◯ Yes ◯ No
Will there be a co-investigator at your site?	◯ Yes ◯ No
Will you [contact_name] be the co-investigator?	◯ Yes ◯ No

Please enter the name and contact details of the site investigator

Name _____
E-mail _____
Phone number _____

Please enter these additional co-investigator details Designation _____ Specialization _____ State Medical Council registration number _____	
Is the co-investigator trained in International Council for Harmonisation, Guideline for Good Clinical Practice (ICH GCP)?	◯ Yes ◯ No

Ethical review details

Does your hospital have an ethics committee registered with CDSCO?	◯ Yes ◯ No
Please enter the ethics committee registration number	_____
In the next three months when is your IEC meeting up?	_____
What is the expected timeline for ethics review at your site?	_____
In which languages do you think the consent form should be translated, considering the languages spoken by the potential participants treated at your hospital?	_____

Departmental logistics

Does your hospital require any additional departmental review besides the ethics? Can you please elaborate on that review?	_____
Are there any potential logistical issues which may interfere with set up or running ofthis project at your site?	_____ (E.g. contract review, adequate space, lack of resources etc.)
What is the expected timeline for contract review, negotiations and execution (in days)?	_____
What is the maximum expected timeline?	_____

Initial trauma care	
How do patients typically arrive to your hospital?	
Who are involved in the management of trauma patients in the emergency department?	
What happens when additional expertise is needed?	
What is the role of the casualty medical officers?	
Are physicians organised in units?	
How big are those units?	
How are those units composed in terms of residents and faculty?	
How often do the units rotate?	
How many units are there working in the emergency department?	
How many trauma patients aged 15 years or older are admitted per day, excluding patients with isolated limb injuries and those who are admitted directly to the ward?	

Intervention and patient inclusion	
How many patients do you think you could include in to the proposed trial per month? (We need to include at least 12 patients per month)	
What is the basis of your patient enrollment estimate?	(For example database review, emergency department record, review of patient records, other)
Do you see any problems with including 12 patients per month at your site? Can you please elaborate on those problems?	

All hospitals in this trial will receive the intervention. The intervention is that we will train approximately 10 physicians providing initial trauma care in ATLS in your hospital. Who do you think we should train to maximise the effect?	_____ (Surgical residents? Emergency medicine residents? Casualty medical officers? Someone else??)
The time point when the training will be implemented will be randomised, but there will be a minimum of three months between the start of the data collection and the training. The training will happen during a one month long "transition period". How long notice do you need to plan the participation of the physicians from your hospital?	_____
Are you aware of any plans to train providers in any formalised trauma life support training programme during the next few years?	_____
Are you aware of any plans to implement other interventions or changes that may radically change how you treat trauma patients at your site?	_____ (For example building a trauma centre, building a new emergency department, shifting the CT)
If we would like to visit your hospital to observe trauma care delivery in the emergency department and talk to providers, how can that be arranged?	_____
General	
How are the patient medical records organised at your site?	☐ Hard-copy ☐ Electronic ☐ Not sure
Do you currently have any competing studies or are you committed to new competing studies?	_____
Do you have access to a computer with high-speed internet access?	◯ Yes ◯ No
Do you currently have the necessary study team including research coordinator and co-investigators to conduct this study? Can you please elaborate on the composition and experience of that team?	_____
What are your expectations of this trial?	_____
Do you have any questions or comments regarding this trial?	_____
Are you interested in participating in this trial?	◯ Yes ◯ No

If you have any questions, feel free to contact Martin Gerdin Wärnberg (martin.gerdin@ki.se), Monty Khajanchi (monta32@gmail.com) or Samriddhi Ranjan (sranjan@georgeinstitute.org.in)

20.3 Case Record Form

Confidential

Screening V1.0.01.10.24

Screening ID	_____
1. Date of screening	_____
2. Date of data entry	_____

Inclusion criteria	
1. Is the patient at least 15 years old?	(Source: Medical record or interview)
2. Did the patient present with a history of trauma defined as having any of the reasons listed in the International Classification of Diseases chapter XX as the reason for presenting? Please see https://icd.who.int/browse10/2019/en#/XX for a complete list of ICD-10 codes	(Source: Medical record or interview)
3. Did the trauma occur less than 48 hours before arrival to the hospital?	(Source: Medical record or interview)
4. Was the patient admitted?	(Source: Medical record)
5. Did the patient die after arrival but before admission?	(Source: Medical record)
6. Was the patient transferred to another hospital for admission?	(Source: Medical record)

Exclusion criteria	
1. Did the patient present with isolated closed extremity fracture?	(Source: Medical record)
2. Was the patient directly admitted to a ward without being seen by a physician in the emergency department?	(Source: Medical record)

Eligibility	
The patient is not eligible for inclusion.	
Is the patient eligible? (Eligible=1; Not Eligible=0)	_____
Any other comments?	_____

Confidential

Consent V1.0.01.10.24

Study Consent

In this trial, consent refers to consent for data collection. It is not possible for patients to opt out from being subjected to the intervention, as the intervention is delivered at the cluster level. Patient participants will be included in this trial under the following modes of consent:
• Opt-out consent for collection of routinely recorded data
• Opt-in consent and assent for non-routinely recorded data, including but not restricted to Quality of Life (EQ5D5L), Disability (WHODAS 2.0) and Return to Work.
• Waiver of informed consent for patients who are unconscious or otherwise unable to provide consent and do not have a legally acceptable representative.
When possible, all patient participants must be approached and provided with information about the study, the option to opt out, and consent for collection of non-routinely recorded data.

Section I: Consent Wavier

Please note that the consent for the collection of the routinely recorded data (in-hospital) will be presumed unless actively declined by the participant/ legally acceptable representative (LAR), using the opt-out form. Information for all forms except for baseline characteristics (marital and work status, education and income), follow-up (Quality of Life (EQ5D5L), Disability (WHODAS 2.0) and Return to Work) will be presumed, unless opted-out.

1. Is this patient included under the waiver of informed consent because the patient is unconscious or otherwise unable to provide consent and do not have a legally acceptable representative?

Section II: Opt in consent for follow up data collection

1. Did the participant/ or legally acceptable representative (LAR) provided consent for collection of non-routinely recorded data

2. Who gave consent for collection of non-routinely recorded data?

3. What is the LAR's relationship with the participant? _____

4. Why was Legally acceptable representative (LAR) approached for consent for collection of non-routinely recorded data?

5. Date when participant or legally acceptable representative (LAR) gave consent for collection of non-routinely recorded data. _____

6. How did the participant or legally acceptable representative (LAR) consent for collection of non-routinely recorded data?

7. Reconsenting done?

7.1 Date when the participant was reconsented?	_____
7.2 Why consenting was not done?	_____

Section III: Assent form	
1. Did the minor give assent for collection of non-routinely recorded data?	
2. Date when the minor gave assent for collection of non-routinely recorded data.	_____
3. In case the minor refused to participate, date when minor refused	_____

Section IV: Opt out form	
1. Is the participant or LAR wants to opt out from study?	
2. Who opted-out of the routinely recorded data (in-hospital)?	
3. Date when participant or legally acceptable representative (LAR) opted-out.	_____
4. Did the participant or legally acceptable representative (LAR) suggested to delete all the previously recorded data?	
Any other comments?	_____

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Consent_Withdrawn V1.0.01.10.24

Consent withdrawal	
1. Does the participant or legally acceptable representative (LAR) want to withdraw the consent?	
2. Date of consent withdrawal for follow-up data collection.	_____
3. Procedure(s) for which consent has been withdrawn	
Any other comments?	_____

Baseline V1.0.01.10.24

1. Age in years	_____ (Source: Medical record of interview)
2. Sex	(Source: Medical record of interview)
3. Current marital status	(Requires opt-in consent, not routinely recorded. Source: Interview)
4. Education level	(Requires opt-in consent, not routinely recorded. Source: Interview)
5. If other, please specify	_____ (Requires opt-in consent, not routinely recorded. Source: Interview)
6. Main work status	(Requires opt-in consent, not routinely recorded. Source: Interview)
7. If other, please specify	_____ (Requires opt-in consent, not routinely recorded. Source: Interview)
8. Income level in INR per month	(Requires opt-in consent, not routinely recorded. Source: Interview)
9. Mechanism of injury description	_____
10. Mechanism of injury ICD 10 code	(Coded using ICD 10. Please see https://icd.who.int/browse10/2019/en#/XX for a complete list of ICD-10 codes. Source: Medical record)

11. Clinical Frailty Scale Very Fit - People who are robust, active, energetic and motivated. These people commonly exercise regularly. They are among the fittest for their age. Fit - People who have no active disease symptoms but are less fit than category 1. Often, they exercise or are very active occasionally, e.g. seasonally. Managing well - People whose medical problems are well controlled, but are not regularly active beyond routine walking. Living with very mild frailty - While not dependent on others for daily help, often symptoms limit activities. A common complaint is being "slowed up", and/or being tired during the day. Living with mild frailty - These people often have more evident slowing, and need help in high order IADLs (finances, transportation, heavy housework, medications). Typically, mild frailty progressively impairs shopping and walking outside alone, meal preparation and housework. Living with moderate frailty - People need help with all outside activities and with keeping house. Inside, they often have problems with stairs and need help with bathing and might need minimal assistance (cuing, standby) with dressing. Living with severe frailty - Completely dependent for personal care, from whatever cause (physical or cognitive). Even so, they seem stable and not at high risk of dying (within ~ 6 months). Living with very severe frailty - Completely dependent, approaching the end of life. Typically, they could not recover even from a minor illness. Terminally ill - Approaching the end of life. This category applies to people with a life expectancy < 6 months, who are not otherwise evidently frail.	(Source: Medical record or treating physician)

12. Comorbidities (Charlson Comorbidity Index) Myocardial infarction - History of definite or probable MI (EKG changes and/or enzyme changes) Congestive heart failure - Exertional or paroxysmal nocturnal dyspnea and has responded to digitalis, diuretics, or afterload reducing agents Peripheral vascular disease - Intermittent claudication or past bypass for chronic arterial insufficiency, history of gangrene or acute arterial insufficiency, or untreated thoracic or abdominal aneurysm (≥6 cm) Cerebrovascular disease - History of a cerebrovascular accident with minor or no residua and transient ischemic attacks Dementia - Chronic cognitive deficit Chronic pulmonary disease - Asthma, chronic bronchitis, emphysema, and other lung disease who have ongoing symptoms such as dyspnea or cough, with mild or moderate activity. Peptic ulcer disease - Any history of treatment for ulcer disease or history of ulcer bleeding Liver disease - Severe = cirrhosis and portal hypertension with variceal bleeding history, moderate = cirrhosis and portal hypertension but no variceal bleeding history, mild = chronic hepatitis (or cirrhosis without portal hypertension) Renal disease - Severe = on dialysis, status post kidney transplant, uremia, moderate = creatinine >3 mg/dL (0.27 mmol/L)	(Source: Medical record, treating physician or interview)
12.1 Please specify	_____
13. Severity of liver disease	(Source: Medical record, treating physician or interview)
14. Severity of diabetes	(Source: Medical record, treating physician or interview)
15. Severity of malignancy	(Source: Medical record, treating physician or interview)
Any other comments?	_____

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Prehospital V1.0.01.10.24

1. Date of injury	_____ (Source: Medical record of interview)
1.1 Time of injury	_____
2. Mode of transport to the participating hospital	(Source: Medical record of interview)
2.1 If other, please specify	_____ (Source: Medical record of interview)
3. Referred or transferred to the participating hospital from another hospital	(Source: Medical record of interview)
4. Date of referring or transferring	_____
5. Time of referring or transferring	_____
Any other comments?	_____

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ATLS adherence V1.1.22.10.24

ATLS adherence checklist

Airway	
1. Airway patency checked	(Source: Observation)
Breathing	
1. Chest wall palpated	(Source: Observation)
2. Breath sounds checked	(Source: Observation)
3. Respiratory rate measured	(Source: Observation)
4. Saturation (SpO2) measured	(Source: Observation)
Circulation	
1. Heart rate measured	(Source: Observation)
2. Blood pressure measured	(Source: Observation)
3. Abdomen palpated	(Source: Observation)
4. Thighs palpated	(Source: Observation)
5. IV access obtained	(Source: Observation)
Disability	
1. GCS checked	(Source: Observation)
2. Pupils checked	(Source: Observation)
Exposure	
1. Patients exposed for assessment	
2. Temperature measured	(Source: Observation)
3. Interventions and adjuncts performed according to ATLS	

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Airway interventions	
1. Which airway interventions were performed?	(Source: Observation)
2. If other airway interventions given, specify	_____
3. Were airway interventions performed while minimising c-spine movement?	(Source: Observation)
Breathing interventions	
1. Which breathing interventions were performed?	(Source: Observation)
2. If other breathing Interventions done, specify	_____
Circulation interventions	
1. Which circulation interventions and adjuncts were performed?	(Source: Observation)
2. If other circulation Interventions done, specify	_____
Disability interventions	
1. Which disability intervention was performed?	(Source: Observation)
2. If other disability interventions done, specify	_____
Exposure interventions	
1. Which exposure intervention was performed?	(Source: Observation)
2. If other exposure interventions done, specify	_____
Any other comments?	_____

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Emergency Department V1.0.01.10.24

1. Date of arrival to the emergency department at the participating hospital.	_____ (Source: Medical record of interview)
2 Time of arrival to the emergency department at the participating hospital	_____
3. First recorded systolic blood pressure (mmHg)	_____ (Source: Medical record)
4. First recorded diastolic blood pressure (mmHg)	_____ (Source: Medical record)
5. First recorded heart rate (beats per minute)	_____ (Source: Medical record)
6. First recorded respiratory rate (breaths per minute)	_____ (Source: Medical record)
7. First recorded Glasgow Coma Scale	_____ (Source: Medical record)
8. First recorded body temperature (°F)	_____ (Source: Medical record)
9. First recorded oxygen saturation (%)	_____ (Source: Medical record)
10. Emergency department disposition	(Source: Medical record)
10.1 Date of death	_____
10.2 Time of death	_____
11. If other, please specify	_____ (Source: Medical record)
12. Date and time of referral or transfer for admission	_____ (Source: Medical record)
Any other comments?	_____

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Hospital V1.0.01.10.24

1. Date of admission to the participating hospital	_____ (Source: Medical record)
2 Time of admission to the participating hospital	_____ (Source: Medical record)
3. Type of admitting ward	(Source: Medical record)
4. Ward name or number	_____ (Source: Medical record)
5. Admitted to intensive care unit during admission	(Source: Medical record)
6. Date of first intensive care unit admission	_____ (Source: Medical record)
7. Time of first intensive care unit admission	_____ (Source: Medical record)
8. Was the participant discharged from intensive care unit?	
9. Date of first intensive care unit discharge	_____ (Source: Medical record)
10. Time of first intensive care unit discharge	_____ (Source: Medical record)
11. Hospital disposition	(Source: Medical record)
12. Was the patient transferred to another hospital for admission?	(Source: Medical record)
13. Date of discharge or transfer from participating hospital	_____ (Source: Medical record)
14. Time of discharge or transfer from participating hospital	_____ (Source: Medical record)
Any other comments?	_____

Surgery V1.0.01.10.24

1. Surgery done?	
2. Date of surgical procedure	_____ (A surgical procedure is defined as any procedure performed in the operating room, interventional dropdownlogy suite, or at the bedside, requiring general or regional anesthesia. Source: Medical record)
3. Time of surgical procedure	_____ (A surgical procedure is defined as any procedure performed in the operating room, interventional dropdownlogy suite, or at the bedside, requiring general or regional anesthesia. Source: Medical record)
4. Preoperative ASA score	(Source: Medical record or treating physician)
5. Description of procedure	_____ (Source: Medical record)
6. Procedure coded according to SNOMED CT	_____ (Source: Medical record)
Any other comments?	_____

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Imaging V1.0.01.10.24

1. Date and time of imaging	_____ (Source: Medical record)
2 Time of imaging	_____ (Source: Medical record)
3. Type of imaging	(Source: Medical record)
Any other comments?	_____

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Transfusion V1.0.01.10.24

1. Transfusion done?	
2. Date of transfusion	_____ (Source: Medical record)
3. Time of transfusion	_____ (Source: Medical record)
4. Type of blood product	(Source: Medical record)
5 Other specify	_____
6. Number of units transfused	_____ (Source: Medical record)
Any other comments?	_____

Injury V1.0.01.10.24

1. Injury description	_____ (Source: Medical record)
2. Injury ICD 10 code	(Coded using ICD 10. Source: Medical record)
3. Injury source data	(Source: Medical record)
4. AIS code	_____
Any other comments?	_____

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Individual Mortality Status V1.0.01.10.24

1. Is the patient dead?	(Source: Medical record or interview)
2. Date of death	_____ (Source: Medical record or interview)
3. Time of death	_____
Any other comments?	_____

Quality of Life (EQ5D5L)

Health Questionnaire

English version

VERSION FOR INTERVIEWER ADMINISTRATION

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Note to interviewer: although allowance should be made for the interviewer's particular style of speaking, the wording of the questionnaire instructions should be followed as closely as possible. In the case of the EQ-5D-5L descriptive system of the questionnaire, the precise wording must be followed.

If the respondent has difficulty choosing a response or asks for clarification, the interviewer should repeat the question word for word and ask the respondent to answer in a way that most closely resembles his or her thoughts about his or her health today.

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INTRODUCTION

(Note to interviewer: please read the following to the respondent.)

We are trying to find out what you think about your health. I will explain what to do as I go along, but please interrupt me if you do not understand something or if things are not clear to you. There are no right or wrong answers. We are interested only in your personal view.

First, I am going to read out some questions. Each question has a choice of five answers. Please tell me which answer best describes your health TODAY.

Do not choose more than one answer in each group of questions.

(Note to interviewer: first read all five options for each question. Then ask the respondent to choose which one applies to him/herself. Repeat the question and options if necessary. Mark the appropriate box under each heading. You may need to remind the respondent regularly that the timeframe is TODAY.)

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EQ-5D DESCRIPTIVE SYSTEM

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1. Date of filling this form

_____

2. First, I would like to ask you about MOBILITY. Would you say that:
(Requires opt-in consent, not routinely recorded. Source: Interview)

◯ You have no problems in walking about?
◯ You have slight problems in walking about?
◯ You have moderate problems in walking about?
◯ You have severe problems in walking about?
◯ You are unable to walk about?

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3. Next, I would like to ask you about SELF-CARE. Would you say that:
(Requires opt-in consent, not routinely recorded. Source: Interview)

◯ You have no problems washing or dressing yourself?
◯ You have slight problems washing or dressing yourself?
◯ You have moderate problems washing or dressing yourself?
◯ You have severe problems washing or dressing yourself?
◯ You are unable to wash or dress yourself?

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4. Next, I would like to ask you about USUAL ACTIVITIES, for example, work, study, housework, family or leisure activities. Would you say that:
(Requires opt-in consent, not routinely recorded. Source: Interview)

◯ You have no problems doing your usual activities?
◯ You have slight problems doing your usual activities?
◯ You have moderate problems doing your usual activities?
◯ You have severe problems doing your usual activities?
◯ You are unable to do your usual activities?

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5. Next, I would like to ask you about PAIN OR DISCOMFORT. Would you say that:
(Requires opt-in consent, not routinely recorded. Source: Interview)

◯ You have no pain or discomfort?
◯ You have slight pain or discomfort?
◯ You have moderate pain or discomfort?
◯ You have severe pain or discomfort?
◯ You have extreme pain or discomfort?

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6. Finally, I would like to ask you about ANXIETY OR DEPRESSION. Would you say that:
(Requires opt-in consent, not routinely recorded. Source: Interview)

◯ You are not anxious or depressed?
◯ You are slightly anxious or depressed?
◯ You are moderately anxious or depressed?
◯ You are severely anxious or depressed?
◯ You are extremely anxious or depressed?

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EQ-5D VAS

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- Now, I would like to ask you to say how good or bad your health is TODAY.

- I would like you to picture in your mind a vertical line that is numbered from 0 to 100.
(Note to interviewer: if interviewing face-to-face, please show the respondent the VAS line.)

- 100 at the top of the line means the best health you can imagine.
0 at the bottom of the line means the worst health you can imagine.

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- I would now like you to tell me the point on this line where you would put your health TODAY. (Note to interviewer: mark the line at the point indicating the respondent's health today.)	0 - The worst health you can imagine	50	100 - The best health you can imagine

(Place a mark on the scale above)

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Any other comments?

Disability (WHODAS 2.0)

Date of form filling	_____
1. Who are you interviewing?	(Requires opt-in consent, not routinely recorded. Source: Interview)
2. What is the relationship between the representative and the participant?	(Requires opt-in consent, not routinely recorded. Source: Interview)
3. If other, please specify	_____

Instructions to the interviewer are written in bold - do not read these aloud.

Text for the respondent to hear is written in italic print in blue. Read this text aloud.

Say to respondent:
The interview is about difficulties people have because of health conditions.

By health condition I mean diseases or illnesses, or other health problems that may be short or long lasting; injuries; mental or emotional problems; and problems with alcohol or drugs.

Remember to keep all of your health problems in mind as you answer the questions. When I ask you about difficulties in doing an activity think about...

- Increased effort
- Discomfort or pain
- Slowness
- Changes in the way you do the activity When answering, I'd like you to think back over the past 30 days. I would also like you to answer these questions thinking about how much difficulty you have had, on average, over the past 30 days, while doing the activity as you usually do it.

Use this scale when responding: None, mild, moderate, severe, extreme or cannot do.

In the past 30 days, how much difficulty did you have in:

1. Standing for long periods such as 30 minutes?	(Requires opt-in consent, not routinely recorded. Source: Interview)
2. Taking care of your household responsibilities?	(Requires opt-in consent, not routinely recorded. Source: Interview)
3. Learning a new task, for example, learning how to get to a new place?	(Requires opt-in consent, not routinely recorded. Source: Interview)

4. How much of a problem did you have joining in community activities (for example, festivities, religious or other activities) in the same way as anyone else can?	(Requires opt-in consent, not routinely recorded. Source: Interview)
5. How much have you been emotionally affected by your health problems?	(Requires opt-in consent, not routinely recorded. Source: Interview)

In the past 30 days, how much difficulty did you have in:	
1. Concentrating on doing something for ten minutes?	(Requires opt-in consent, not routinely recorded. Source: Interview)
2. Walking a long distance such as a kilometre [or equivalent]?	(Requires opt-in consent, not routinely recorded. Source: Interview)
3. Washing your whole body?	(Requires opt-in consent, not routinely recorded. Source: Interview)
4. Getting dressed?	(Requires opt-in consent, not routinely recorded. Source: Interview)
5. Dealing with people you do not know?	(Requires opt-in consent, not routinely recorded. Source: Interview)
6. Maintaining a friendship?	(Requires opt-in consent, not routinely recorded. Source: Interview)
7. Your day-to-day work/school?	(Requires opt-in consent, not routinely recorded. Source: Interview)

Instructions to the interviewer are written in bold - do not read these aloud.

Text for the respondent to hear is written in italic print in blue. Read this text aloud.

Say to respondent:
The interview is about difficulties people have because of health conditions.

By health condition I mean diseases or illnesses, or other health problems that may be short or long lasting; injuries; mental or emotional problems; and problems with alcohol or drugs.

Remember to keep all of your health problems in mind as you answer the questions. When I ask you about difficulties in doing an activity think about...

- Increased effort
- Discomfort or pain
- Slowness
- Changes in the way you do the activity When answering, I'd like you to think back over the past 30 days and, to the best of your knowledge, answer these questions thinking about how much difficulty your friend, relative or carer had while doing the following activities. I will use

the term "relative" to mean "friend", "relative" or "carer". For each question, please give only one response. In the past 30 days, how much difficulty did your relative have in:	
1. Standing for long periods such as 30 minutes?	(Requires opt-in consent, not routinely recorded. Source: Interview)
2. Taking care of his or her household responsibilities?	(Requires opt-in consent, not routinely recorded. Source: Interview)
3. Learning a new task, for example, learning how to get to a new place?	(Requires opt-in consent, not routinely recorded. Source: Interview)
4. How much of a problem did he or she have joining in community activities (for example, festivities, religious or other activities) in the same way as anyone else can?	(Requires opt-in consent, not routinely recorded. Source: Interview)
5. How much has your relative been emotionally affected by his or her health condition?	(Requires opt-in consent, not routinely recorded. Source: Interview)
In the past 30 days, how much difficulty did your relative have in:	
1. Concentrating on doing something for ten minutes?	(Requires opt-in consent, not routinely recorded. Source: Interview)
2. Walking a long distance such as a kilometre [or equivalent]?	(Requires opt-in consent, not routinely recorded. Source: Interview)
3. Washing his or her whole body?	(Requires opt-in consent, not routinely recorded. Source: Interview)
4. Getting dressed?	(Requires opt-in consent, not routinely recorded. Source: Interview)
5. Dealing with people he or she does not know?	(Requires opt-in consent, not routinely recorded. Source: Interview)
6. Maintaining a friendship?	(Requires opt-in consent, not routinely recorded. Source: Interview)
7. His or her day-to-day work/school?	(Requires opt-in consent, not routinely recorded. Source: Interview)
Number of days	
1. Overall, in the past 30 days, how many days were these difficulties present?	_____ (Requires opt-in consent, not routinely recorded. Source: Interview)
2. In the past 30 days, for how many days were you totally unable to carry out your usual activities or work because of any health condition?	_____ (Requires opt-in consent, not routinely recorded. Source: Interview)

3. In the past 30 days, not counting the days that you were totally unable, for how many days did you cut back or reduce your usual activities or work because of any health condition?	_____ (Requires opt-in consent, not routinely recorded. Source: Interview)
Any other comments?	_____

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Return To Work V1.0.01.10.24

1. Date of form filling	_____
2. Did the participant return to work?	
3. Date and time of return to work	_____ (Requires opt-in consent, not routinely recorded. Source: Interview)
4. Work status	(Requires opt-in consent, not routinely recorded. Source: Interview)
Any other comments?	_____

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Safety Events V1.0.01.10.24

1. Is any safety event reported?	
2. Date reported to trial management team of safety event	_____
3. Type of safety event	(Source: Medical record or treating physician)
4. Elaborate on other safety event	_____ (Source: Medical record or treating physician)
5. Investigator assessment of safety event	_____ (Source: Investigator)
Any other comments?	_____

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End Of Study V1.0.01.10.24

1. What is the reason for the end of study?

2. Date of end of study

_____

Any other comments?

_____

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ATLS
Page 28

Scheduling

Day 7

_____

Day 30

_____

Day 90

_____

Scheduling Page

Event Date
Date of arrival to the emergency departments [initial_assessment_arm_1][arrival_datetime]
Day 7 [day7]
Day 30 [day30]
Day 90 [day90]

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Review

Status

Reviewed by

Review comments

(For example, comments about main quality issues, if any.)